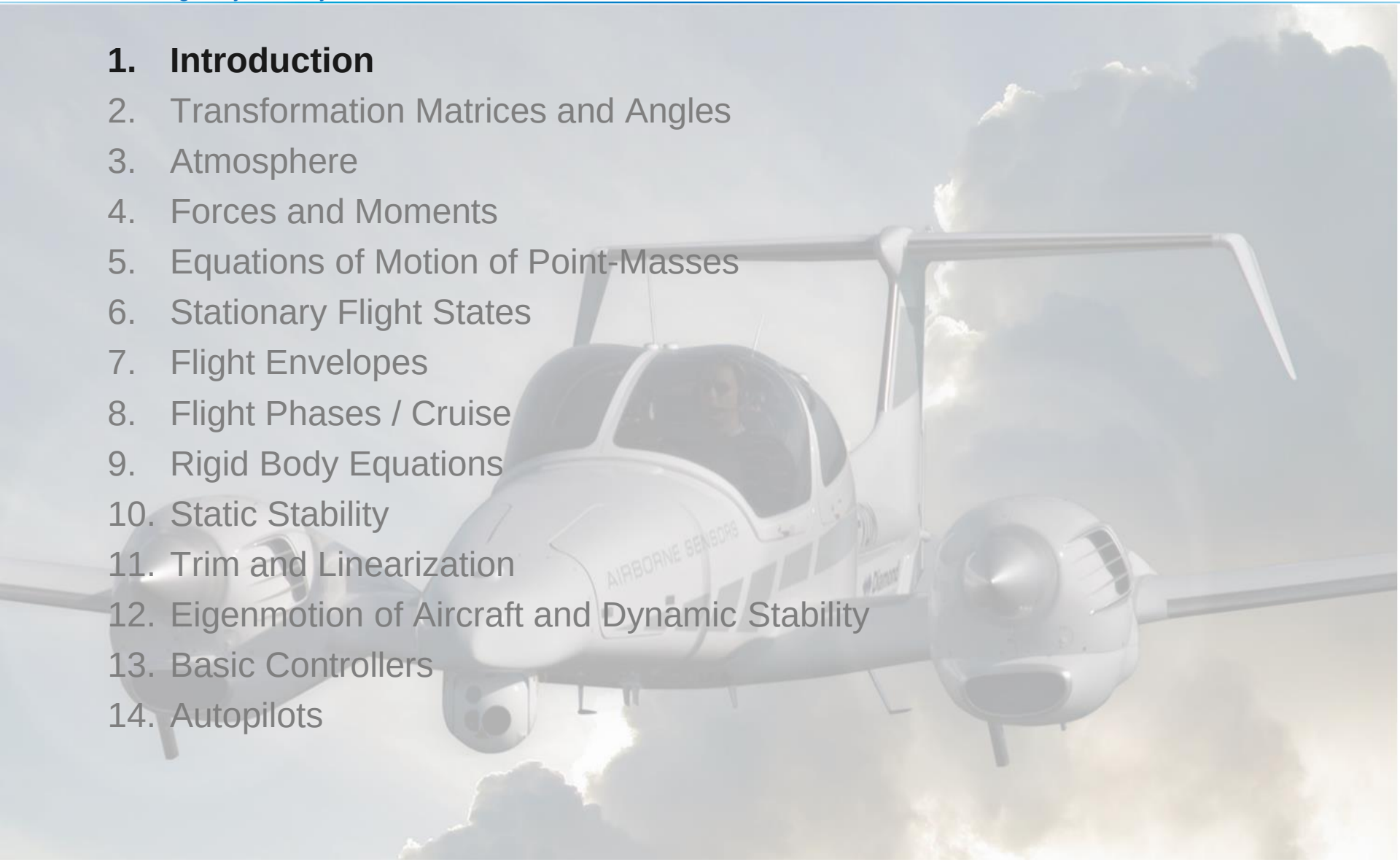
Introduction To Flight Mechanics

Recap



source: flugrevue.de



1. **Introduction**
2. Transformation Matrices and Angles
3. Atmosphere
4. Forces and Moments
5. Equations of Motion of Point-Masses
6. Stationary Flight States
7. Flight Envelopes
8. Flight Phases / Cruise
9. Rigid Body Equations
10. Static Stability
11. Trim and Linearization
12. Eigenmotion of Aircraft and Dynamic Stability
13. Basic Controllers
14. Autopilots

Important units in aerospace engineering

Knots <-> meter/second

1 m/s = 1.944 knots

or

1 knots = 0.514 m/s

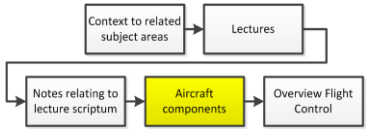
Feet <-> meter

1 m = 3.281 ft

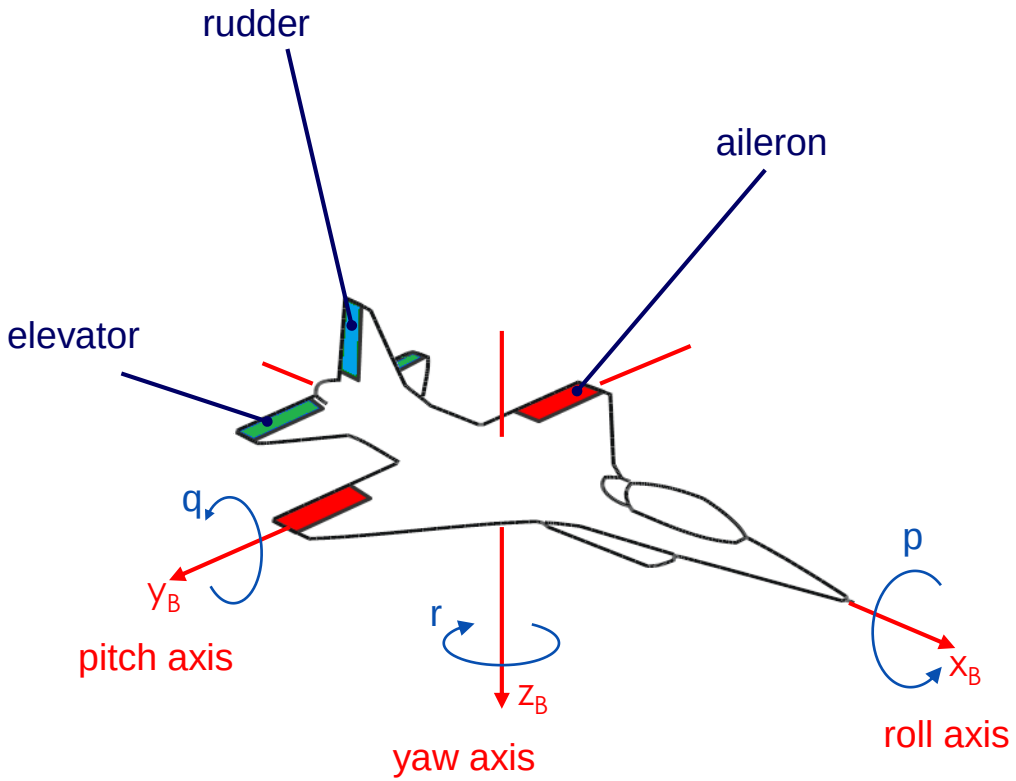
or

1 ft = 0.305 m

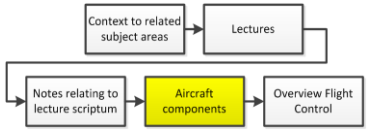
Knowing these conversions **could** be helpful in the exam!



Main Control Surfaces of a classic Fixed Wing Aircraft

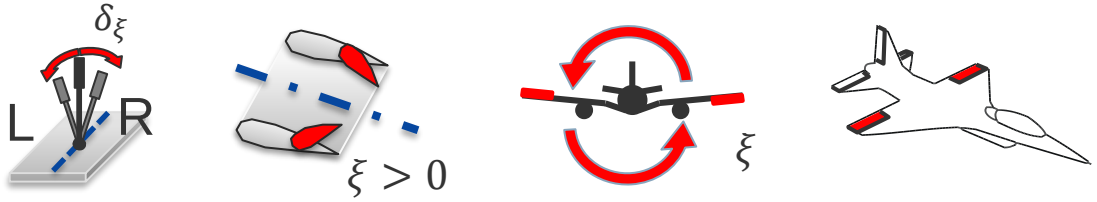


Control Variables of a classic Fixed Wing Aircraft



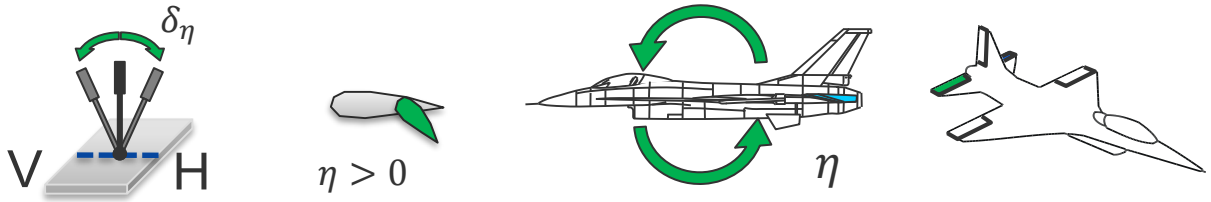
- **Yoke / Stick**

  Stick left / right: aileron, roll axis, roll



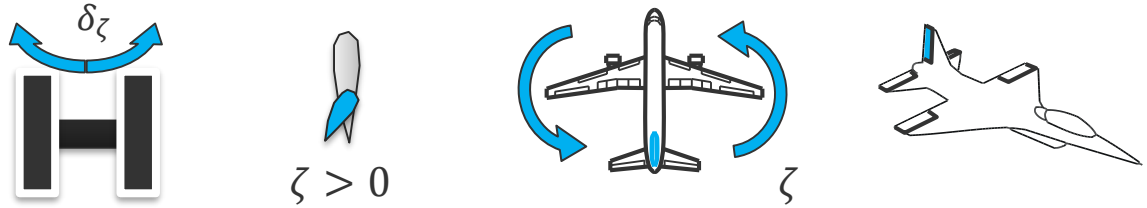
ξ Lateral Motion

  Stick forward / backward: elevator, pitch axis, pitch



η Longitudinal Motion

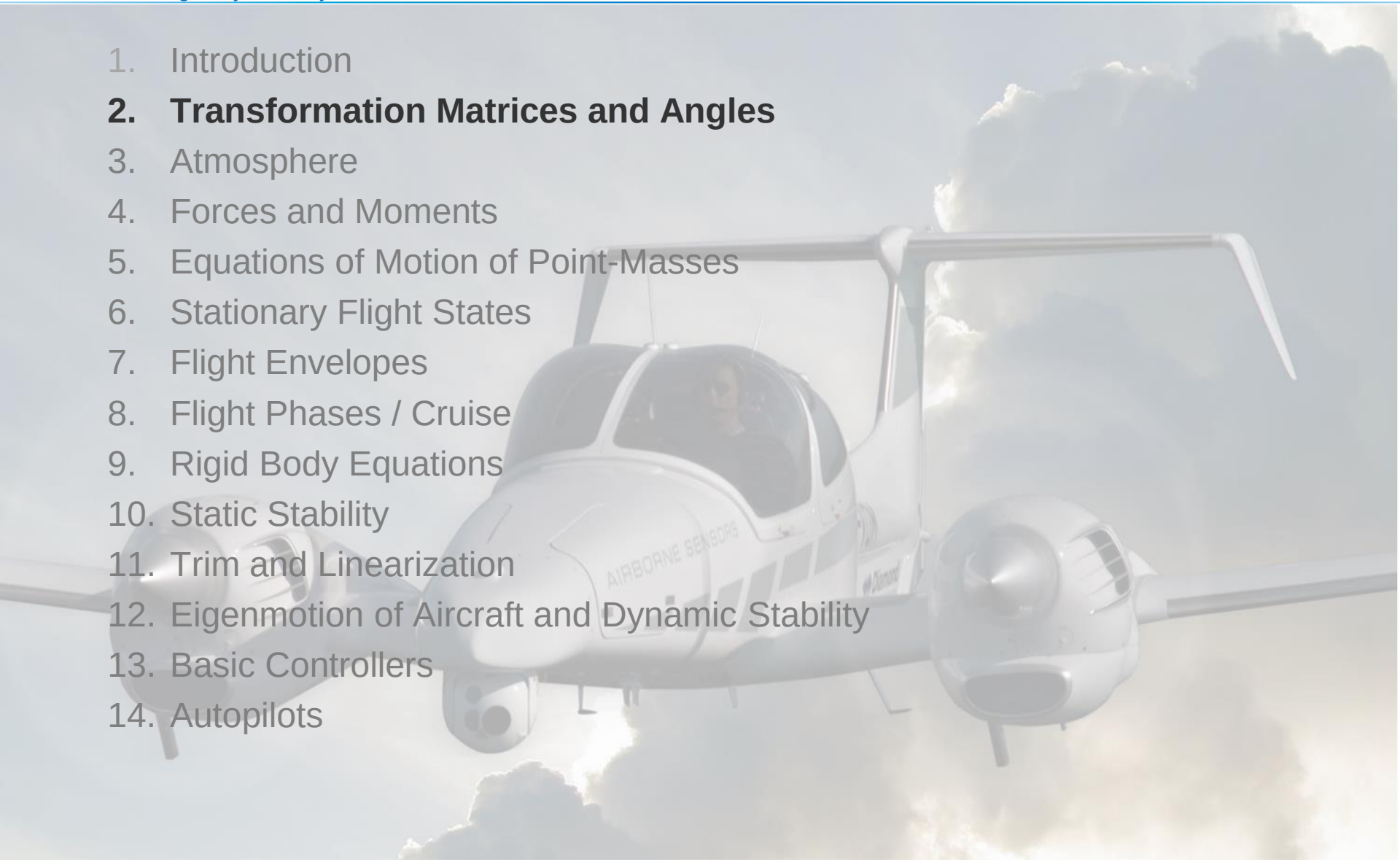
- **Pedals** left / right: rudder, yaw axis, yaw



ζ Lateral Motion

- **Throttle** δ_T: Engine thrust

δ_T Longitudinal Motion



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- **Position of point *G* relative the center of Earth** (*absolute position*)

$$(\vec{r}^{G})_B = \begin{bmatrix} x^G \\ y^G \\ z^G \end{bmatrix}_B$$

Components in *body fixed* system

- **Position of point *P* relative to point *G*** (*relative position*)

$$(\vec{r}^{GP})_B = \begin{bmatrix} x^{GP} \\ y^{GP} \\ z^{GP} \end{bmatrix}_B$$

Components in *body fixed* system

Positions are always given in m

- **Velocity of point *G* relative to the center of Earth *O***

$$\left(\frac{d}{dt}\right)^{E} (\vec{r}^{G})_B = (\vec{V}_K^G)_B^E = \begin{bmatrix} u_K^G \\ v_K^G \\ w_K^G \end{bmatrix}_B^E$$

Velocity relative to the *ECEF-frame*

Components in body fixed frame

- **Velocity of point *P* relative to point *G***

$$\left(\frac{d}{dt}\right)^{I} (\vec{r}^{GP})_B = (\vec{V}_K^{GP})_B^I = \begin{bmatrix} u_K^{GP} \\ v_K^{GP} \\ w_K^{GP} \end{bmatrix}_B^I$$

Velocity relative to the *ECI-frame*

Components in body fixed frame

Types of velocity

***K* – kinematic** **_W_ – wind** **_A_ – aerodynamic**

Velocities are always given in m/s

- **Acceleration of point *G***

Velocity relative to the *E*-frame differentiated relative to the B-frame

$$\left(\vec{a}_K^G\right)_B^{EB} = \left(\frac{d}{dt}\right)^B \left(\vec{V}_K^G\right)_B^E = \left(\frac{d}{dt}\right)^B \left[\left(\frac{d}{dt}\right)^E \left(\vec{r}^G\right)_B\right] = \begin{bmatrix} \dot{u}_K^G \\ \dot{v}_K^G \\ \dot{w}_K^G \end{bmatrix}_B^{EB}$$

Components in body fixed frame

- **Acceleration of point P relative to point G**

$$\left(\vec{a}_K^{GP}\right)_B^{IE} = \left(\frac{d}{dt}\right)^E \left(\vec{V}_K^{GP}\right)_B^I = \left(\frac{d}{dt}\right)^E \left[\left(\frac{d}{dt}\right)^I \left(\vec{r}^{GP}\right)_B\right] = \begin{bmatrix} \dot{u}_K^{GP} \\ \dot{v}_K^{GP} \\ \dot{w}_K^{GP} \end{bmatrix}_B^{IE}$$

Types of acceleration

***K* – kinematic** ***W* – wind** ***A* – aerodynamic**

Accelerations are always given in m/s²

- **Angular rate of *E*-frame relative to *I*-frame**

$$(\vec{\omega}_K^{IE})_O = \begin{bmatrix} \omega_{K,x}^{IE} \\ \omega_{K,y}^{IE} \\ \omega_{K,z}^{IE} \end{bmatrix}_O$$

Components in O-Frame

- **Angular rate of *B*-frame relative to *0*-frame**

$$\left(\vec{\omega}_K^{OB}\right)_B = \begin{bmatrix} \omega_{K,x}^{OB} \\ \omega_{K,y}^{OB} \\ \omega_{K,z}^{OB} \end{bmatrix}_B = \begin{bmatrix} p_K \\ q_K \\ r_K \end{bmatrix}_B$$

Components in body fixed system

Types of angular rate
***K* – kinematic** ***W* – wind** ***A* – aerodynamic**

Angular rates are always given in rad/s

- **Time derivative of angular rate in B-frame relative to I-frame, time derivation is relative to *O*-System (angular acceleration)**

Angular rate differentiated relative to the*O*-frame

$$\left(\dot{\vec{\omega}}_K^{IB}\right)_B^O = \left(\frac{d}{dt}\right)^O \left(\vec{\omega}_K^{IB}\right)_B = \begin{bmatrix} \dot{\omega}_{K,x}^{IB} \\ \dot{\omega}_{K,y}^{IB} \\ \dot{\omega}_{K,z}^{IB} \end{bmatrix}_B^O$$

Components in body fixed system

Types of angular acceleration

***K* – kinematic** ***W* – wind** ***A* – aerodynamic**

Angular accelerations are always given in rad/s²

Force is applied to reference point R

$$\left(\vec{F}_A^R\right)_B = \begin{bmatrix} X_A^R \\ Y_A^R \\ Z_A^R \end{bmatrix}_B$$

Force components are called "X,Y,Z"

Components in body fixed system

Physical force origin

T	*–total*
A	*–aerodynamic*
P	–propulsive
G	–gravitational

Forces are always given in N

Moments applied to reference point R

$$\left(\vec{M}_A^R\right)_B = \begin{bmatrix} L_A^R \\ M_A^R \\ N_A^R \end{bmatrix}_B$$

Components are called „L,M,N“

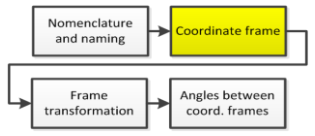
Components in body fixed system

Type of moment

T	– *total*
A	– *aerodynamic*
P	–*propulsive*
G	–*gravitational*

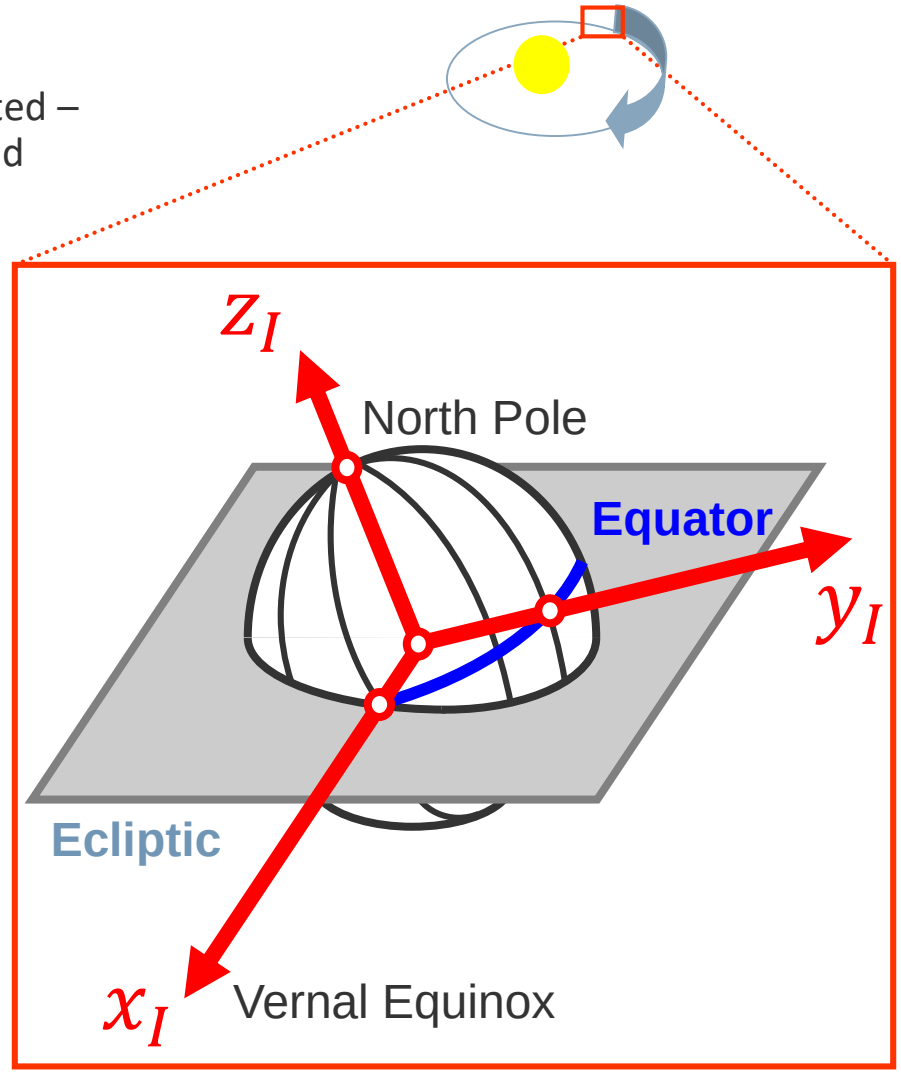
Moments are always given in Nm

ECI - Frame (Earth-Centered Inertial)



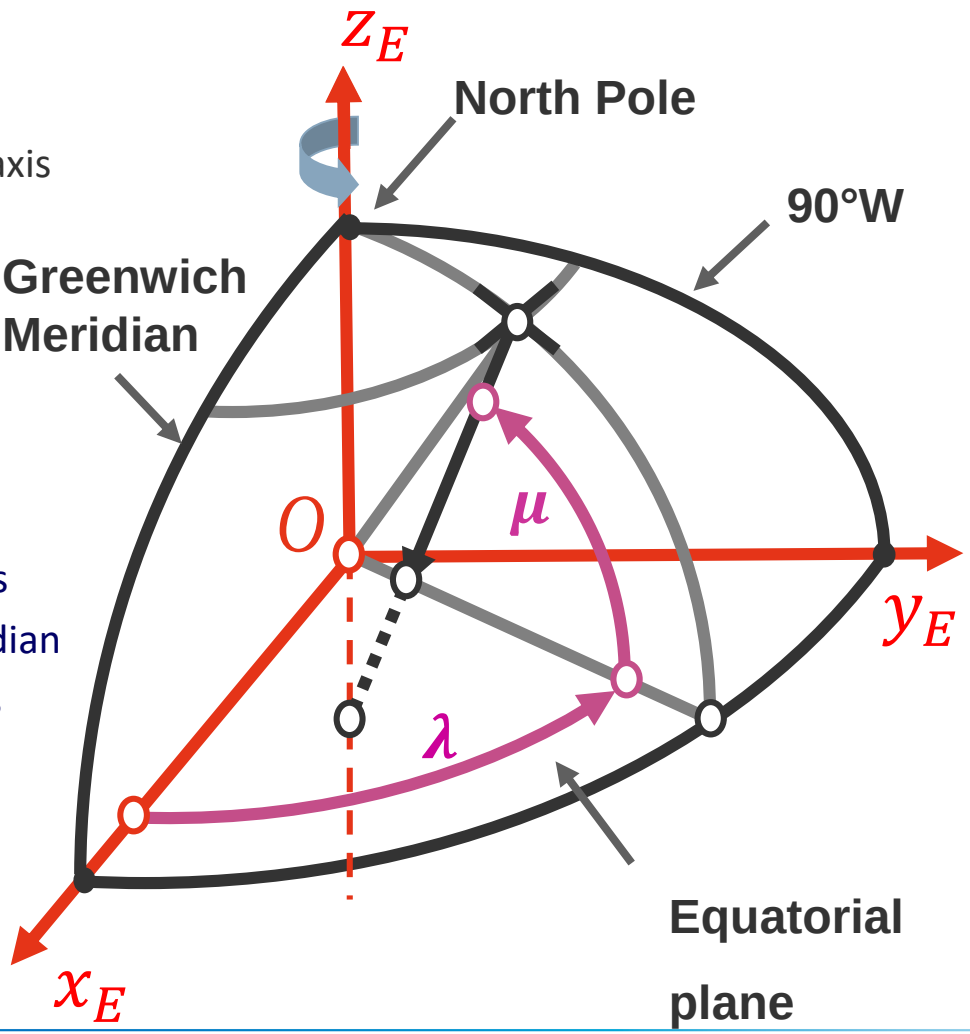
Index:	I
Role:	Euclidean Frame Inertial system, not accelerated – Newton's Laws can be applied
Origin:	Center of Earth
Translation:	Around the sun, with the solar system
Rotation:	none
x-Axis:	In equatorial plane, pointing towards the vernal equinox
y-Axis:	In equatorial plane, forms right-handed coordinate system with z- and x-axis
z-Axis:	Axis of rotation of Earth

Ecliptic: Plane of the trajectory of Earth around the sun



ECEF - Frame (Earth Centered Earth Fixed)

Index:	$\boldsymbol{E}$
Role:	Navigational/positional frame
Origin:	Center of Earth
Translation:	same as ECI-frame
Rotation:	Earth rotation (around z-axis with Earth rate, i.e. about 1x (=2π) per 24h) $(\vec{\omega}^{IE})$
x-Axis:	in Equatorial plane, points towards Greenwich Meridian
y-Axis:	In Equatorial plane, forms right-handed coordinate system with x- and z-axis
z-Axis:	Same as ECI frame



The position of a point in ECEF-frame is either given in cartesian coordinates x, y, z or in WSG-84 coordinates, which are composed of two angles and the height over the reference ellipsoid

WGS-84 are relative to ECEF-frame.

Geodetic Longitude: λ

angle, measured in Equatorial plane, between prime meridian (xz-plane in ECEF-frame) and the meridial plane of point *P*.

Range: $-\pi \leq \lambda < \pi$

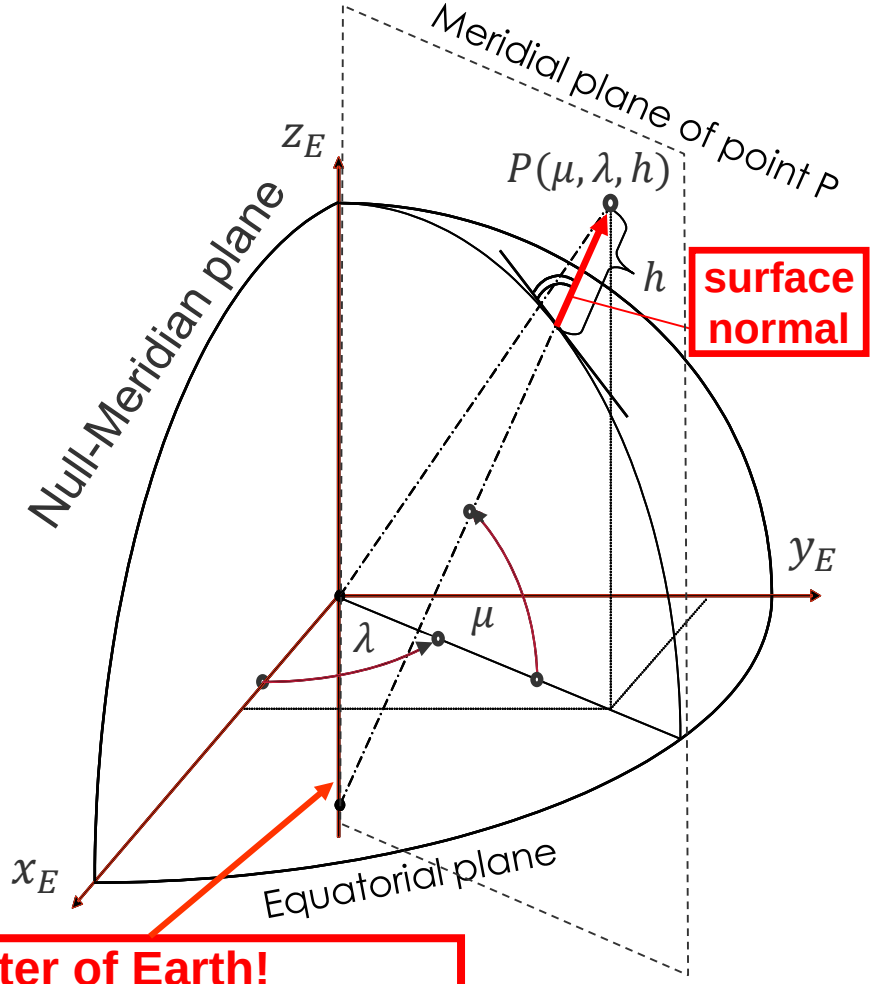
Geodetic Latitude: μ

angle, measured on meridial plane of point P, between Equatorial plane (xy-plane of ECEF-frame) and the surface normal point P.

Range: $-\pi/2 \leq \mu \leq \pi/2$

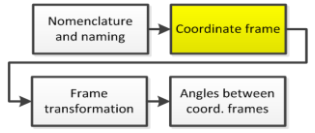
Geodetic Height: h

Height over WGS-84 ellipsoid along surface normal.

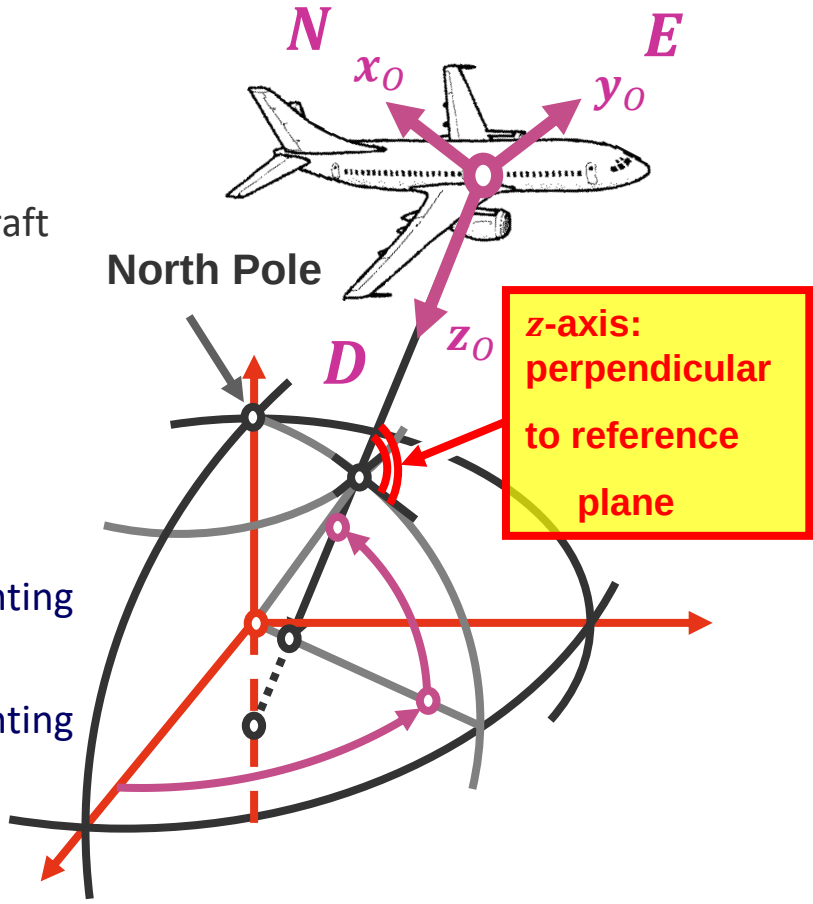


Surface normal does not pass the center of Earth!

NED-Frame (North-East-Down Frame)

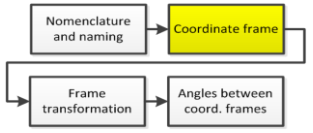


Index:	O
Role:	Attitude /Orientation system – Used to determine attitude of aircraft
Origin:	Reference point of aircraft
Translation:	Moves with reference point of aircraft
Rotation:	Rotates with transport rate of aircraft in order to fullfill NED requirement $(\vec{\omega}^{EO})$

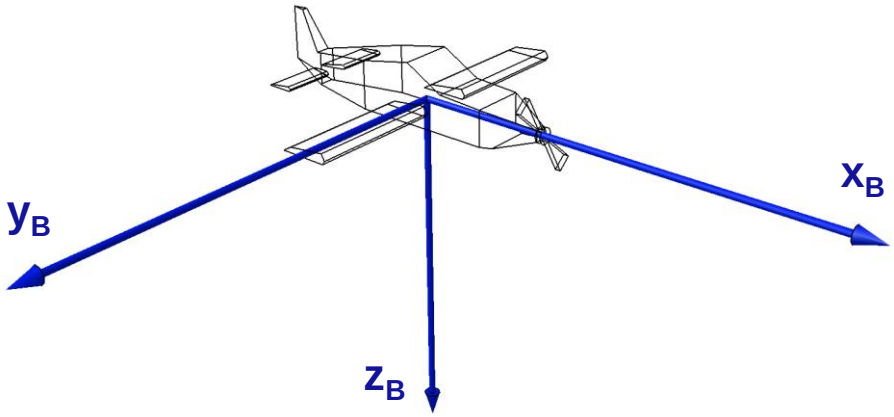


x-axis:	Parallel to local Geoid surface, pointing toward geographical North Pole
y-axis:	Parallel to local Geoid surface, pointing towards East to form right-handed system with x-axis and and z-axis
z-axis:	Points downward, perpendicular to local Geoid surface

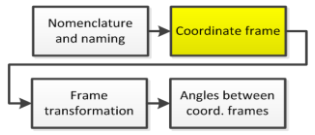
Body Fixed Frame



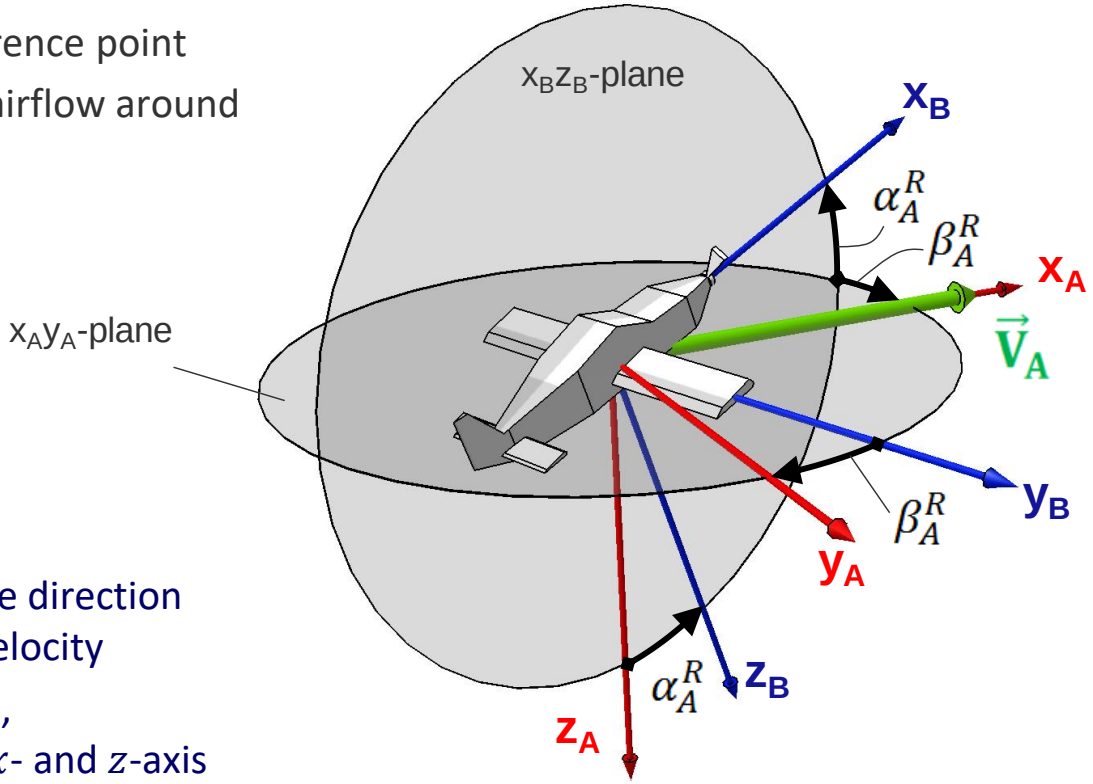
Index:	$\boldsymbol{B}$
Role:	noting forces and moments, aircraft geometry
Origin:	Reference point of aircraft
Translation:	With aircraft reference point
Rotation:	With aircraft reference point



x-axis:	Points to the nose of the aircraft xz plane of symmetry
y-axis:	Points to the starboard wing, forms an orthogonal right-handed system
z-axis:	Points downwards in symmetry plane of aircraft, perpendicular to x- and y-axis

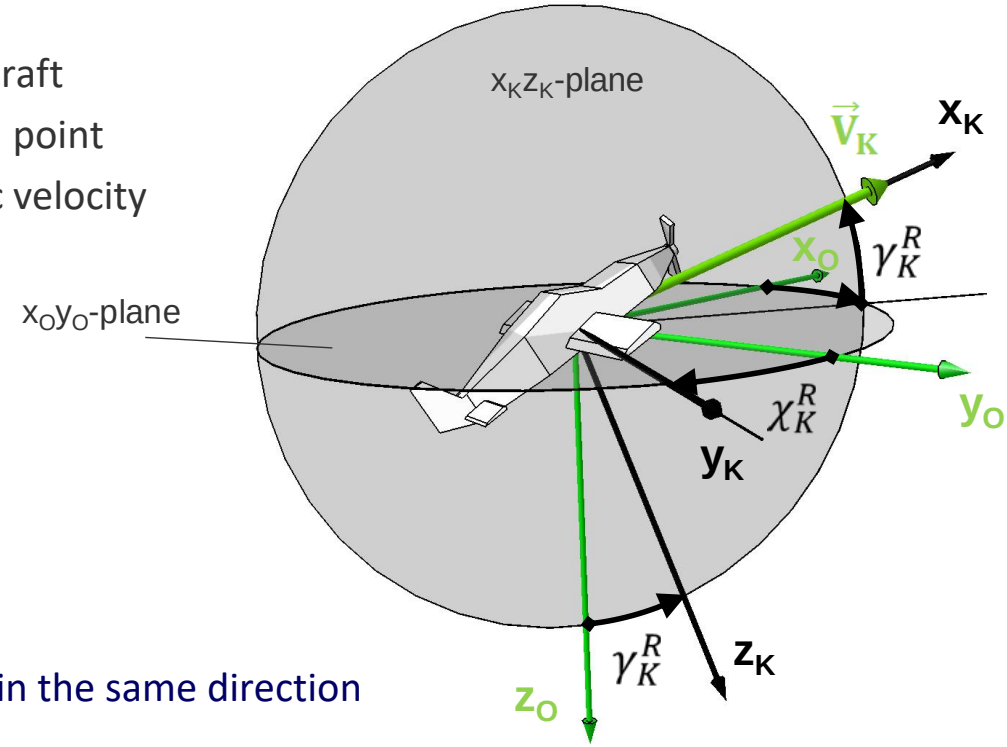


Index:	A
Role:	For noting aerodynamic forces and moments
Origin:	Reference point of aircraft
Translation:	With aircraft reference point
Rotation:	Rotates with the airflow around the aircraft



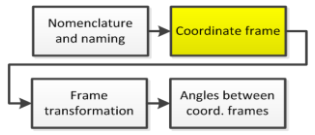
x-axis:	Collinear and same direction as aerodynamic velocity
y-axis:	Points to the right, perpendicular to x- and z-axis
z-axis:	Points downward in symmetry plane of the aircraft, perpendicular to xy-plane

Index:	$\boldsymbol{K}$
Role:	Flight path frame
Origin:	Reference point of aircraft
Translation:	With aircraft reference point
Rotation:	Rotates with kinematic velocity of aircraft

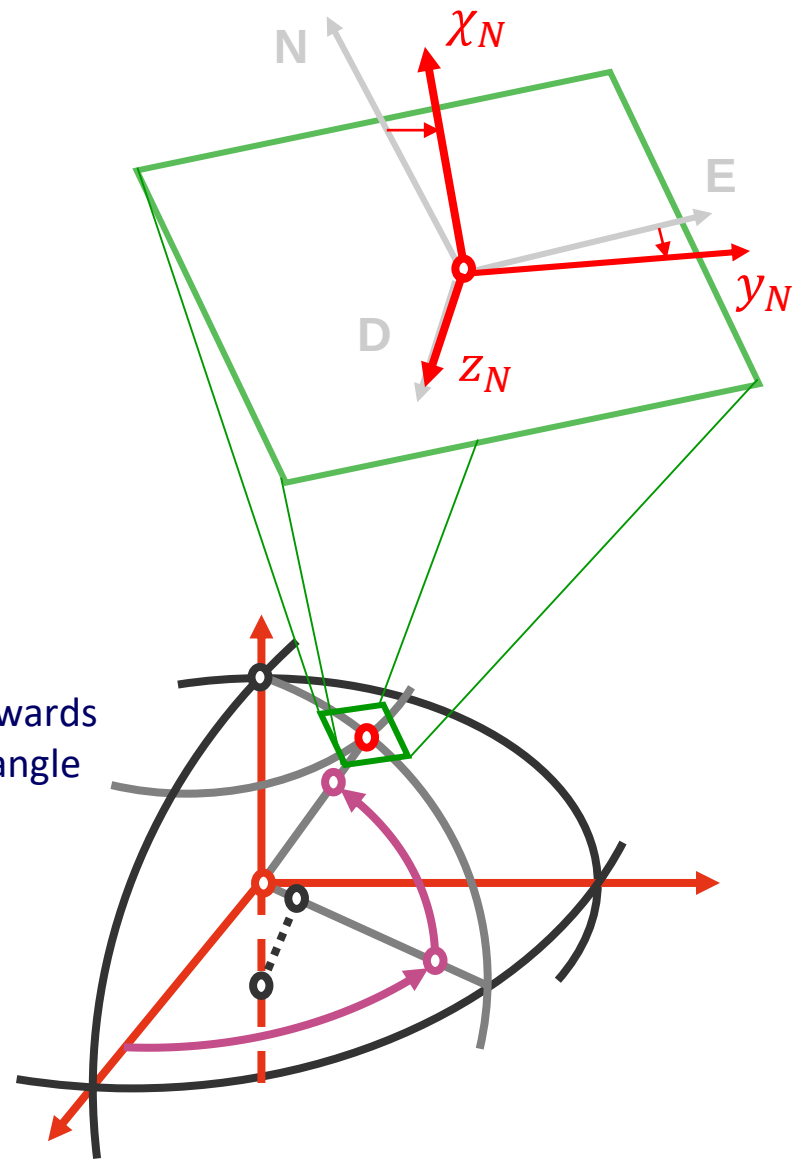


x-axis:	Collinear and pointing in the same direction as kinematic velocity
y-axis:	Pointing to the right-hand side, perpendicular to x- and z-axis
z-axis:	Pointing downwards, parallel to projection of local surface normal of WGS-84 reference ellipsoid in a plane perpendicular to x-axis (i.e. the kinematic velocity)

Navigation Frame

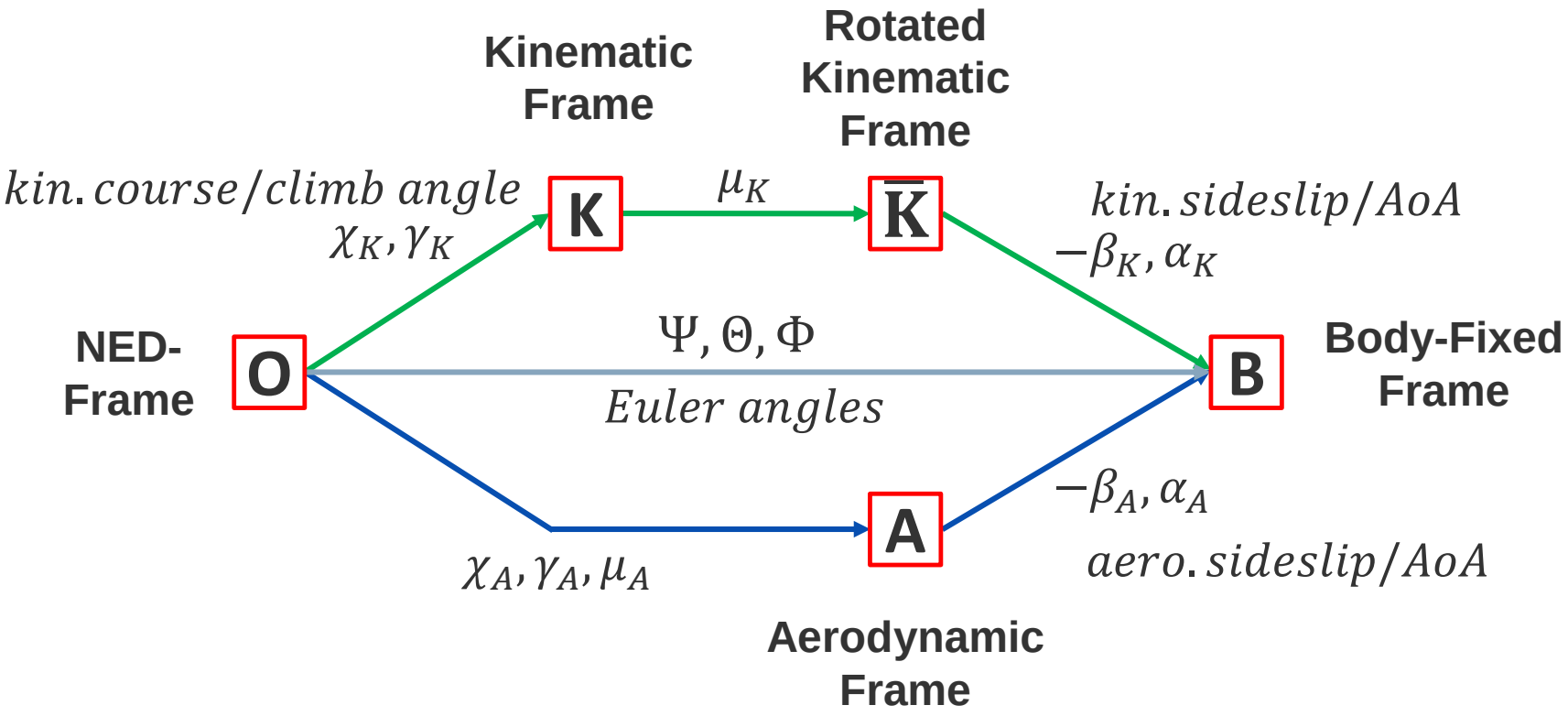


Index:	N
Role:	Navigation system, derived from NED frame in order to determine local position
Origin:	Point on Earth‘s surface
Translation:	None
Rotation:	Only Earth‘s rate, since it is fixed to it (does not turn with transport rate)

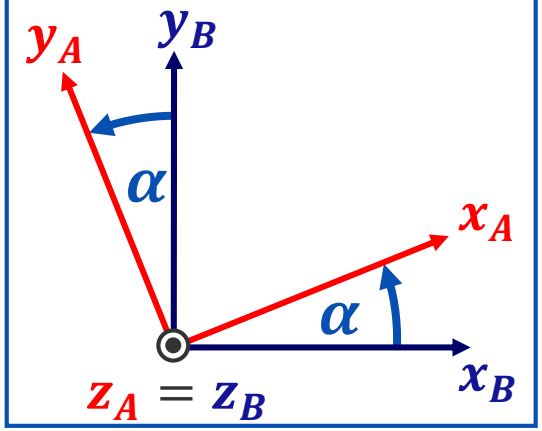


x-axis:	Parallel to local Geoid surface, points towards the direction that is determined by the angle χ_N off the line pointing North.
y-axis:	Parallel to local Geoid surface, forms right-hand system with x- and z-axis
z-axis:	Pointing downward, perpendicular to local Geoid surface

(no wind -> A = $\bar{K}$ frame)



See Exercise 4 for more information on what is relevant wrt. the exam.



Rotation by angle α:

$$\begin{bmatrix} x_A \\ y_A \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} x_B \\ y_B \end{bmatrix}$$

Properties of rotation matrices (3x3); orthonormal:

$$\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I} \quad \boldsymbol{A}^T = \boldsymbol{A}^{-1} \quad det(\boldsymbol{A}) = 1$$

Inverted by transposing

Orthogonal Matrix $\boldsymbol{M}_{BA} = \boldsymbol{M}_{AB}^T$

Rot. angle: $\cos\varphi = 0{,}5(Sp\,\boldsymbol{A} - 1)$

Rot. axis: $(\boldsymbol{A} - \boldsymbol{E}) \cdot \boldsymbol{a} = 0$

Rotation around x-axis

$$\boldsymbol{M}_{AB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{bmatrix}$$

Rotation around y-axis

$$\boldsymbol{M}_{AB} = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix}$$

Rotation around z-axis

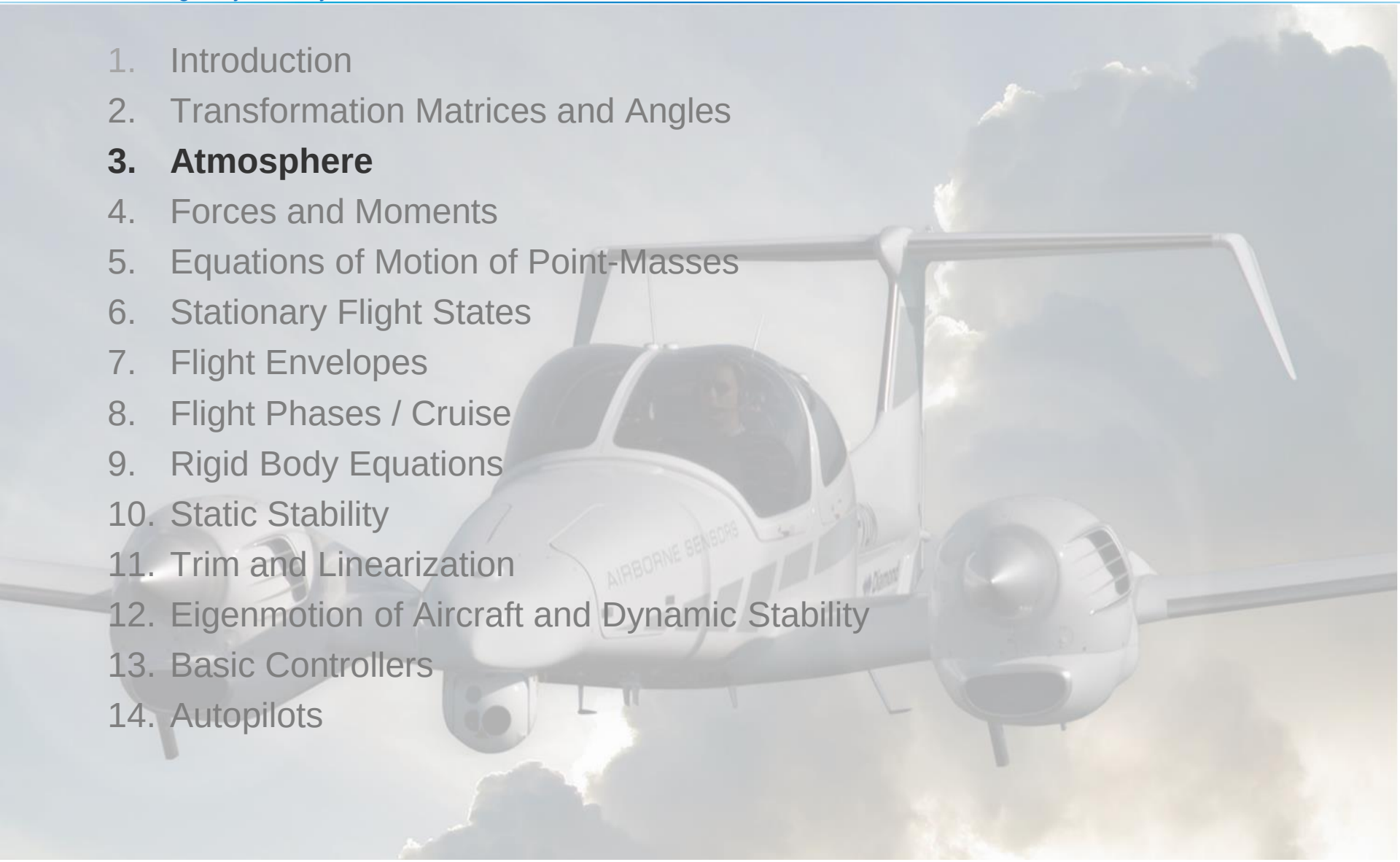
$$\boldsymbol{M}_{AB} = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Successive rotations around coordinate system axis are <u>not</u> commutative!

Pay attention to the order of rotation when constructing a rotation matrix created through multiplication!

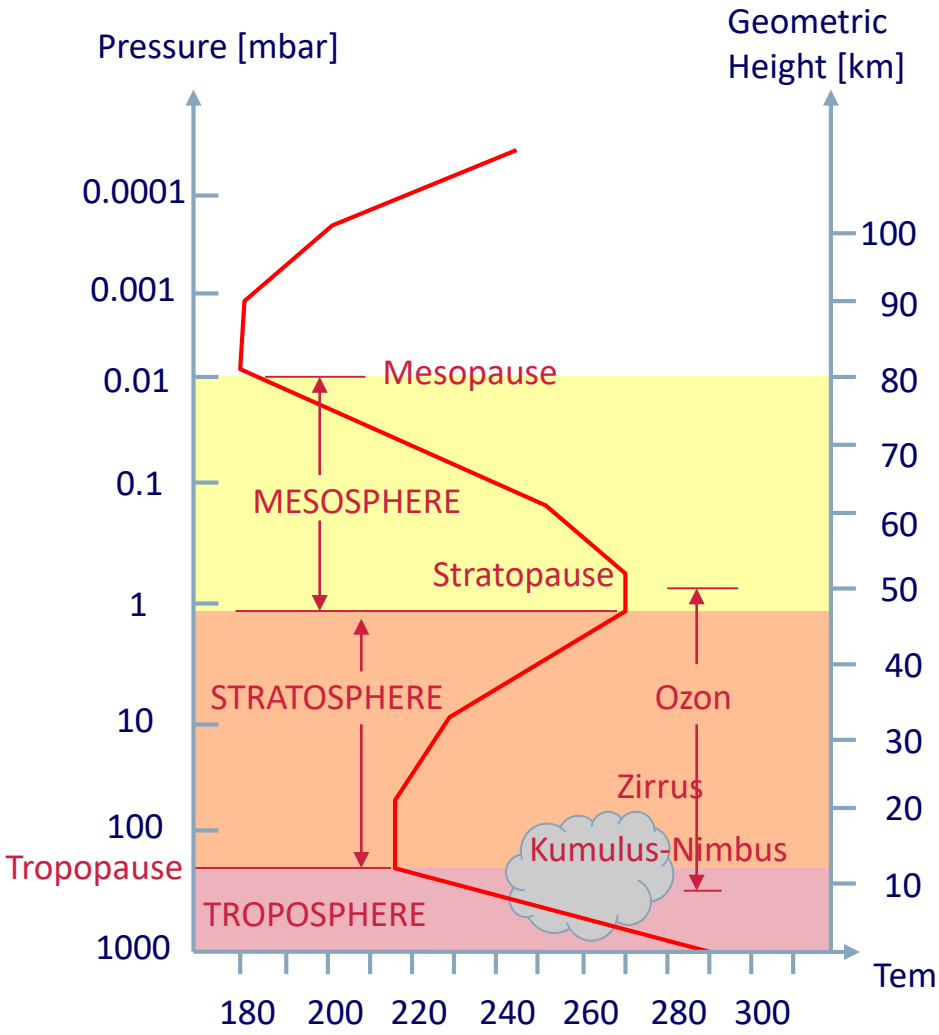
$$\boldsymbol{M}_{321} = \boldsymbol{M}_3 \cdot \boldsymbol{M}_2 \cdot \boldsymbol{M}_1 \neq \boldsymbol{M}_1 \cdot \boldsymbol{M}_2 \cdot \boldsymbol{M}_3$$

Order of rotation: 1 - 2 - 3



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Air Layers Atmosphere

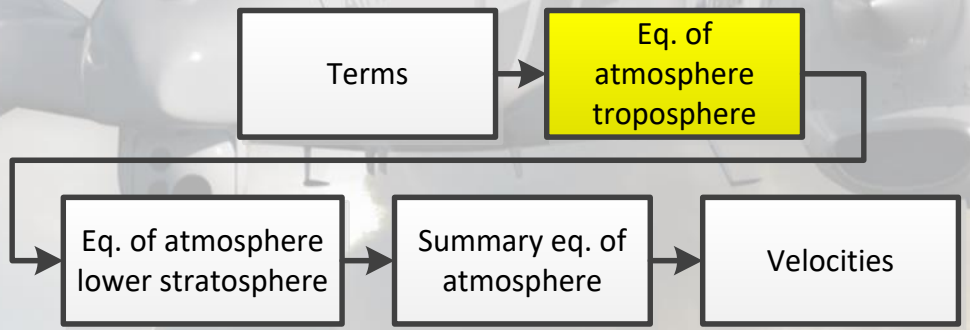


- Possible ways to describe the atmosphere:
  - analytically
  - numerically
  - statistically

- Components of dry air at sea level (relative to volume)
  - Nitrogen 78.1 %
  - Oxygen 20.9 %
  - Argon 0.9 %
  - Carbon dioxide 0.04 %
  - Other elements only as traces (ppm)

3. Atmosphere

3.1 Terms
3.2 Equations of atmosphere troposphere / upper stratosphere
3.3 Equations of atmosphere lower stratosphere
3.4 Summary equations of atmosphere
3.5 Velocities



Geopotential height H_G

Defined as the height that takes changes of gravity with height into account.

$$H_G = \frac{r_E \cdot h}{r_E + h}$$

Earth radius according to ISO 2533: $r_E = 6356766m$
Geodetic height: h

By using the geopotential height within the equations, we can assume the earth acceleration $\boldsymbol{g_s}$ constant. The error of this assumption (in reality $\boldsymbol{g = g(h)}$ decreases with height), is compenstated by using $\boldsymbol{H_G}$.

$$\frac{dp}{p} = -\frac{\rho\, g(h) dh}{\rho\, R\, T} = -\frac{g(h)}{R\, T} dh \xrightarrow{g(h)dh = g_s\, dH_G} \boxed{\frac{dp}{p} = -\frac{\rho\, g_s\, dH_G}{\rho\, R\, T} = -\frac{g_s}{R\, T} dH_G}$$

Results equations of atmosphere

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\frac{n}{n-1}}$$

With the upper functions, we can write now the equations for temperature, pressure and density by using the polytrope exponent:

1. **Temperature:** Assumption of a **linear curve**

$$T(H_G) = T_R + \gamma(H_G - H_{GR}) \qquad \gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

$$\frac{T(H_G)}{T_R} = \left[1 - \left(\frac{n-1}{n}\right)\frac{g_S}{RT_R}(H_G - H_{GR})\right]$$

2. **Pressure:** The relation between pressure and temperature leads to:

$$\frac{p(H_G)}{p_R} = \frac{T(H_G)}{T_R}^{-\frac{g_S}{\gamma R}} \qquad \gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

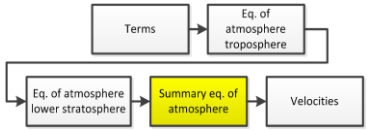
$$\frac{p(H_G)}{p_R} = \left[1 - \left(\frac{n-1}{n}\right)\frac{g_S}{RT_R}(H_G - H_{GR})\right]^{\frac{n}{n-1}}$$

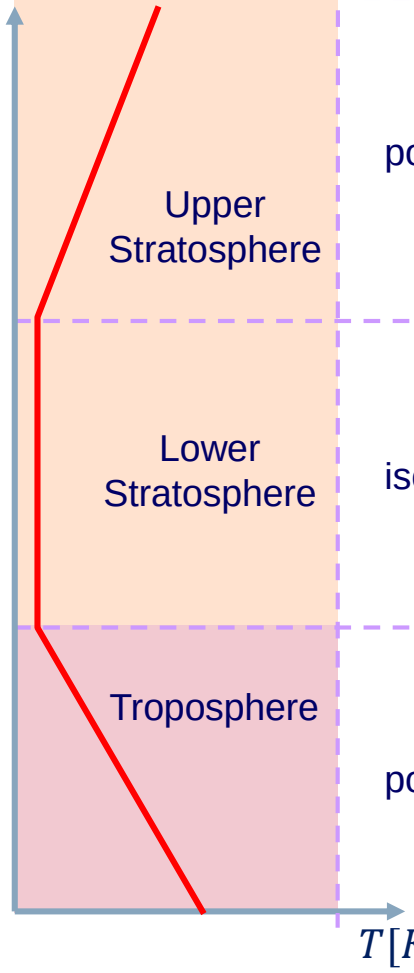
3. **Density:** The relation between density and temperature leads to:

$$\frac{\rho(H_G)}{\rho_R} = \frac{T(H_G)}{T_R}^{-\left(\frac{g_S}{\gamma R}+1\right)} \qquad \gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

$$\frac{\rho(H_G)}{\rho_R} = \left[1 - \left(\frac{n-1}{n}\right)\frac{g_S}{RT_R}(H_G - H_{GR})\right]^{\frac{1}{n-1}}$$

Standard Atmosphere– ISO 2533





Layer	Features Of Layer	Atmosphere Equations	Data	Height
Upper Stratosphere	polytrope	$\frac{T}{T_{20}} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_{20}}(H_G - H_{G20})\right] = \left[1 + \gamma_{OStr}\frac{(H_G - H_{G20})}{T_{20}}\right]$ $\frac{p}{p_{20}} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_{20}}(H_G - H_{G20})\right]^{\frac{n}{n-1}}$ $\frac{\rho}{\rho_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}(H_G - H_{GR})\right]^{\frac{1}{n-1}}$ $n_{OStr} = 0{,}972$ $\gamma_{OStr} = +1{,}0e-3$	$T_{20} = 216{,}65K$ $\rho_{20} = 0{,}088\frac{kg}{m^3}$ $p_{20} = 547{,}488Pa$ $H_{G20} = 20000m$	20 to 32 km
Lower Stratosphere	isotherm	$\frac{T}{T_{11}} = 1$ $\frac{p}{p_{11}} = e^{-\frac{g_S}{R\,T_{11}}(H_G - H_{G11})}$ $\frac{\rho}{\rho_{11}} = e^{-\frac{g_S}{R\,T_{11}}(H_G - H_{G11})}$ $n_{UStr} = 1$ $\gamma_{UStr} = 0$	$T_{11} = 216{,}65K$ $\rho_{11} = 0{,}364\frac{kg}{m^3}$ $p_{11} = 22632Pa$ $H_{G11} = 20000m$	11 to 20 km
Troposphere	polytrope	$\frac{T}{T_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}H_G\right] = \left[1 + \frac{\gamma_{Tr}\,H_G}{T_S}\right]$ $\frac{p}{p_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}H_G\right]^{\frac{n}{n-1}}$ $\frac{\rho}{\rho_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}H_G\right]^{\frac{1}{n-1}}$ $n_{Tr} = 1{,}235$ $\gamma_{Tr} = -6{,}5e-3$	$T_S = 288{,}15K$ $\rho_S = 1{,}225\frac{kg}{m^3}$ $p_S = 101325Pa$ $H_{GS} = 0m$	-2 to 11 km

$g_S = 9{,}80665\frac{m}{s^2}$ $\quad R = 287{,}05\frac{J}{kg\,K}$

$$n = \frac{1}{1 + \frac{\gamma\,R}{g_S}} = \frac{g_S}{g_S + \gamma\,R}$$

$$\gamma = \frac{dT}{dH_G} = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

Sample Exercise E.1

Where do 100 meters of height difference have a higher impact on the static pressure, at mean sea level or at 3000m height? Justify your response with a calculation and use the standard atmosphere.

$$\gamma = -0.0065\ \mathrm{K/m}$$
$$r_E = 6356766\ m$$
$$g_S = 9{,}80665\ \frac{m}{s^2}$$

$$T(H_G) = T_R + \gamma(H_G - H_{GR})$$

$$\gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

$$\frac{p(H_G)}{p_R} = \frac{T(H_G)}{T_R}^{-\frac{g_S}{\gamma R}}$$

$$H_G = \frac{r_E \cdot h}{r_E + h}$$

1) At MSL

$$\frac{p(H_{G,h=100})}{p_{MSL}} = \frac{T(H_{G,h=100})}{T_{MSL}}^{-\frac{g_S}{\gamma R}} = \frac{287.50K}{288{,}15K}^{5.26} = 0.9882$$

$$H_{G,MSL} = 0m \qquad T_{MSL} = 288{,}15K$$

$$H_{G,h=100} = \frac{r_E \cdot 100m}{r_E + 100m} = 100.00m \qquad T(H_{Gh=100}) = T_{MSL} + \gamma(H_G - H_{G,MSL}) = 287.50K$$

Where do 100 meters of height difference have a higher impact on the static pressure, at mean sea level or at 3000m height? Justify your response with a calculation and use the standard atmosphere.

$$\gamma = -0.0065\ \mathrm{K/m}$$
$$r_E = 6356766\ m$$
$$g_S = 9{,}80665\ \frac{m}{s^2}$$

$$T(H_G) = T_R + \gamma(H_G - H_{GR})$$

$$\gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

$$\frac{p(H_G)}{p_R} = \frac{T(H_G)}{T_R}^{-\frac{g_S}{\gamma R}}$$

$$H_G = \frac{r_E \cdot h}{r_E + h}$$

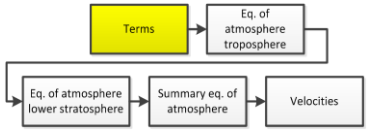
2) At MSL

$$\frac{p(H_{G,h=3100})}{p(H_{G,h=3000})} = \frac{T(H_{G,h=3100})}{T(H_{G,h=3000})}^{-\frac{g_S}{\gamma R}} = \frac{268.01K}{268.67K}^{5.26} = 0.9873$$

$$H_{G,h=3000} = \cdots = 2998.58m \qquad T(H_{G,h=3000}) = T_{MSL} + \gamma\left(H_{G,h=3000} - H_{G,MSL}\right) = 268.66\mathrm{K}$$

$$H_{G,h=3100} = \cdots = 3098{,}49m \qquad \mathrm{T(H_{G,h=3100})} = \mathrm{T(H_{G,h=3000})} + \gamma\left(\mathrm{H_{G,h=3100}} - \mathrm{H_{G,h=3000}}\right) = 268.01\mathrm{K}$$

Airspeeds

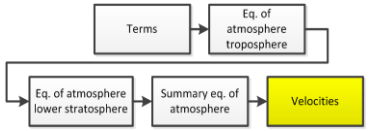


- **Indicated Airspeed (IAS):**
  - Flight speed that is measured in the airplane
  - Used in Flight operations (stall, max. speed, etc.)

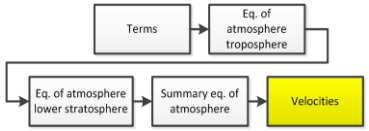
- **Calibrated Airspeed (CAS):**
  - CAS corrects instrument errors in IAS
  - Errors that occur because of a wrong placement of the „Static Port“ (measuring static pressure) are compensated

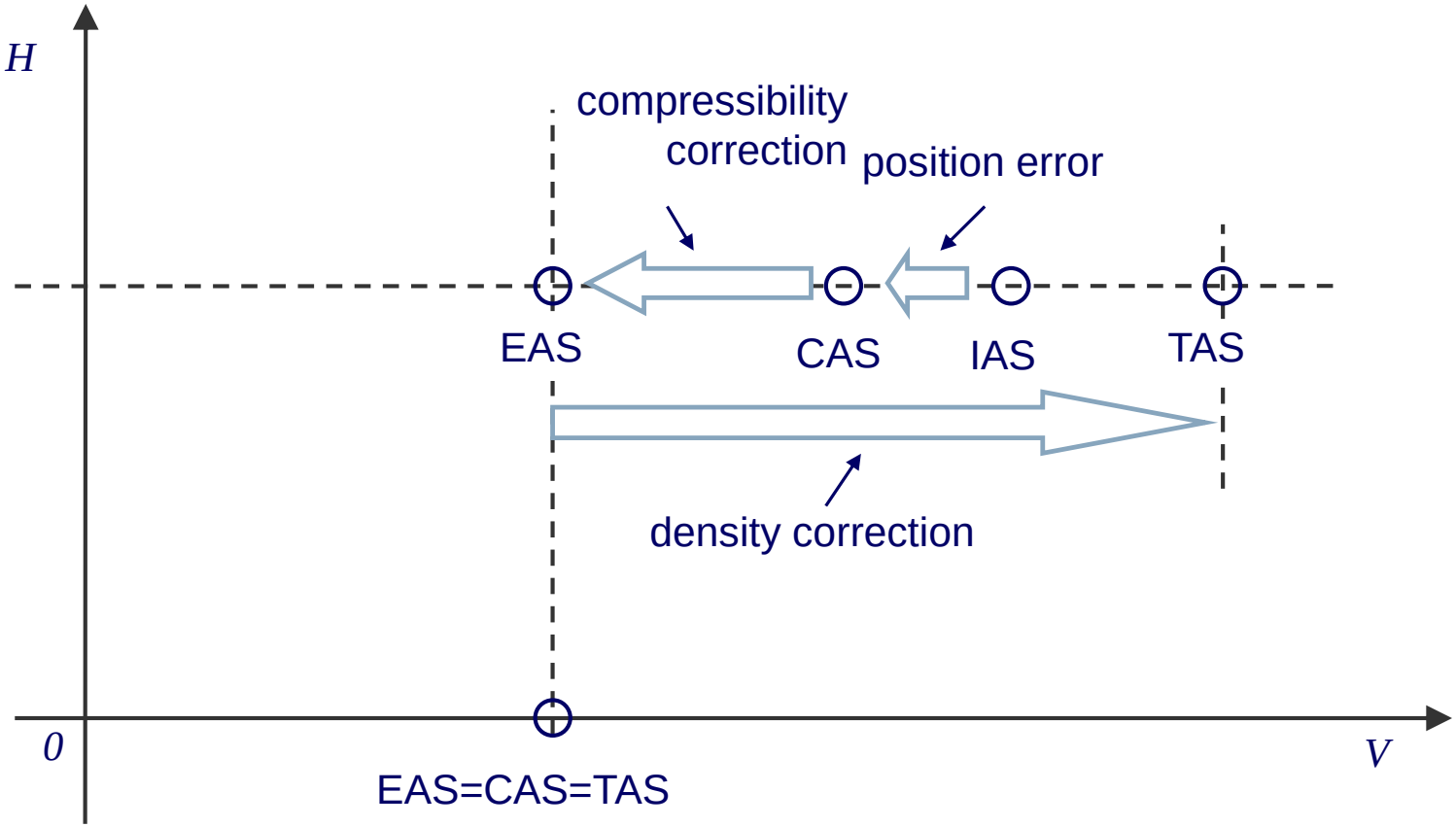
- **Equivalent Airspeed (EAS):**
  - EAS corrects compressibility effects
  - EAS doesn‘t change with height when flying with a constant dynamic pressure, used in R&D (flight envelopes)

- **True Airspeed (TAS):**
  - EAS with additional compensation of density variation with altitude
  - Used for performance calculations (curve radii)

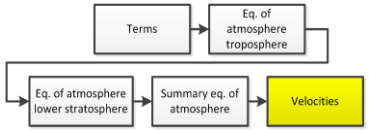


	Incompressible	Compressible
CAS	$V_{CAS} = \sqrt{\frac{2 \cdot (p_T - p)}{\rho_{MSL}}}$	$V_{CAS} = a_{MSL} \sqrt{\frac{2}{\kappa - 1}\left(\left(\frac{(p_T - p)}{p_{MSL}} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)}$
EAS	$V_{EAS} = V_{CAS}$	$V_{EAS} = a \sqrt{\frac{2}{\kappa - 1} \cdot \frac{\rho}{\rho_{MSL}}\left(\left(\frac{(p_T - p)}{p} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)} = V_{TAS}\sqrt{\frac{\rho}{\rho_{MSL}}}$
TAS	$V_{TAS} = V_{EAS}\sqrt{\frac{\rho_{MSL}}{\rho}}$	$V_{TAS} = a \sqrt{\frac{2}{\kappa - 1}\left(\left(\frac{(p_T - p)}{p} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)} = V_{EAS}\sqrt{\frac{\rho_{MSL}}{\rho}}$


Terms
Eq. of atmosphere troposphere
Eq. of atmosphere lower stratosphere
Summary eq. of atmosphere
Velocities


H
compressibility correction
position error
EAS
CAS
IAS
TAS
density correction
0
V
EAS=CAS=TAS

Mach Number and Mach Regions (1)

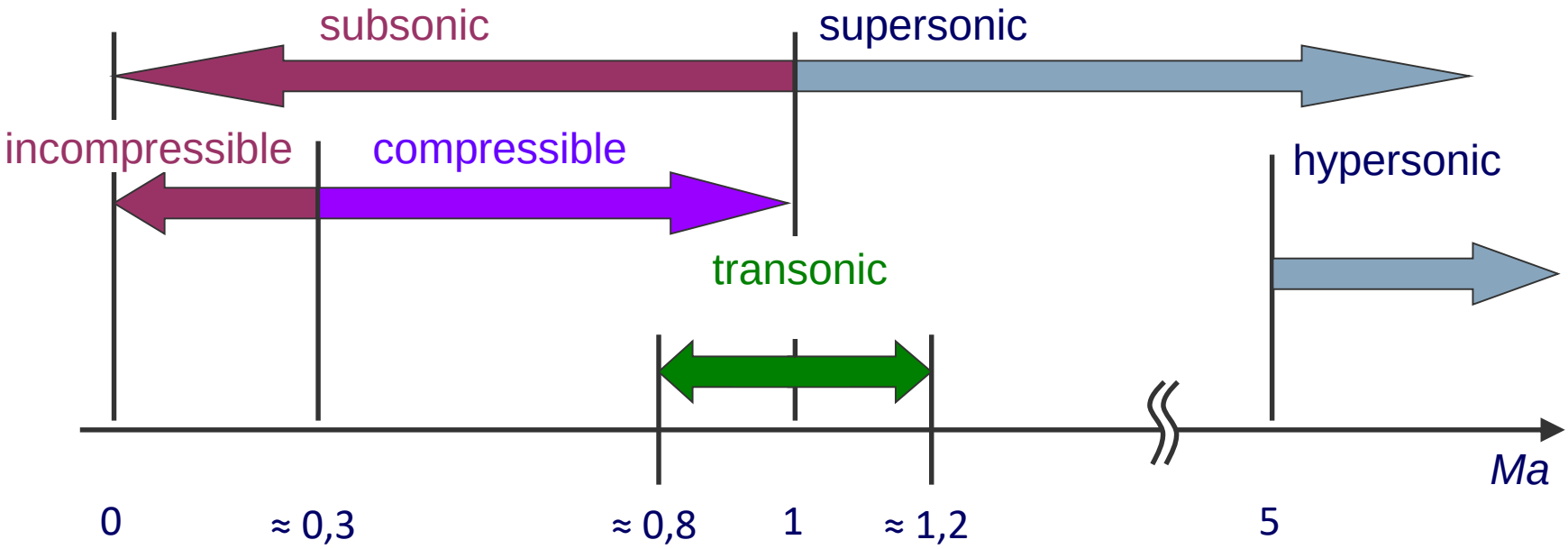


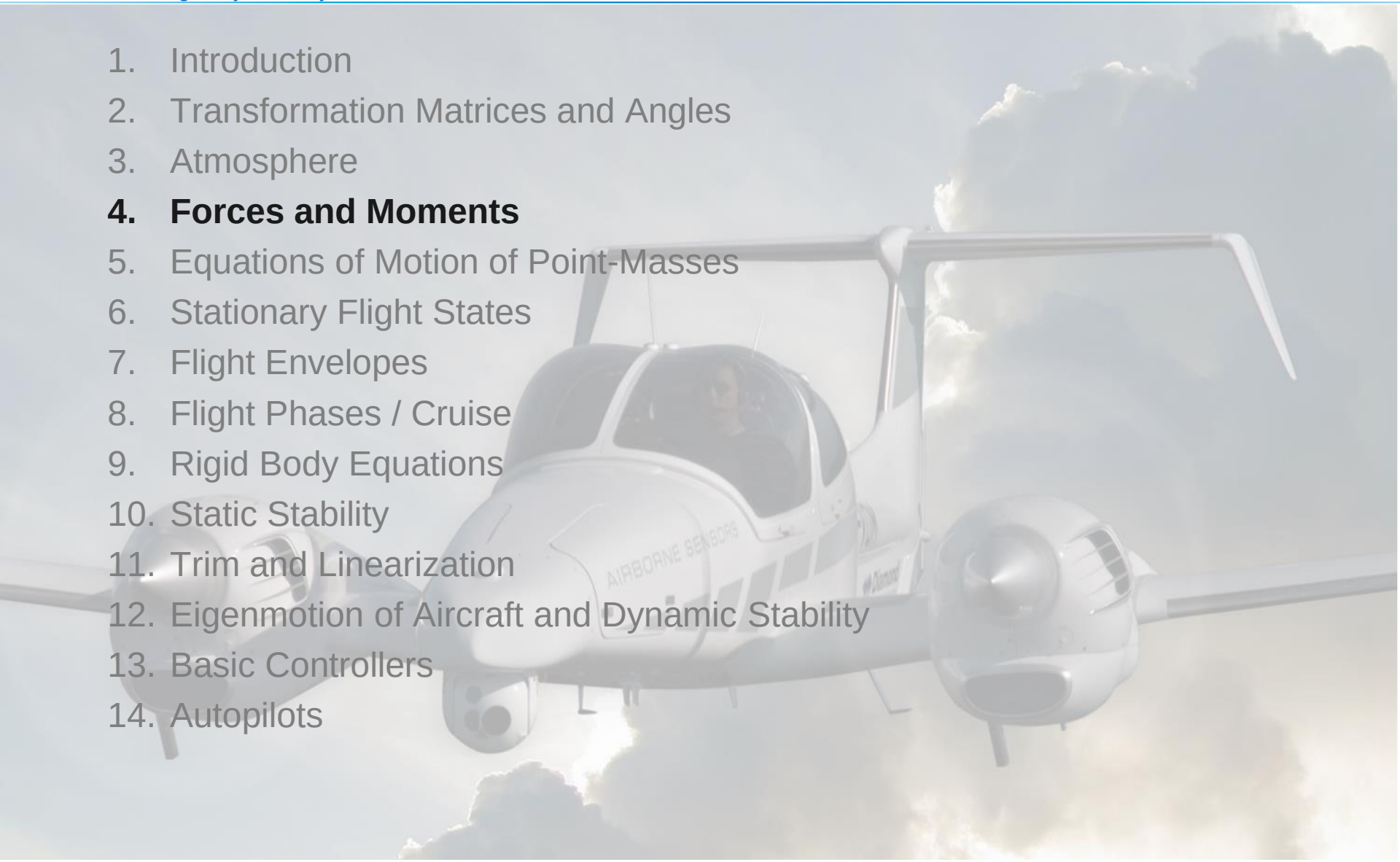
- Definition: Ratio of airspeed to speed of sound

$$Ma = \frac{v}{a}$$

Note:
We use the true airspeed to compute the Mach number!

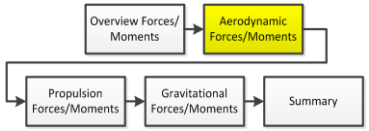
- Mach Regions:

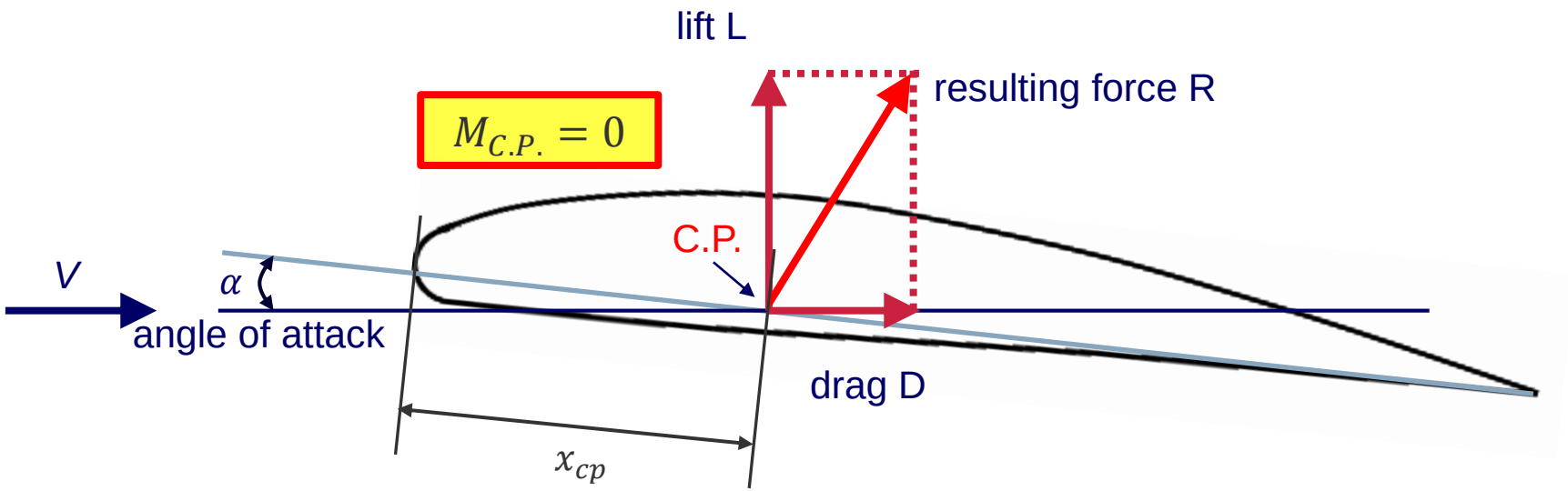




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12. Eigenmotion of Aircraft and Dynamic Stability
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14. Autopilots

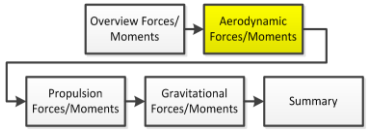
Center of Pressure

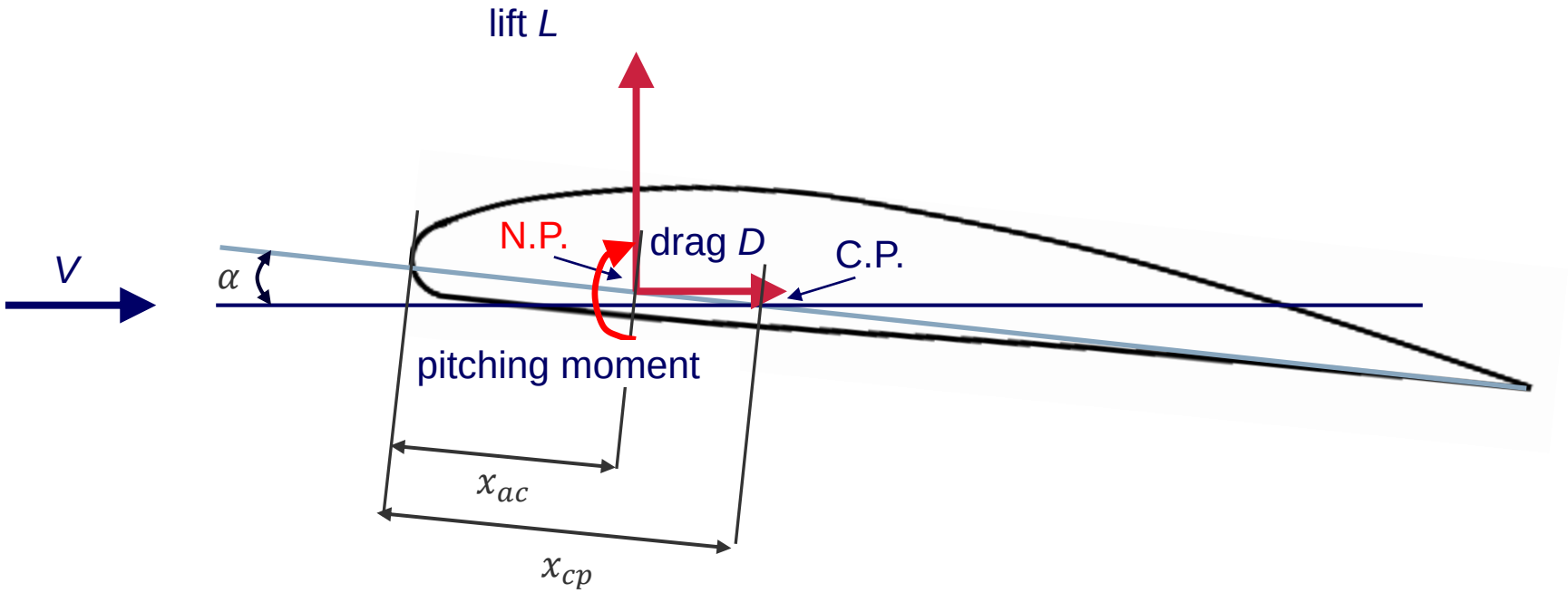




- The center of pressure is the point of attack of the total aerodynamic force due to the pressure distribution around the profile
- There is **no** total resulting moment around the center of pressure
- **C.P. changes with angle of attack**
- The resulting aerodynamic force can be split into two components
  - Lift (*perpendicular to the direction of airflow*)
  - Drag (*parallel to the direction of airflow*)

Neutral Point



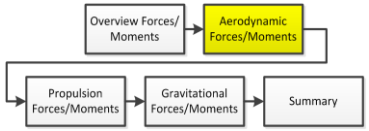


- The neutral point is the point where the pitching moment does not change with the angle of attack

$$M_{NP} \neq 0 \; ; C_{M\alpha,NP} = 0$$

- The N.P. is located around 25% of the cord line in subsonic flight, and at around 50% in supersonic flight

Overview: Aerodynamic Forces and Moments



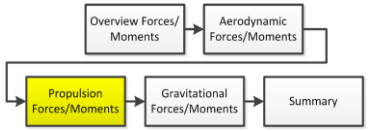
- Aerodynamic force coefficients in **aerodynamic frame A**

$$\left(\vec{\boldsymbol{F}}_A^A\right)_A = \begin{bmatrix} X_A^A \\ Y_A^A \\ Z_A^A \end{bmatrix}_A = \begin{bmatrix} -D \\ Q \\ -L \end{bmatrix}_A = \bar{q}\, S \begin{bmatrix} -C_D \\ C_Q \\ -C_L \end{bmatrix}_A$$

- Aerodynamic moment coefficients in **body-fixed frame *B***

$$\left(\vec{\boldsymbol{M}}_A^A\right)_B = \begin{bmatrix} L_A^A \\ M_A^A \\ N_A^A \end{bmatrix}_B = \bar{q}\, S \begin{bmatrix} s\, C_l \\ \bar{c}\, C_m \\ s\, C_n \end{bmatrix}_B$$

Engine Types – Mathematical Modelling



- The behavior of an engine is described by two parameters: thrust and specific fuel consumption
- Both depend on speed and height. The following is usually used as an approximation:

$$\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$$

- The dependency of thrust on speed is modelled through n_V:

$n_V = -1$: classic reciprocrating engine with propeller	$n_V = -0.75$: Turboprop
$n_V = 0$: Turbojet engine (subsonic)	$n_V = -0.25$: Turbofan

- The dependency of thrust on height is modelled through n_ρ:

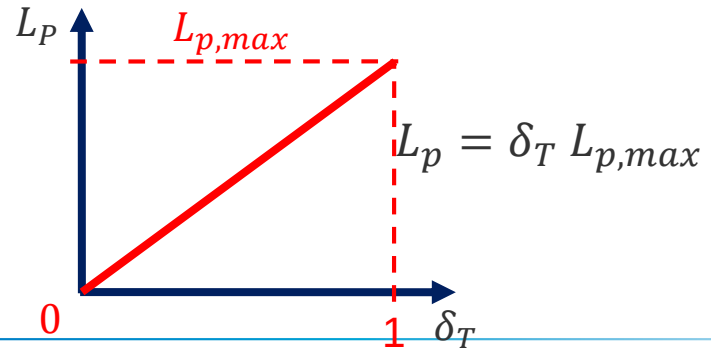
$n_\rho = 0{,}7 - 0{,}8$:	for air breathing engines at heights of $0 < H < 11\ km$
$n_\rho = 1{,}0$:	for air breathing engines at heights of $11\ km < H < 20\ km$

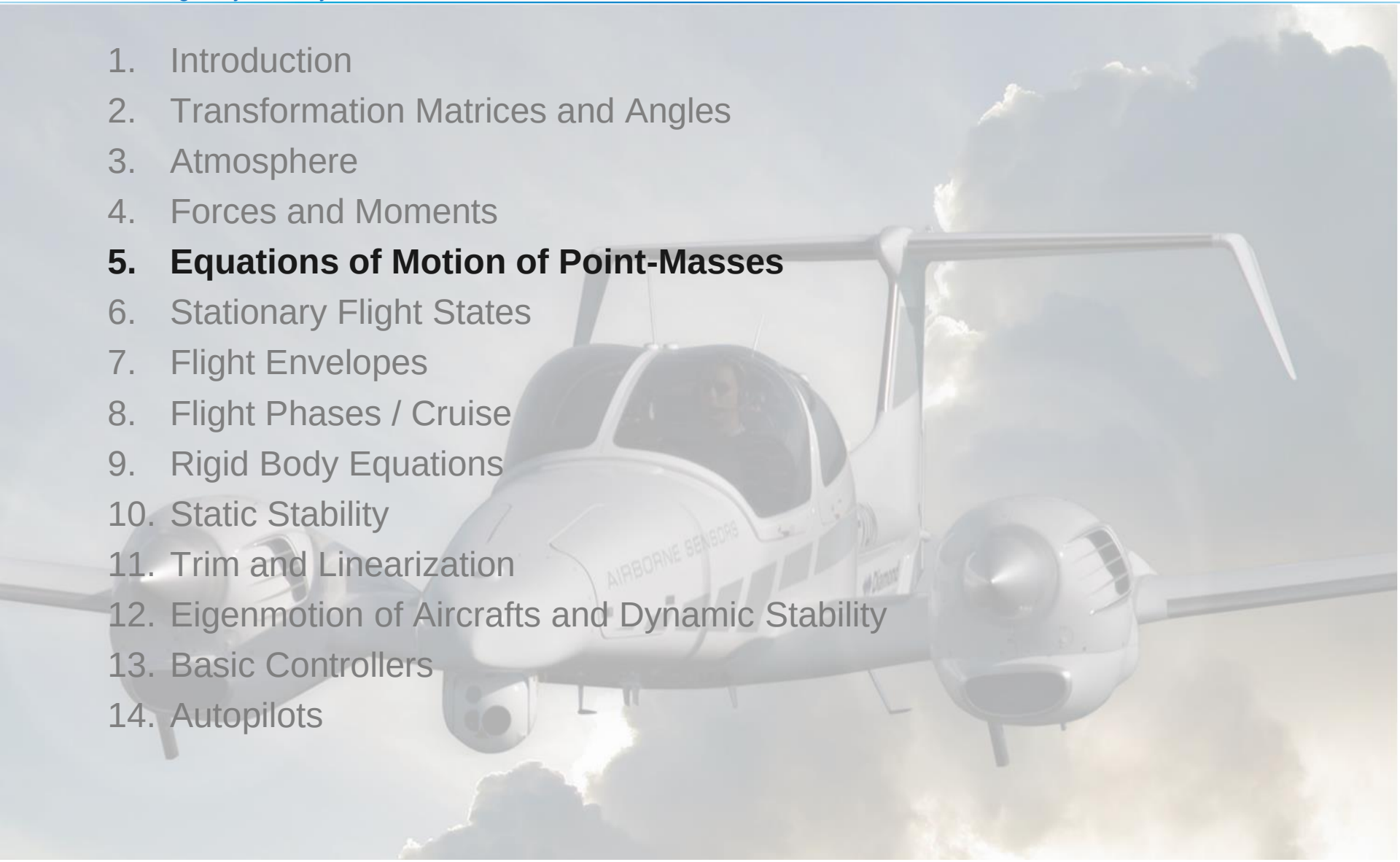
- Engine performance number:

$$L_p = \frac{T}{V^{n_V}}$$

- Engine throttle position:

$$\delta_T = \frac{L_p}{L_{p,max}}$$





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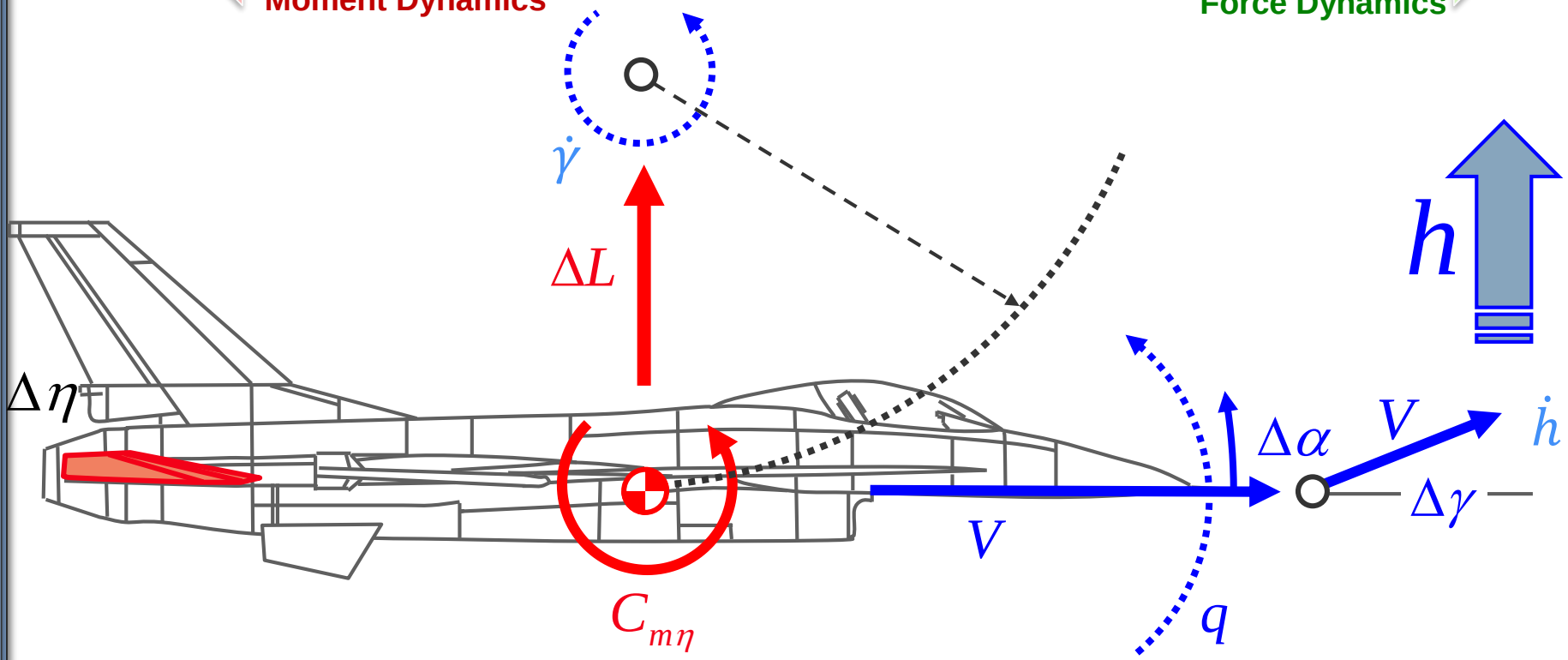
Example: Reaction of Altitude h to an elevator deflection η

Neglected: small allpass influence

$$\Delta\eta \rightarrow M_\eta \rightarrow \dot{q} \xrightarrow{\int} q \rightarrow \dot{\alpha} \xrightarrow{\int} \Delta\alpha \rightarrow \Delta L \rightarrow \dot{\gamma} \xrightarrow{\int} \gamma \rightarrow \dot{h} \xrightarrow{\int} h$$

Rotation — **Moment Dynamics**

Translation — **Force Dynamics**



Summary

Translation: Linear momentum
Force equations
3 Equations

$$\left(\vec{V}_K^G\right)_B^E = \begin{bmatrix} u_K^G \\ v_K^G \\ w_K^G \end{bmatrix}_B^E \quad \begin{bmatrix} V_K^G \\ \alpha_K^G \\ \beta_K^G \end{bmatrix} \quad \begin{bmatrix} V_K^G \\ \gamma_K^G \\ \chi_K^G \end{bmatrix}$$

Rotation: Angular momentum
Moment equations
3 Equations

$$\left(\vec{\boldsymbol{\omega}}_K^{IB}\right)_B = \begin{bmatrix} \omega_{K,x}^{IB} \\ \omega_{K,y}^{IB} \\ \omega_{K,z}^{IB} \end{bmatrix}_B \quad \left(\vec{\boldsymbol{\omega}}_K^{OB}\right)_B = \begin{bmatrix} \omega_{K,x}^{OB} \\ \omega_{K,y}^{OB} \\ \omega_{K,z}^{OB} \end{bmatrix}_B = \begin{bmatrix} p_K \\ q_K \\ r_K \end{bmatrix}_B$$

Attitude:
Attitude propagation equations
3 Equations

$$\begin{bmatrix} \Phi \\ \Theta \\ \Psi \end{bmatrix}$$

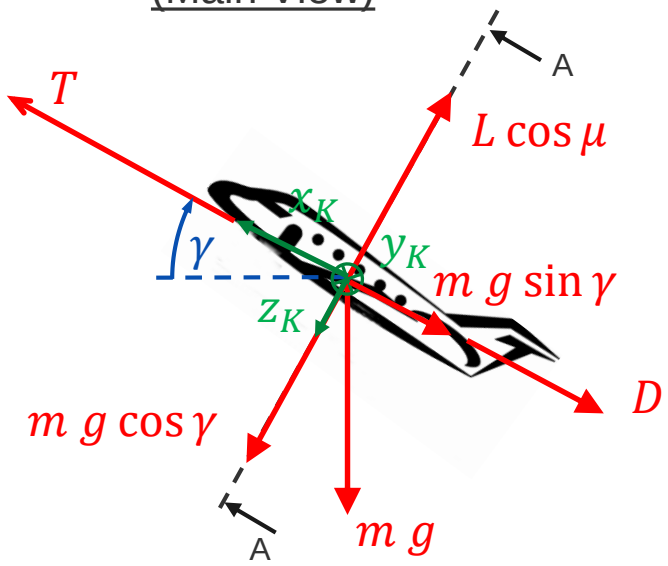
Position:
Position propagation
3 Equations

$$\begin{bmatrix} \lambda \\ \mu \\ h \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix}_N$$

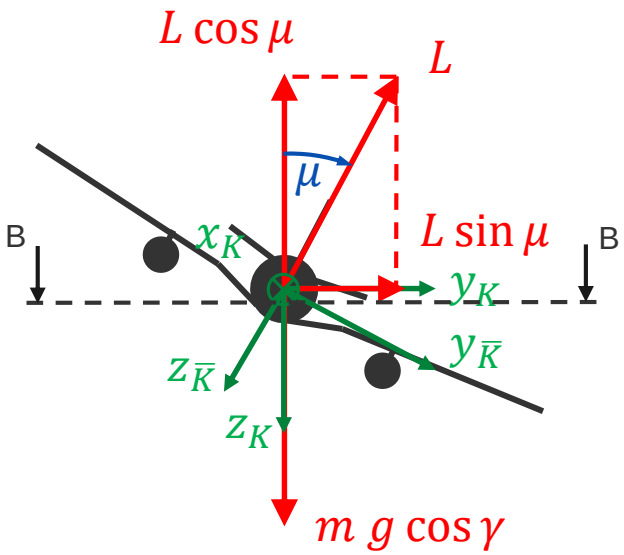
Stationary Balance of Forces

All forces are marked on the aircraft:
In the first step, dynamic changes are not considered yet:

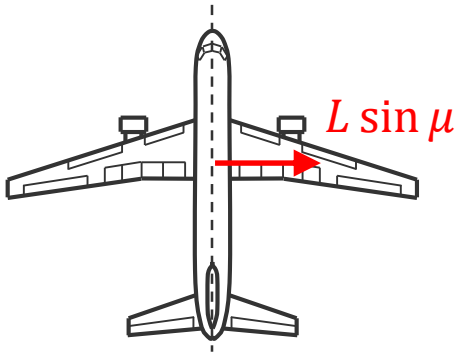
Side View
(Main View)



View A-A

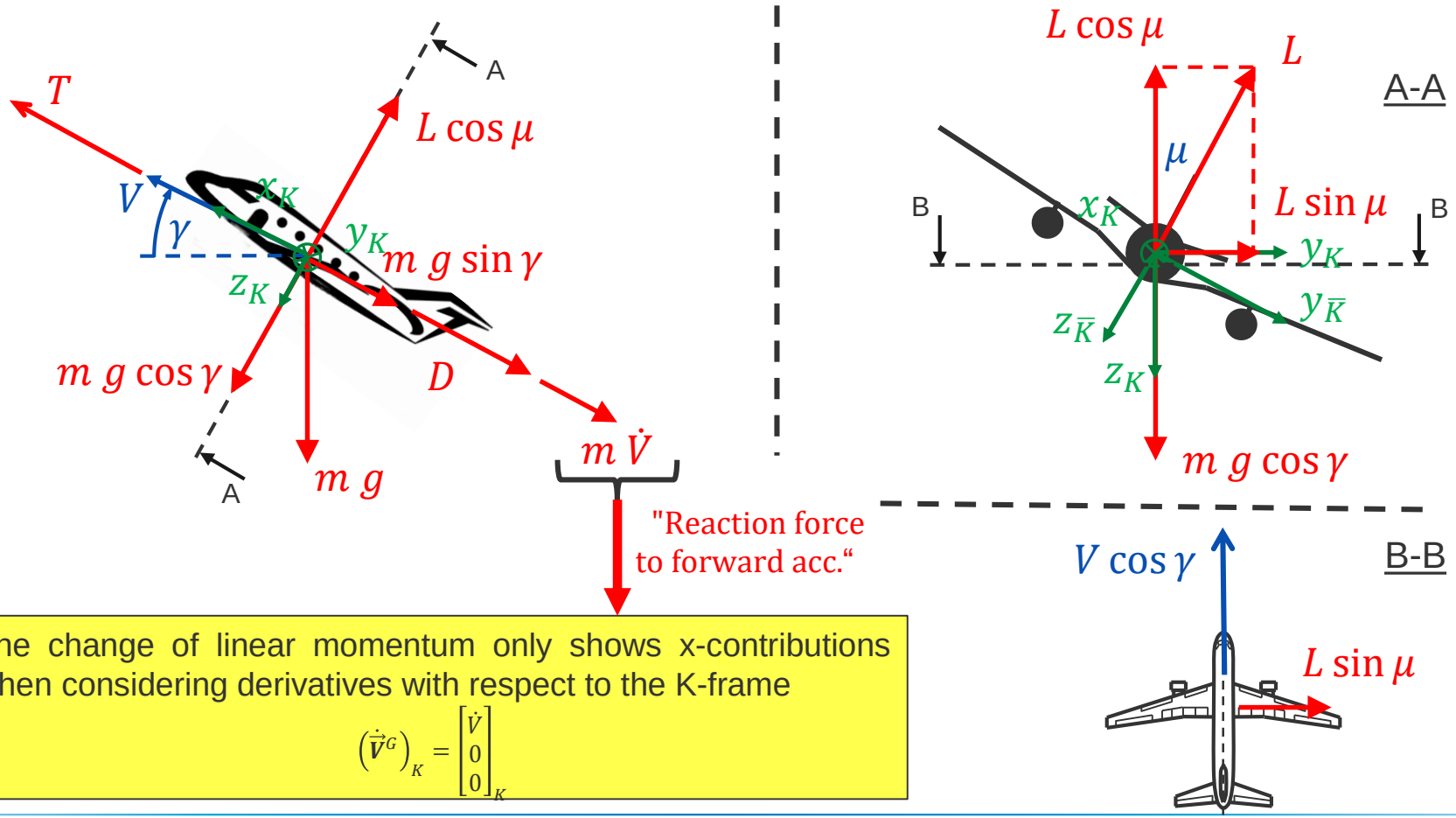


View B-B



Dynamic Balance of Forces

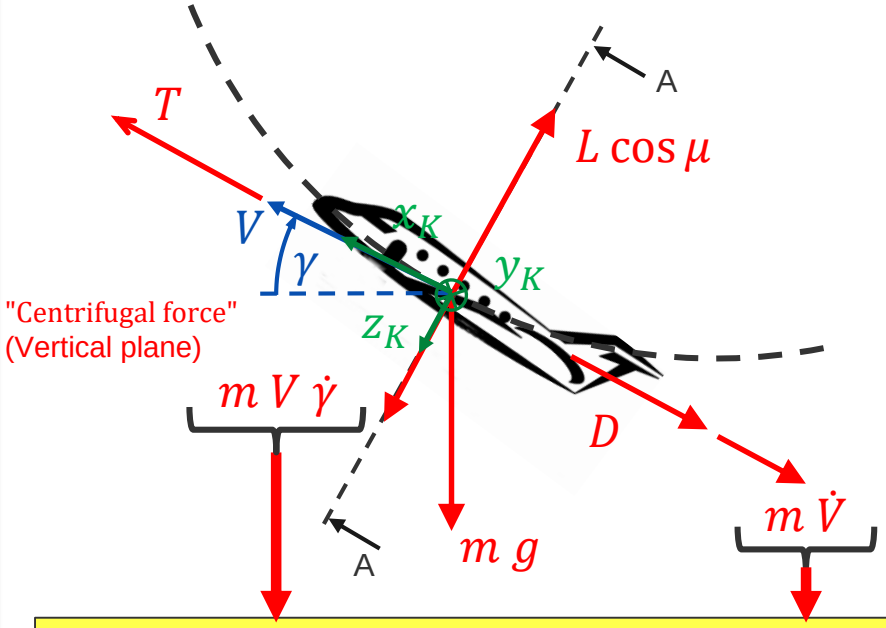
For simplification flight path changes (turn and flight path angle change) are considered in the next step.

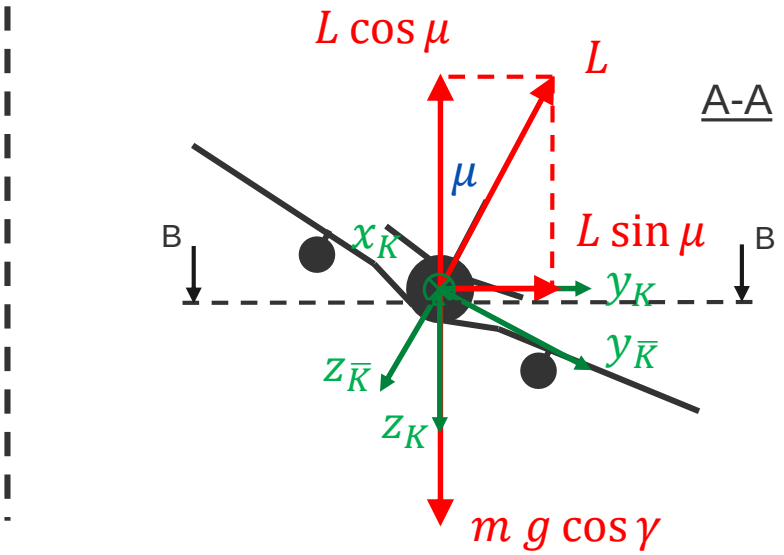


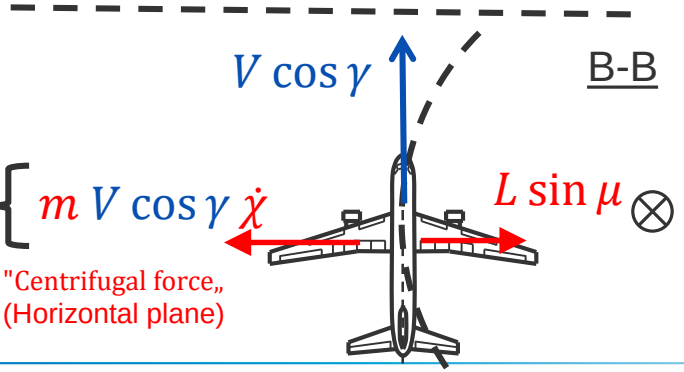
The change of linear momentum only shows x-contributions when considering derivatives with respect to the K-frame

$$\left(\dot{\vec{V}}^{G}\right)_{K} = \begin{bmatrix} \dot{V} \\ 0 \\ 0 \end{bmatrix}_{K}$$

For defining the full motion of the aircraft, time derivative of course and flight path angle have to be taken into account.







Deriving the equations of motion in the kinematic system K:
Time derivative of the momentum only includes the quantities V, γ und χ and the time derivatives.

$$\left(\dot{\vec{v}}^{G}\right)_K = \begin{bmatrix} \dot{V} \\ 0 \\ 0 \end{bmatrix}_K + \begin{bmatrix} 0 \\ V\cos\gamma\,\dot{\chi} \\ -V\cdot\dot{\gamma} \end{bmatrix}_K$$

Equation of Motion Translation

Taking the equations from the last slide into account leads to:

$$T - D - m\,g\sin\gamma - m\,\dot{V} = 0$$
$$L\sin\mu - m\,V\cos\gamma\;\dot{\chi} = 0$$
$$m\,V\,\dot{\gamma} + m\,g\cos\gamma - L\cos\mu = 0$$

Solving above equations for the quantities $\dot{V}$, $\dot{\chi}$ und $\dot{\gamma}$ leads to the equations of motion.

Note: You have to know these!

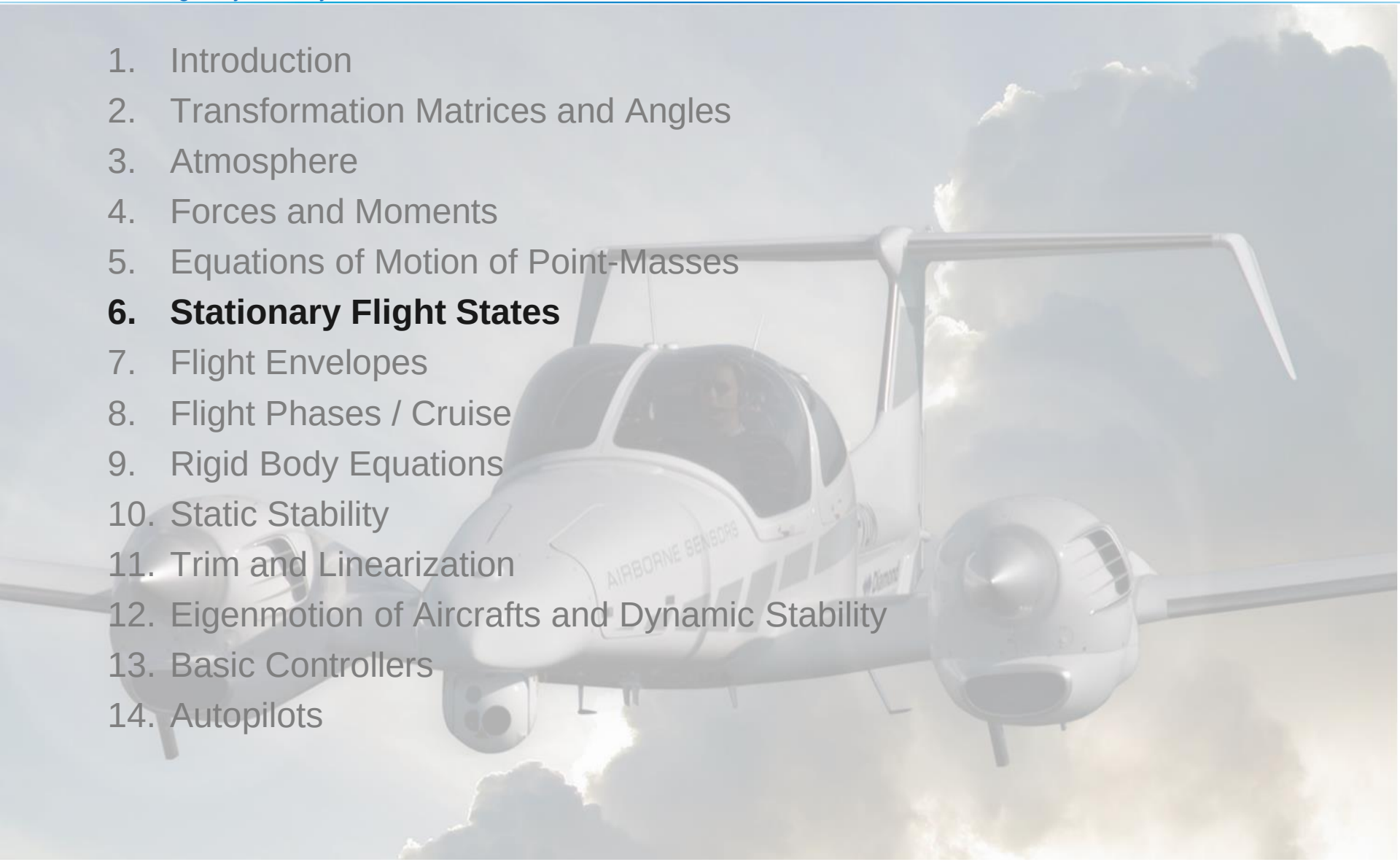
$$\Longrightarrow \dot{V} = \frac{T - D}{m} - g\sin\gamma$$

$$\dot{\chi} = \frac{L\sin\mu}{m\,V\cos\gamma}$$

$$\dot{\gamma} = \frac{L\cos\mu}{m\,V} - \frac{g}{V}\cos\gamma$$

$$L = \bar{q}\,S\,C_L$$
$$\mathrm{D} = \bar{q}\,S\,C_D$$
$$\bar{q} = \frac{1}{2}\rho\,V^2$$
$$C_D = C_{D_o} + k\,C_L^2$$
$$T = L_P\,V^{n_V}$$

Note: Know which n_V is used for which type of aircraft!



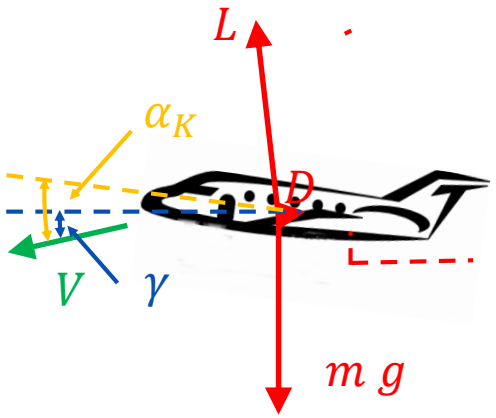
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Hint: If you know the **PM-EOM** and the **conditions of each flight phase**, you can easily derive all formulas!!

Definition of Gliding:

Straight flight without thrust but constant velocity.

Straight Flight:	$\dot{\chi} = 0$	
Stationary:	$\dot{\gamma} = 0$	$\dot{V} = 0$
No Thrust:	$T = 0$	



source: marinebuzz.com

Parameters of Interest:

Gliding is effective if for a lost meter of altitude a high horizontal distance can be covered (small glide number).

1. *Glide Number for most effective flight?*	→	ε^*
2. *Most efficient velocity for gliding?*	→	V^*
3. *Coefficient of lift for most efficient gliding?*	→	C_L^*
4. *Coefficient of drag for most efficient gliding?*	→	C_D^*

Equations of Motion and Glide Number

We use the conditions for gliding in the equations of motion:

Hint: If you know the **PM-EOM** and the **conditions of each flight phase**, you can easily derive all formulas!!

$$\dot{V} = \frac{T - D}{m} - g\sin\gamma \overset{!}{=} 0 \quad (T = 0) \longrightarrow D = -m\,g\sin\gamma$$

⇦ Steady state and without thrust

$$\dot{\chi} = \frac{L\sin\mu}{m\,V\cos\gamma} \overset{!}{=} 0 \longrightarrow \mu = 0$$

⇦ Steady state and straight flight

$$\dot{\gamma} = \frac{L\cos\mu}{m\,V} - \frac{g}{V}\cos\gamma \overset{!}{=} 0 \longrightarrow L = m\,g\cos\gamma$$

⇦ Steady state and straight flight

Ratio of drag and lift

$$\frac{-D}{L} = \frac{m\,g\sin\gamma}{m\,g\cos\gamma} = \tan\gamma = -\frac{\frac{1}{2}\rho V^2 S C_D}{\frac{1}{2}\rho V^2 S C_L} = -\frac{C_D}{C_L}$$

$$\varepsilon = \frac{C_D}{C_L} = -\tan\gamma$$

for gliding

The ratio of drag and lift is called **Glide Number ε.** It does not depend on the aircraft mass (or wing loading $\frac{m\,g}{S}$) and altitude ($\widehat{=}\ \rho$). It only depends on the coefficients C_D and C_L. For gliding the glide number equals $-\tan\gamma$.

Calculation of the Performance Parameters V^*, C_L^*, C_D^* und ε^*:

From Equations of Motion $$L = m\,g\underbrace{\cos\gamma}_{\approx 1} = \frac{1}{2}\,\rho\,V^2\,S\,C_L$$

$\cos\gamma \approx 1 - \frac{1}{2}\gamma^2 + \frac{1}{24}\gamma^4 - \cdots$
$\Rightarrow \cos\gamma \approx 1$ for small γ

→ Velocity in gliding: $$V = \sqrt{2}\cdot \underbrace{\sqrt{\frac{m\,g}{S}}}_{\substack{wing\\loading}} \cdot \underbrace{\sqrt{\frac{1}{\rho}}}_{altitude} \cdot \underbrace{\sqrt{\frac{1}{C_L}}}_{aerodynamics}$$

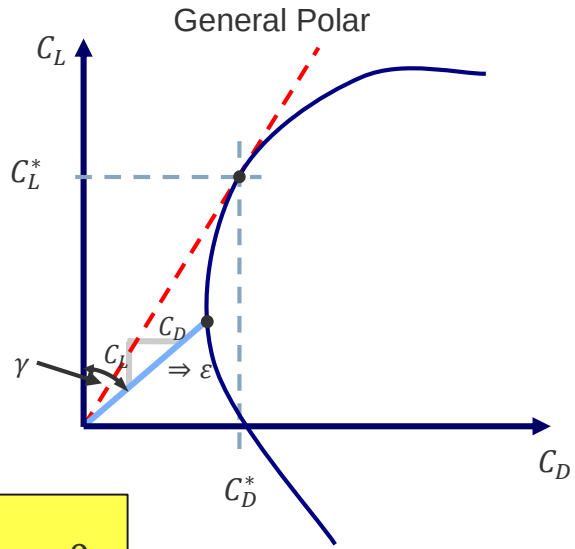
$V \sim \sqrt{\frac{m\,g}{S}}$, $V \sim \sqrt{\frac{1}{\rho}}$
Velocity of gliding depends on wing load and altitude

Idea of Calculations :

Gliding is most efficient, when the ratio $\frac{C_D}{C_L}$ is at minimum, meaning that the glide number and glide angle are also at minimum:

Assumption:
Symmetric quadratic Polar:

$$\varepsilon = \frac{C_D}{C_L} = \frac{C_{D0} + k\,C_L^2}{C_L} = \frac{C_{D0}}{C_L} + kC_L$$



Condition for the minimum:

$$\frac{\partial\varepsilon}{\partial C_L} = 0 \quad \Rightarrow -\frac{C_{D0}}{C_L^2} + k = 0$$

$$\Rightarrow -\frac{C_{D0}}{C_L^2} + k = 0$$

The C_L solving these equations is C_L^*

$$\Rightarrow -\frac{C_{D0}}{(C_L^*)^2} + k = 0$$

Solving for C_L^* and inserting in the equation yields:

$$\frac{C_{D0}}{(C_L^*)^2} = k \quad \xrightarrow{\text{Solving for } C_L^*} \quad C_L^* = \sqrt{\frac{C_{D0}}{k}}$$

$$C_D^* = C_{D0} + kC_L^{*2} \quad \xrightarrow{C_L^* \text{ inserted}} \quad C_D^* = 2C_{D0}$$

$$V = \sqrt{2\frac{m\,g \cdot 1}{S\,\rho\,C_L}} \quad \xrightarrow{C_L^* \text{ inserted}} \quad V^* = \sqrt{2\frac{m\,g \cdot 1}{S\,\rho\sqrt{\frac{C_{D0}}{k}}}}$$

$$\varepsilon = \frac{C_D}{C_L} \quad \xrightarrow{C_L^* \text{ and } C_D^* \text{inserted}} \quad \varepsilon^* = 2\sqrt{k\,C_{D0}}$$

Velocity Polar

The velocity can be split into a horizontal and a vertical part:

$$V_H = \dot{x} = V \cdot \cos\gamma \qquad V_V = \dot{h} = V \cdot \sin\gamma$$

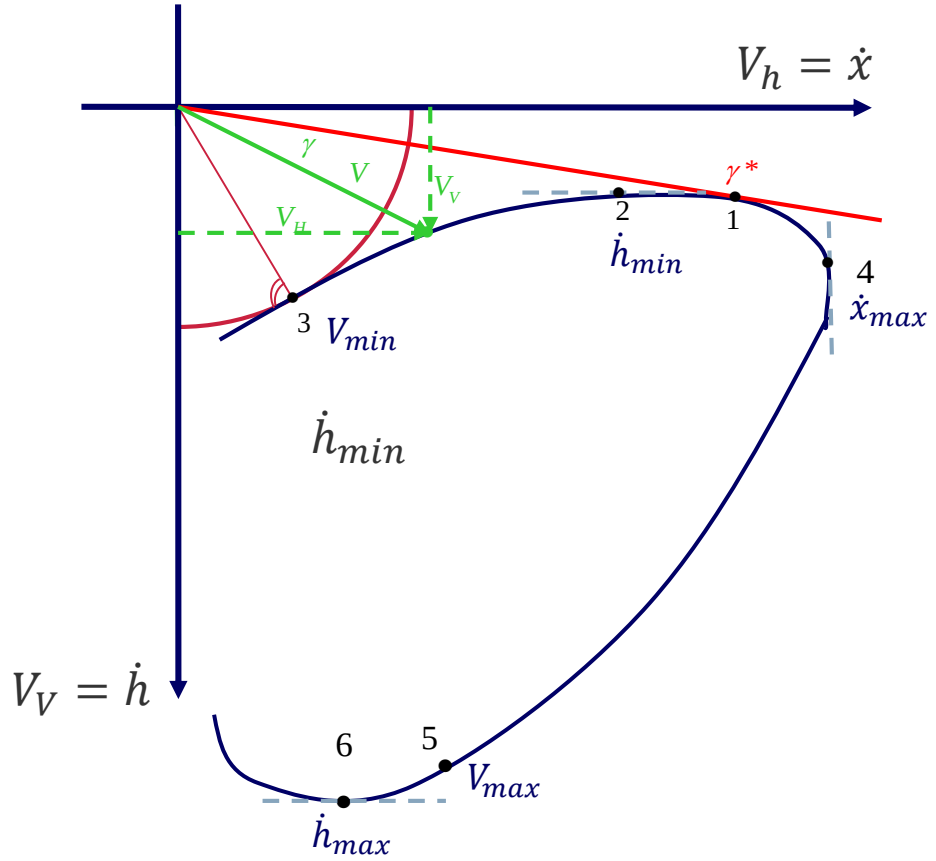
The ratio of both velocities is shown in the **„Velocity Polar“**.

For the glide angle it holds:

$$\frac{V_V}{V_H} = \frac{\dot{h}}{\dot{x}} = \frac{V \cdot \sin\gamma}{V \cdot \cos\gamma}$$

$$\frac{\Delta h}{\Delta x} = \tan\gamma$$

The glide angle is a measure for how much horizontal distance is covered after losing a specified amount of altitude.



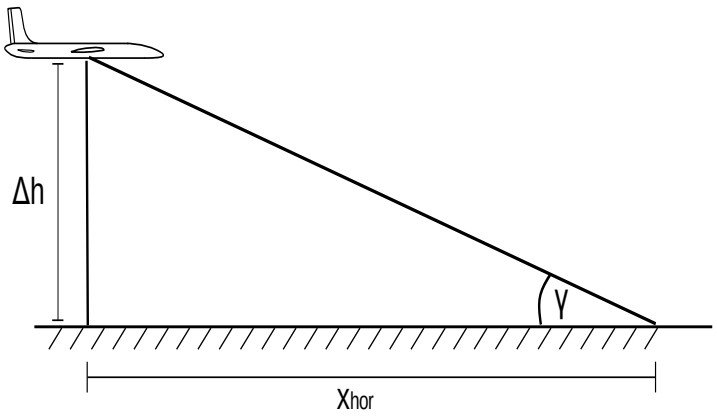
Note:
*The importance of the *-quantities goes far beyond gliding flight! For example, horizontal flight with velocity V* is drag-minimal. V* and ε* can also be used to set up equations for drag, velocity, or the climb angle. Since the *-quantities depend only on the aerodynamics of the aircraft, this results in very compact representations.*

$$C_L^* = \sqrt{\frac{C_{D0}}{k}}$$

$$C_D^* = 2C_{D0}$$

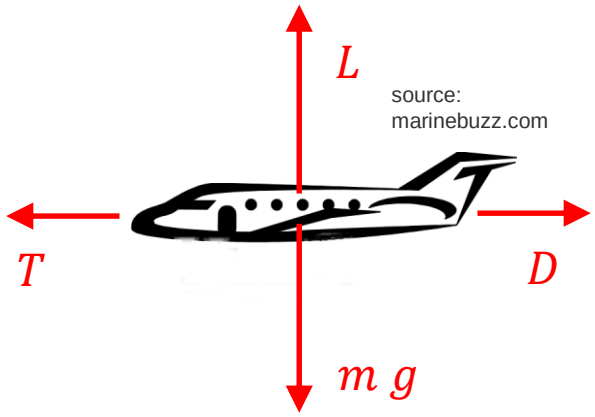
$$V^* = \sqrt{2\frac{m\,g\cdot 1}{S\,\rho\sqrt{\frac{C_{D0}}{k}}}}$$

$$\varepsilon^* = 2\sqrt{k\,C_{D0}}$$



$$\frac{\Delta h}{\Delta x} = \tan\gamma$$

Definition straight horizontal flight:



Straight:	$\dot{\chi} = 0$	
Stationary:	$\dot{\gamma} = 0$	$\dot{V} = 0$
Horizontal:	$\gamma = 0$	

Performance parameters straight horizontal flight:
Parameters measuring the economic effiecencies are important

1. *Which velocity leads to the minimum drag?* → $V_{C_D,min}$
2. *General drag for horizontal flight* → D
3. *Required thrust for horizontal flight* → T
4. *Minimum drag for horizontal flight* → D_{min}
5. *Minimum required thrust for horizontal flight* → T_{min}

Using the definition of the horizontal straight flight in the point mass equations of motion yields:

$$\dot{V} = \frac{T - D}{m} - g \underbrace{\sin\gamma}_{=0} = 0 \qquad \longrightarrow D = T$$

⇦ Stationary and Horizontal

$$\dot{\chi} = \frac{L \sin\mu}{m\, V \cos\gamma} = 0 \qquad \longrightarrow \mu = 0$$

⇦ Stationary, Straight and Horizontal

$$\dot{\gamma} = \frac{L \cos\mu}{m\, V} - \frac{g}{V} \underbrace{\cos\gamma}_{=1} = 0 \qquad \longrightarrow L = m\, g$$

⇦ Stationary, Straight and Horizontal

$$\longrightarrow \frac{T}{m\, g} = \frac{D}{L} = \frac{\bar{q}\, S\, C_D}{\bar{q}\, S\, C_L} = \frac{C_D}{C_L} = \varepsilon$$

In horizontal flight, the thrust-to-weight ratio equals the glide number ε and the ratio of C_D and C_L.

Insertion: Glide number and aerodyn. efficiency

For defining aircraft performance characteristics, the aerodynamic efficiency is used frequently. It is the inverse of the glide number for the most efficient flight ε^*:

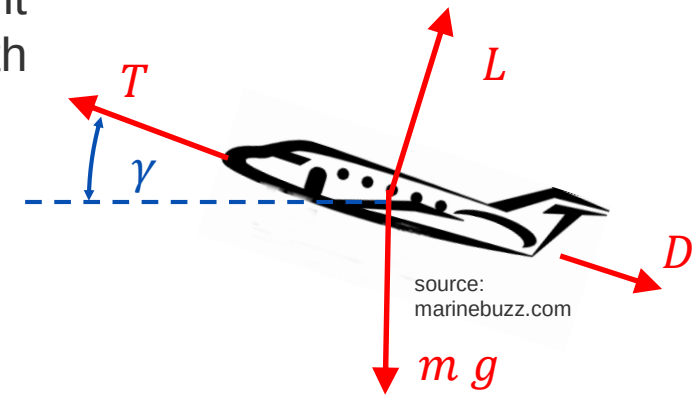
$$E = \frac{1}{\varepsilon^*}$$

Examples for aerodynamic efficiency:

- Albatross: 20
- Gull: 10
- Sparrow: 4
- Helicopter: ≈ 4
- Wright Flyer: 8.3
- Concorde: 8
- Airbus A320: 17
- Airbus A340-600: 19.5
- Boeing B747: 17.5
- Boeing B707: 16.8
- Cessna 172: 10
- Glider (Discus): 43
- Glider (ASW 22): 60

Definition Climb and Descent:

Pure climb and descent are defined as straight flight without maneuvers but constant flight path angle γ.



Straight flight:	$\dot{\chi} = 0$	
Stationary:	$\dot{\gamma} = 0$	$\dot{V} = 0$
Constant flight path:	$\gamma = \text{const.}$	

Important Performance Parameters for Climb and Descent:

1. *Which thrust is desired for climbing?*	→	T *bzw.* $\frac{T}{m \cdot g}$
2. *Climb anlge dependency on velocity?*	→	$\gamma\ (V)$
3. *Thrust reserve for maneuvers?*		

Using the definition of climb and descent in the point mass equations of motion:

$$\dot{V} = \frac{T - D}{m} - g \sin\gamma = 0 \quad \longrightarrow T = mg \sin\gamma + D \quad \Leftarrow \text{Stationary}$$

$$\dot{\chi} = \frac{L \sin\mu}{m\, V \cos\gamma} = 0 \quad \longrightarrow \mu = 0 \quad \Leftarrow \text{Stationary and straight}$$

$$\dot{\gamma} = \frac{L\cos\mu}{m\, V} - \frac{g}{V}\cos\gamma = 0 \quad (\cos\mu = 1) \quad \longrightarrow L = m\, g \cos\gamma \quad \Leftarrow \text{Stationary, straight and without maneuvers}$$

$$\longrightarrow \tan\gamma = \frac{T - D}{L}$$

$$\longrightarrow \sin\gamma = \frac{T - D}{mg}$$

Goal: Deriving equation for thrust-to-weight ratio.

The slides for gliding already contained the relations between change of altitude $\dot{h}$ and flight path angle γ:

$$V_V = \dot{h} = V \cdot \sin\gamma \xrightarrow{\sin\gamma = \frac{T}{m\,g} - \frac{D}{mg}} \boxed{\dot{h} = V \cdot \left(\underbrace{\frac{T}{m\,g}}_{spec.\,Thrust} - \underbrace{\frac{D}{mg}}_{spec.\,Drag} \right)}$$

General equation for power: $\boxed{P = F \cdot V}$

Equation for specific power: $\boxed{P_{spec} = \frac{F \cdot V}{mg}}$

The above equation for $\dot{h}$ equals the definition of a specific power with:

$$\dot{h} = \left(\underbrace{\frac{T \cdot V}{m\,g}}_{spec.\,propulsion\,power} - \underbrace{\frac{D \cdot V}{mg}}_{spec.\,power\,dissipated\,by\,drag} \right)$$

This power excess is also called **„specific excess power (SEP)“** and is a measure for the power available for climbing

Using the definition of turn in the point mass equations of motion

$$\dot{V} = \frac{T - D}{m} - g \underbrace{\sin\gamma}_{=0} = 0 \quad \longrightarrow T = D \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\dot{\chi} = \frac{L \sin\mu}{m\, V \underbrace{\cos\gamma}_{=1}} \quad \longrightarrow L \sin\mu = mV\dot{\chi} \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\dot{\gamma} = \frac{L\cos\mu}{m\,V} - \frac{g}{V}\underbrace{\cos\gamma}_{=1} = 0 \quad \longrightarrow L\cos\mu = m\,g \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\frac{L\sin\mu}{L\cos\mu} = \frac{m\,V\,\dot{\chi}}{m\,g} \longrightarrow \tan\mu = \frac{V\,\dot{\chi}}{g}$$

$$L^2(\sin^2\mu + \cos^2\mu) = (m\,V\dot{\chi})^2 + (m\,g)^2 \longrightarrow L = m\,g\sqrt{1 + \left(\frac{V\,\dot{\chi}}{g}\right)^2}$$

Question: Is there a parameter defining the turn performance of an aircraft?

Introduction of the loadfactor n_z

For defining turn performance, we introduce the load factor n_z:

$$(n_z)_K = \frac{L}{m\,g}$$

Acceleration due to lift

Normalization with respect to weight

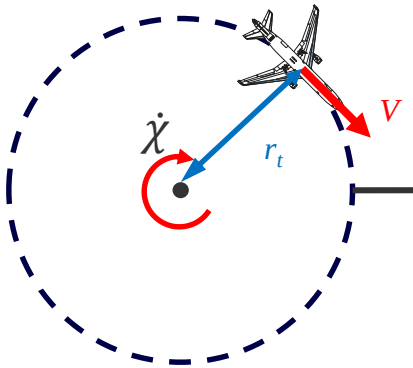
The load factor is the lift force normalized by the gravitational force. This normalization leads to $n_z = 1$ in straight horizontal flight.

$$\left(n_{z,Horizontal}\right)_K = 1$$

With this definition of the load factor, the available turn performance can be described, as the turn performance results from an excess of lift over weight.

Annotation: This interpretation of load factor is fit to the following turn performance. However, the benefit and usage of the load factor can be applied to more problems as well.

How can the turn radius be obtained?



The turn radius depends on aircraft velocity and turn rate

$$r_t = \frac{V}{\dot{\chi}}$$

Goal: dependency between turn rate, turn radius and n_z, in order to answer the question of minimum possible turn rate

$$n_z = \frac{L}{m\,g} \quad \xrightarrow{L = m\,g\sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2}} \quad n_z = \sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2} \quad \xrightarrow{\dot{\chi} = \frac{V}{r_t}} \quad n_z = \sqrt{1+\left(\frac{V^2}{g\,r_t}\right)^2}$$

$$n_z = \frac{1}{\cos\mu}$$

$$\tan\mu = \frac{V\,\dot{\chi}}{g} \quad \xrightarrow{\text{Solving for } \dot{\chi}} \quad \dot{\chi} = \frac{g}{V}\tan\mu \quad \xrightarrow{n_z = \sqrt{1+\tan^2\mu} \rightarrow \tan\mu = \sqrt{n_z^2-1}} \quad \dot{\chi} = \frac{g}{V}\sqrt{n_z^2-1}$$

Minimum Turn Rate

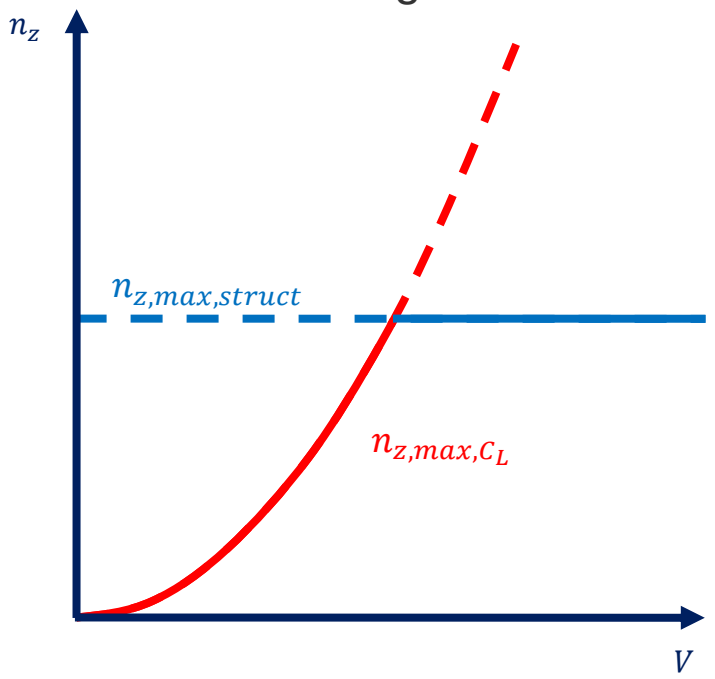
$$\dot{\chi} = \frac{g}{V}\sqrt{n_z^2 - 1} \qquad r_t = \frac{V}{\dot{\chi}} \qquad r_t = \frac{V^2}{g\sqrt{n_z^2 - 1}}$$

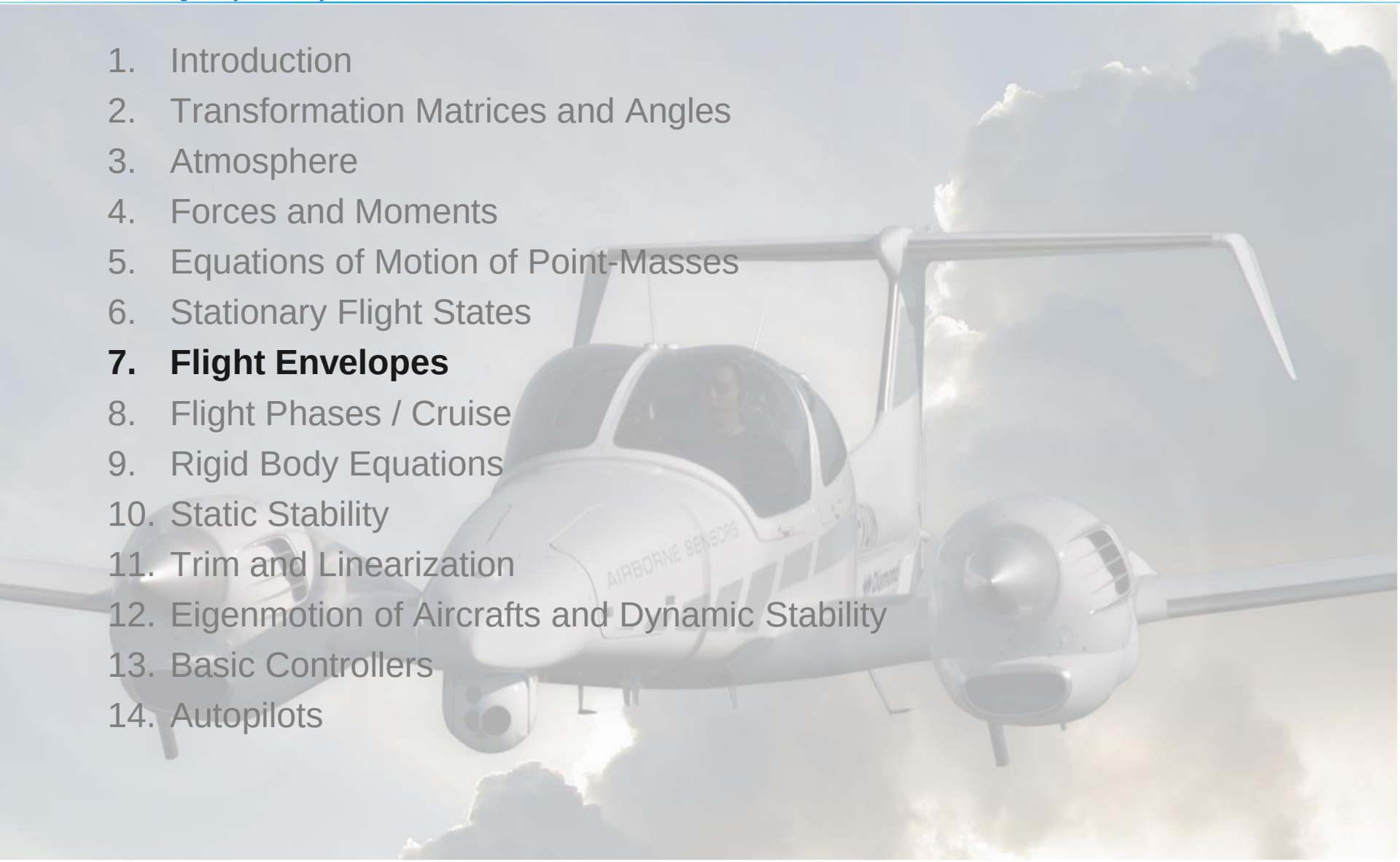
The turn radius is physically bounded by the maximum available load factor. A maximum load factor exists due to two reasons:

1. Pilot aircraft structure only withstand a maximum structural load factor.
2. The lift required for a the load factor is limited:

The minimum possible turn rate is defined by Lift, capability of structure and pilot to withstand loads, and maximum available thrust.

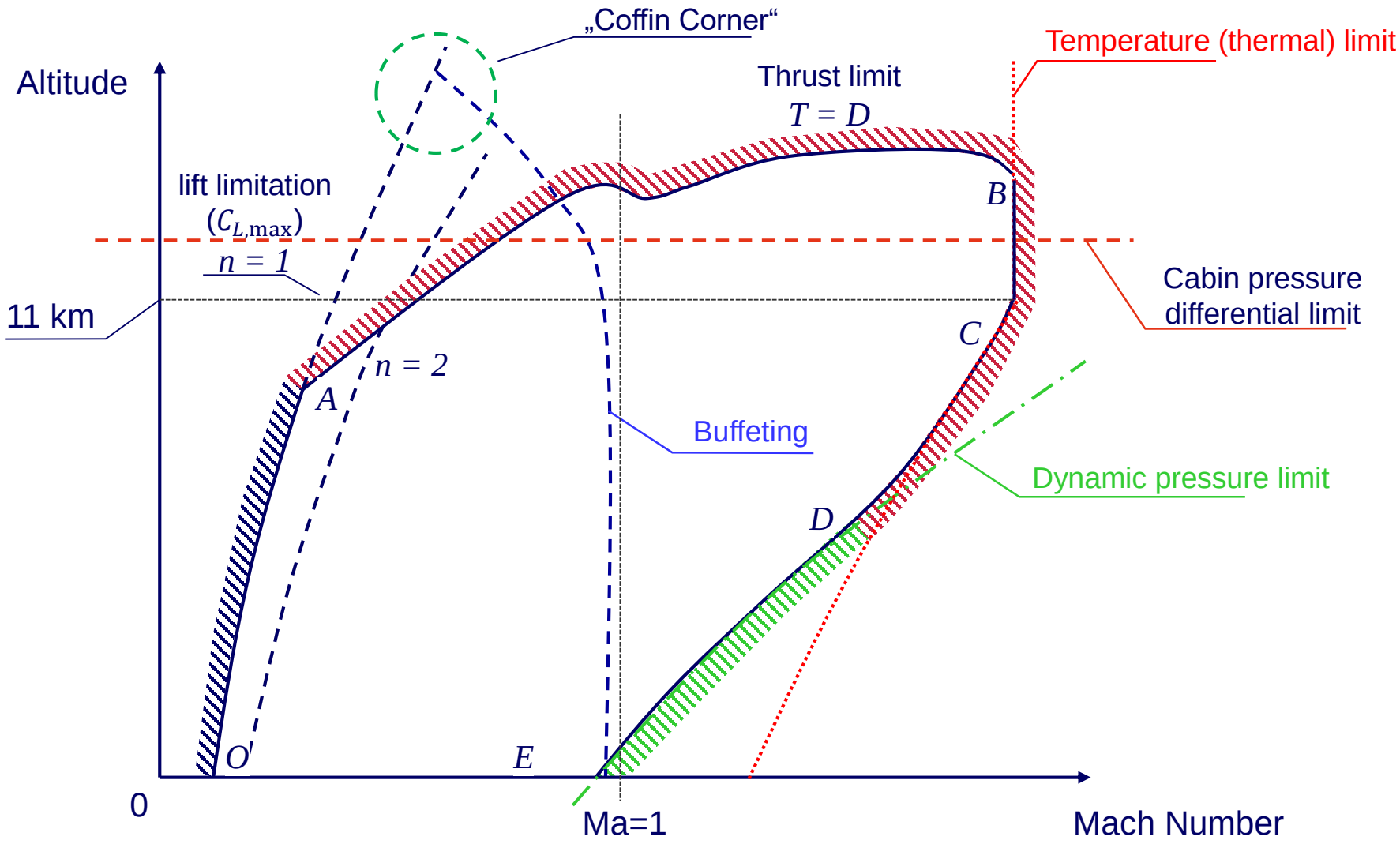
V-n Diagramm

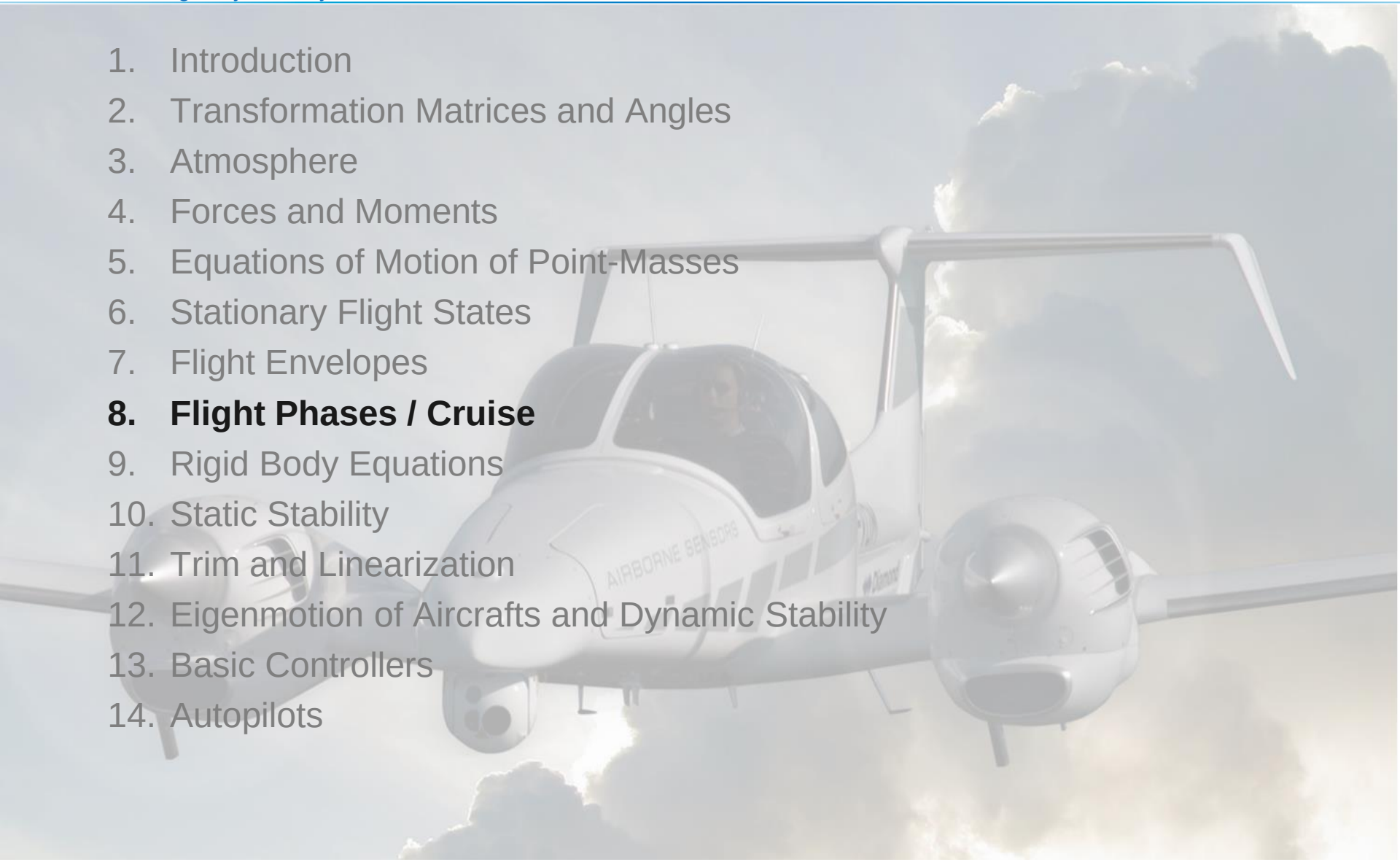




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Flight Envelope





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Definition Cruise

- Flight phase for covering distance
- Quasi stationary
- Mass change due to fuel consumption has to be considered

Quasi stationary: $\dot{\chi} = 0, \dot{\gamma} = 0$ $\dot{V}$ *without influence*

continuous decrease of mass: $\dot{m} = \frac{dm}{dt} = -\dot{m}_F$

Performance Parameters Cruise

1. *Which time is required for the flight* → t
2. *Which range can be covered?* → s

With the specific fuel consumption $b_L = \frac{\dot{m}_F}{L_P}$ the possible flight time and range can be calculated:

$$-\dot{m}_F = \frac{dm}{dt} \longrightarrow dt = -\frac{dm}{b_L\, L_P} \qquad ds = V\, dt = -\frac{V}{b_L\, L_P} dm$$

Stationary flight state and small flight path angle: L_P can be calculated with the drag equation

$$L_P = T\, V^{-n_V} = D\, V^{-n_V}$$

For stationary flight ($L = m\, g$), the drag equals:

$$D = \varepsilon\, m\, g = \frac{C_D}{C_L} m\, g$$

Inserting in the equations for :

$$dt = -\frac{V^{n_V}}{g\, b_L} \frac{C_L}{C_D} \frac{dm}{m} \qquad L_P = \frac{C_D}{C_L} m\, g\, V^{-n_V} \qquad ds = -\frac{V^{n_V+1}}{g\, b_L} \frac{C_L}{C_D} \frac{dm}{m}$$

Different Types of Cruise

$$dt = -\frac{V^{n_v}}{g\ b_L}\frac{C_L}{C_D}\frac{dm}{m}$$

$$ds = -\frac{V^{n_v+1}}{g\ b_L}\frac{C_L}{C_D}\frac{dm}{m}$$

Condition for lift:

$$\frac{1}{2}\rho\, V^2\, S\, C_L = m\, g$$

→ Three equations for cruise

⟹ 3 independent variables C_L, V, ρ

Integration of the first two equations yields the flight time and range. In order to simplify the integration, two of the three variables V, ρ und C_L are set to a constant value. The third quantity results from the lift equation.

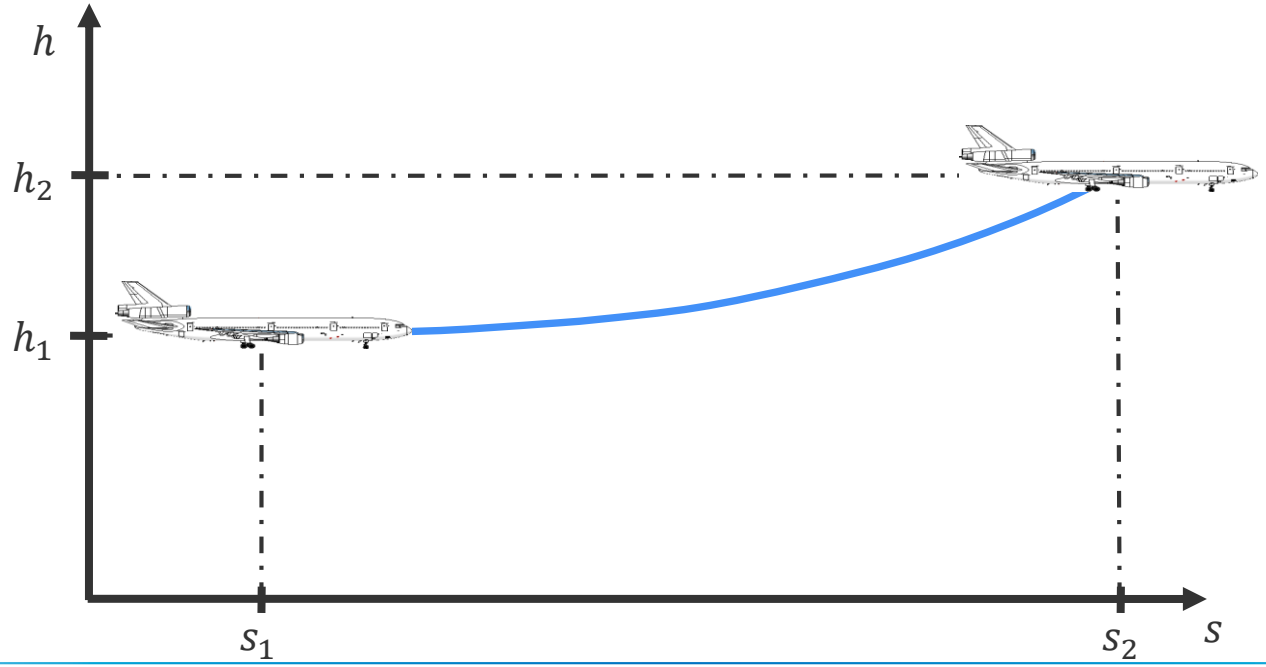
The following combinations are possible:

Type	V	C_L	ρ	Variable
I	const.	const.	-----	$\rho \sim m$
II	const.	-----	const.	$C_L \sim m$

Annotation: The third possibility of changing velocity is not part of this class.

Type I – Velocity *V* and coefficient of lift C_L as constants

- Flight time: $t = \int_{m_1}^{m_2} -\frac{V^{n_V}}{g\, b_L}\frac{C_L}{C_D}\frac{dm}{m} = \frac{V^{n_V}}{g\, b_L}\frac{C_L}{C_D}\ln\frac{m_1}{m_2}$
- Range: $s = \int_{m_1}^{m_2} -\frac{V^{n_V+1}}{g\, b_L}\frac{C_L}{C_D}\frac{dm}{m} = \frac{V^{n_V+1}}{g\, b_L}\frac{C_L}{C_D}\ln\frac{m_1}{m_2}$
- Direct integration of the equations, as C_L and V are constant and can be put in front of the integration



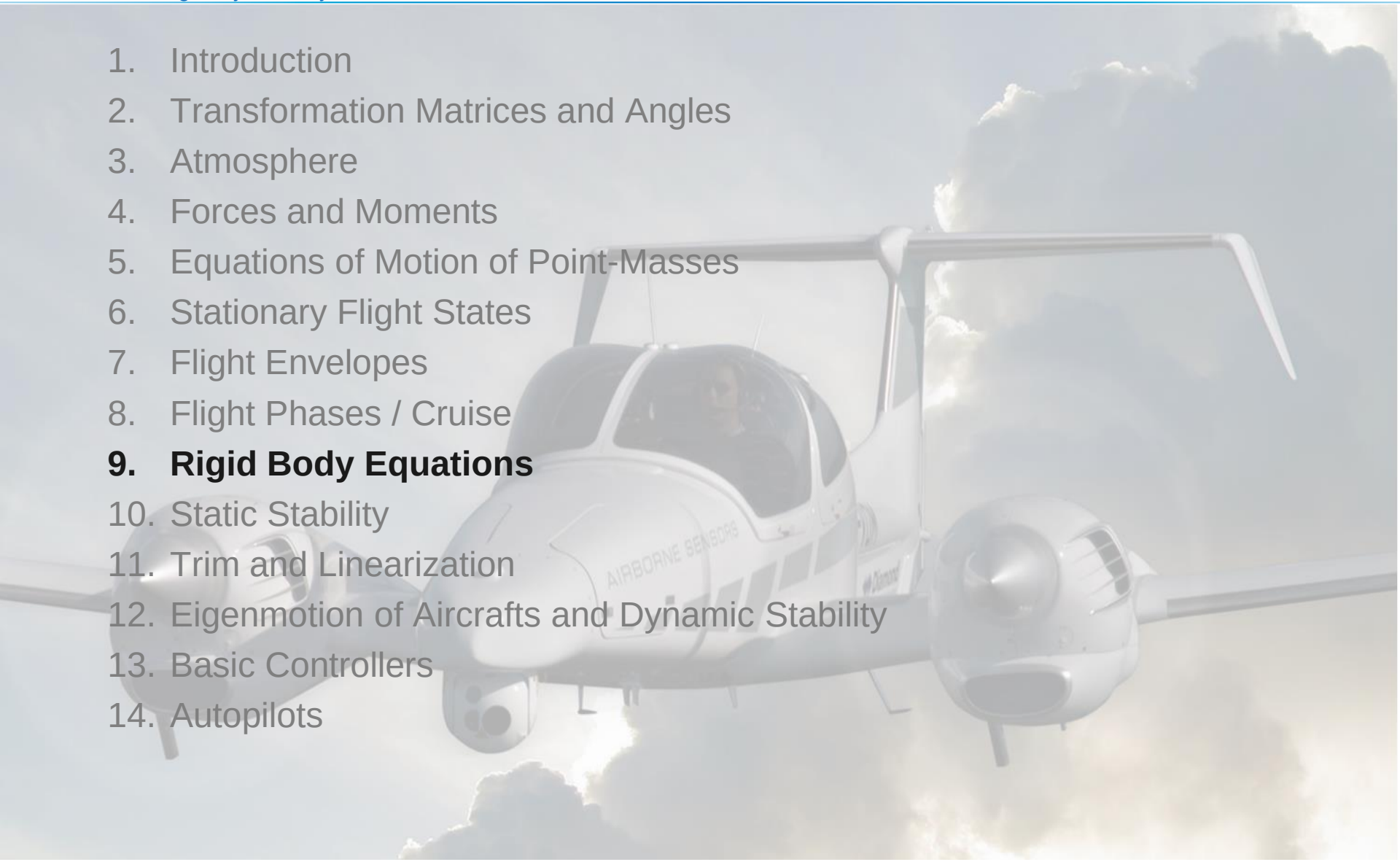
Sample Exercise E.2

Order the following flight states:

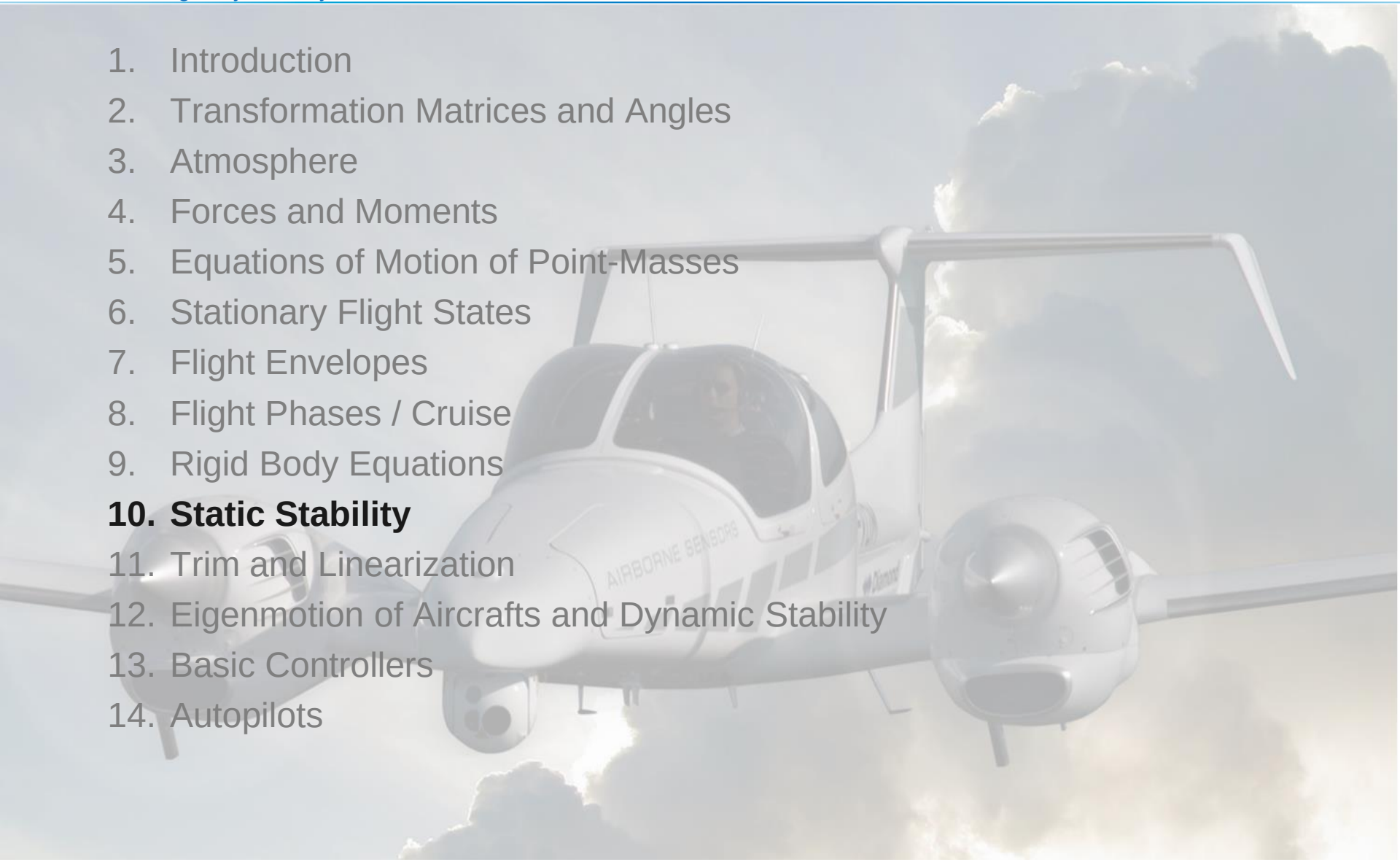
- Cruise
- Descending
- Climb

according to the following quantities:

- Required Thrust
- Required Lift



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Position of the Neutral Point with respect to the center of gravity

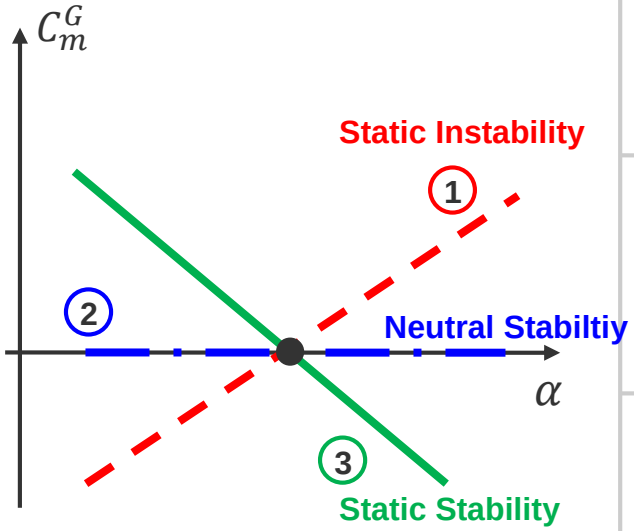
$$SM = -\frac{C_{m\alpha}^{G}}{C_{L\alpha}} = -\frac{x^{N} - x^{G}}{\bar{c}}$$

For Static Stability: **$SM > 0$**

$C_{L\alpha}$ *always positive (a more positive angle of attack leads to a more positive lift)*

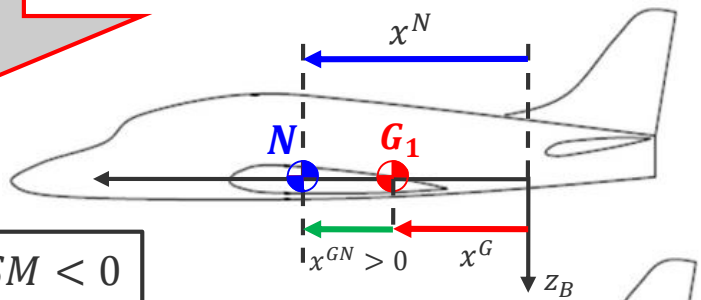
chord always positive

Dependency of the Position of the Neutral Point



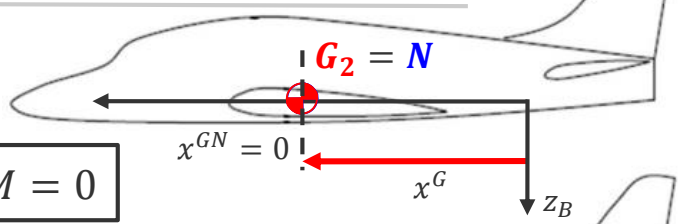
① Static Instability

$x^G < x^N$ | $C_{m\alpha}^G > 0$ | $SM < 0$



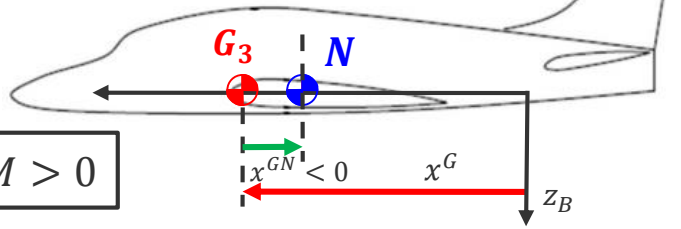
② Neutral Stability

$x^G = x^N$ | $C_{m\alpha}^G = 0$ | $SM = 0$

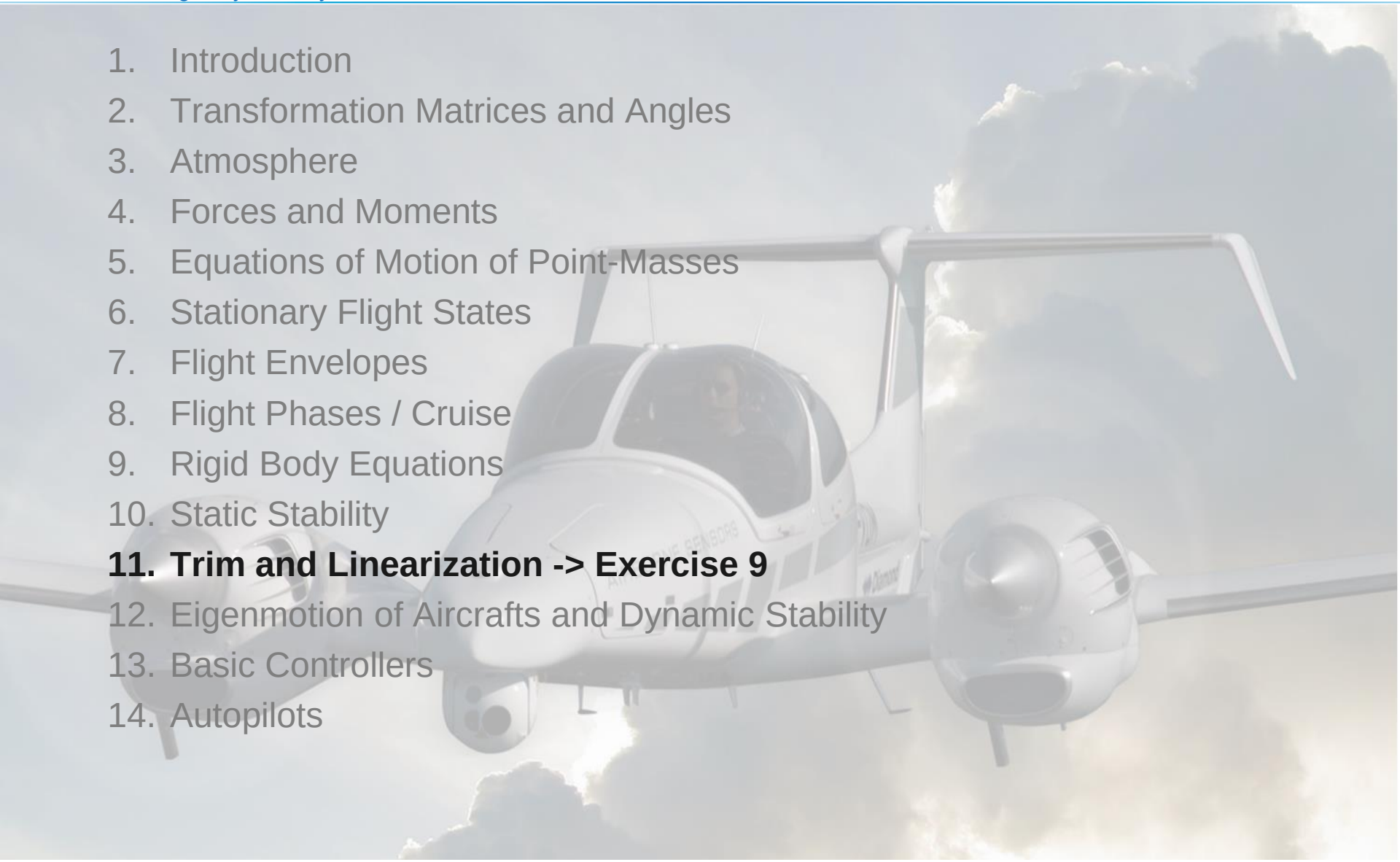


③ Static Stability

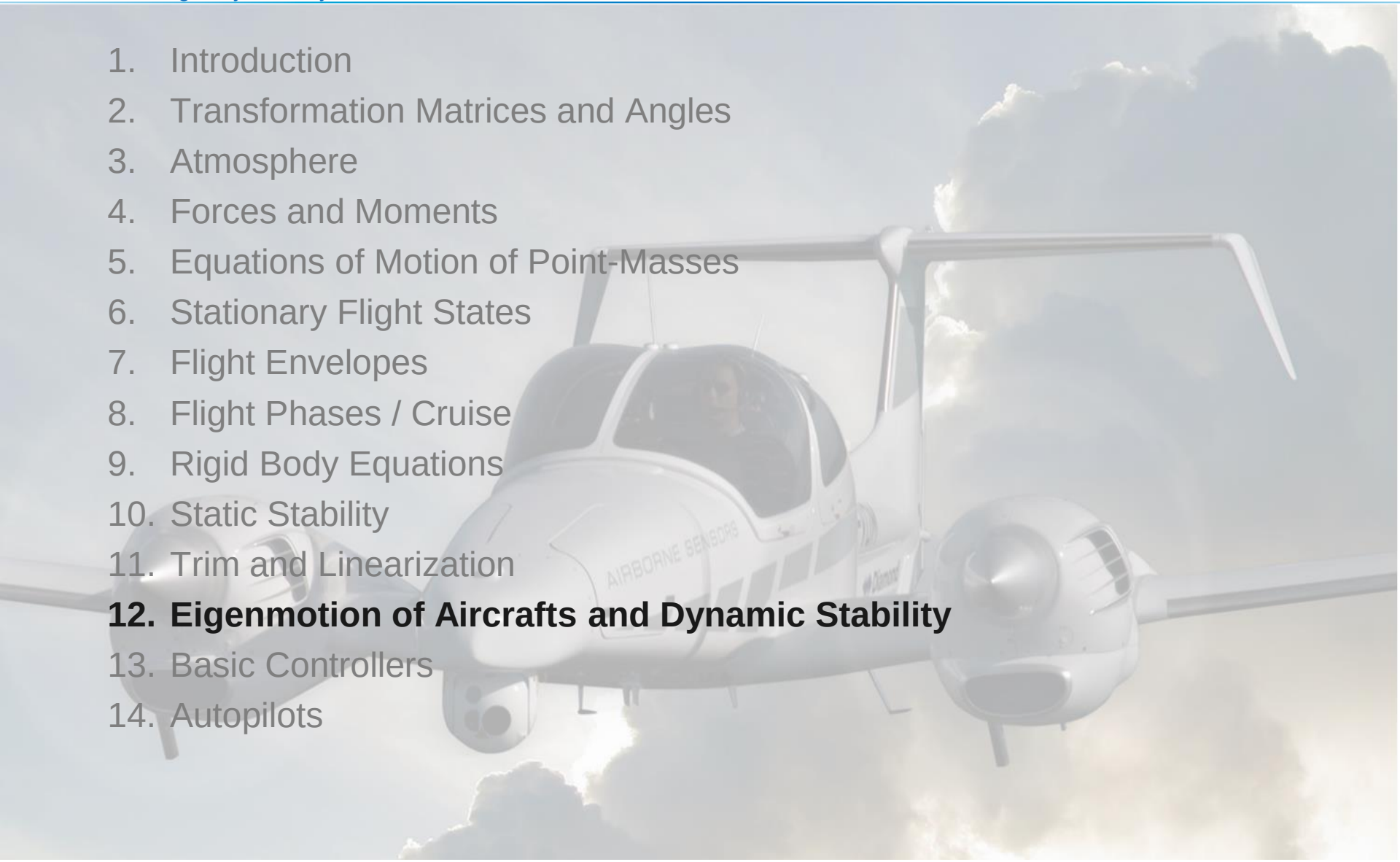
$x^G > x^N$ | $C_{m\alpha}^G < 0$ | $SM > 0$



In order for the aircraft to be statically stable around the pitch axis, the Neutral Point must be behind the Center of Gravity



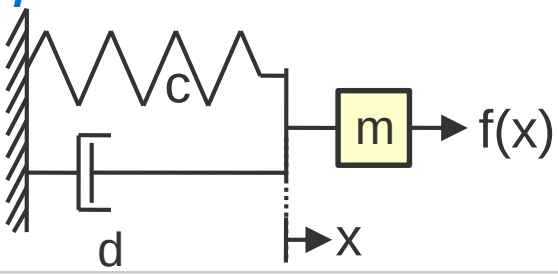
1. Introduction
2. Transformation Matrices and Angles
3. Atmosphere
4. Forces and Moments
5. Equations of Motion of Point-Masses
6. Stationary Flight States
7. Flight Envelopes
8. Flight Phases / Cruise
9. Rigid Body Equations
10. Static Stability
11. **Trim and Linearization -> Exercise 9**
12. Eigenmotion of Aircrafts and Dynamic Stability
13. Basic Controllers
14. Autopilots



1. Introduction
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Example: Mass-Spring-Damper System

- ***Equation of Motion:***



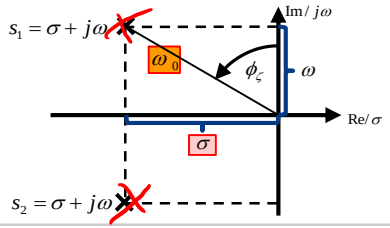
Spring constant	c
Damping coefficient	d
Mass	m
Deflection	x

$$m\ddot{x} + d\dot{x} + cx = 0$$

$$\ddot{x} + \frac{d}{m}\dot{x} + \frac{c}{m}x = 0$$

The differential equation of a linear 2nd order system can be written as:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = 0 \qquad \text{or} \qquad \ddot{x} + 2\sigma\dot{x} + \omega_0^2 x = 0$$



Comparing the coefficients leads to:

Natural Frequency: $\omega_0 = \sqrt{\frac{c}{m}}$ **Absolute Damping:** $\sigma = -\frac{1}{2}\frac{d}{m}$ **Relative Damping:** $\zeta = -\frac{\sigma}{\omega_0} = \frac{d}{2\cdot\sqrt{m\cdot c}}$

Characteristic Polynomial:

$$\lambda^2 + \frac{d}{m}\cdot\lambda + \frac{c}{m} = 0 \quad \Rightarrow \lambda^2 + 2\zeta\,\omega_0\cdot\lambda + \omega_0^2 = 0 \quad \Rightarrow \quad \lambda_{1,2} = \sigma \pm j\,\omega = -\zeta\,\omega_0 \pm j\,\omega_0\sqrt{1-\zeta^2} \quad \Rightarrow \quad \lambda_{1,2} = -\frac{d}{2m} \pm \sqrt{\left(\frac{d}{2m}\right)^2 - \frac{c}{m}}$$

Eigenvalues

- ***State Space Model:***

$$\begin{bmatrix}\dot{x}\\ \ddot{x}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{c}{m} & -\frac{d}{k}\end{bmatrix}\begin{bmatrix}x\\ \dot{x}\end{bmatrix}$$

Typical Eigenvalues und Eigenvectors (1)

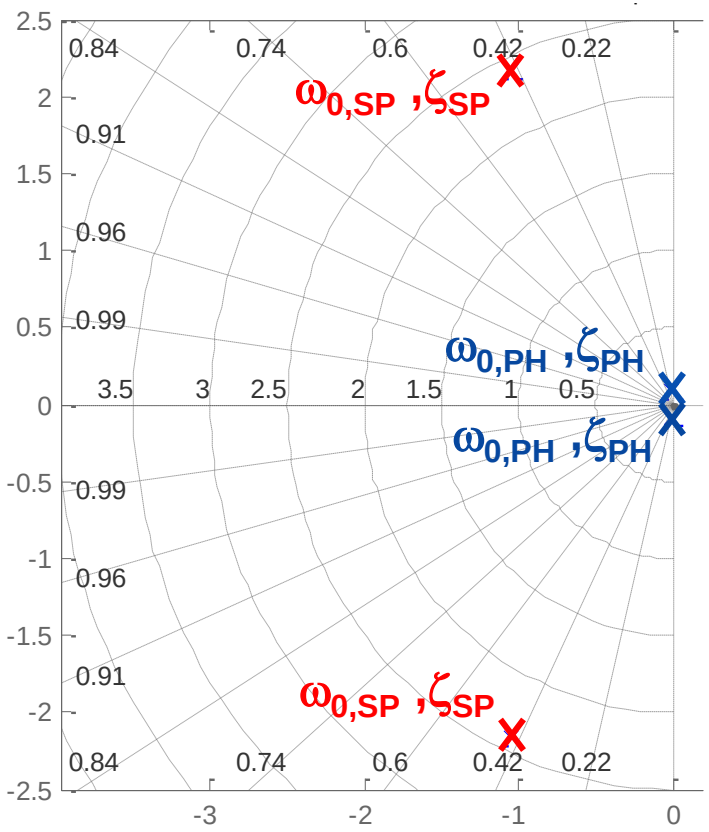
Typical position of the eigenvalues of a conventional aircraft in the complex plane:



source: www.airliners.net

Example Aircraft and Trim Data:

Aircraft:	DC 8
Mach Number:	0.443
Velocity:	142,6464 [m/s]
Mass:	86182,55 [kg]
Altitude:	4,57200 [km]



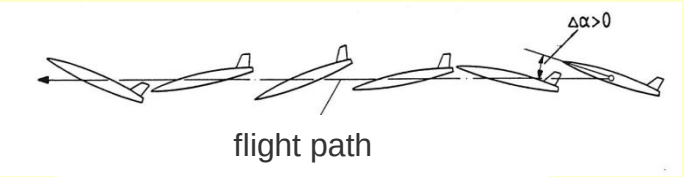
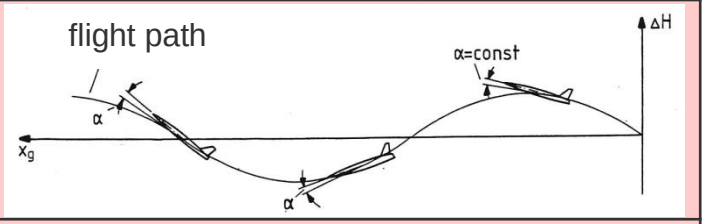
- **Two pairs of complex conjugate eigenvalues**
  → Two periodic and stable eigenmodes
- **The two different eigenmodes have highly different natural frequencies**

	Re(λ) [-]	Im(λ) [-]	ζ [-]	ω_0 [rad/s]
1	-0.0027447	0.089143	0.030776	0.089185
	-0.0027447	-0.089143	0.030776	0.089185
2	-1.043	2.1613	0.43462	2.3998
	-1.043	-2.1613	0.43462	2.3998

Slow periodic (low frequency), weakly damped
Phugoid (PH)

Fast periodic (high frequency), strongly damped
Short Period (SP)

Comparison of Eigenmodes (1)

Eigenmodes of the Longitudinal Motion	**Short Period**	**Phugoid**
		
Damping	Strongly damped	Weakly damped
Time Period (frequency)	Small period time ($T_{DC\,8\ Short\ Period} \approx 3$ sec) (higher frequency)	High period time ($T_{DC\,8\ Phygoide} \approx 70$ sec) (small frequency)
Contribution of the velocity V	Small contribution (corresponding elements are small in the eigenvector)	High contribution (corresponding elements are high in the eigenvector)
Contribution of the flight path angle γ	Small contribution (corresponding elements are small in the eigenvector)	High contribution (corresponding elements are high in the eigenvector)
Contribution fo the angle of attack α	High contribution (corresponding elements are high in the eigenvector)	Small contribution (corresponding elements are small in the eigenvector)
Contribution of the pitch rate q	High contribution (corresponding elements are high in the eigenvector)	Small contribution (corresponding elements are small in the eigenvector)

Complementary Attributes of Short Period and Phugoid

Conclusion

Comparison of Eigenmodes (2)

	Short Period	Phugoid
	The period time of the short period is significantly smaller than the period time of the Phugoid, the contribution of the primary states of the Phugoid on the short period V and γ are very small	The period time of the short period is significantly smaller than the period time of the Phugoid, the contribution of the primary states of the Phugoid on the short period V and γ are very small The states α and q decay very fast when looking at the Phugoid time scale. Because of this, α and q can be neglected for a simplified model of the Phugoid.
Used states for the simplification	α and q	V and γ

⇒ Simplification of the eigenmodes of the longitudinal motion via dividing the state space model

$$\begin{bmatrix}\dot{V}\\ \dot{\gamma}\\ \dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}X_V & -g & X_\alpha & X_q\\ -Z_V & 0 & -Z_\alpha & -Z_q\\ Z_V & 0 & Z_\alpha & Z_q+1\\ M_V & 0 & M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}V\\ \gamma\\ \alpha\\ q\end{bmatrix} + \begin{bmatrix}X_\eta & X_{\delta T}\\ -Z_\eta & -Z_{\delta T}\\ Z_\eta & Z_{\delta T}\\ M_\eta & 0\end{bmatrix}\cdot\begin{bmatrix}\eta\\ \delta_T\end{bmatrix}$$

Short Period – Simplification

Using above conclusions leads to:

$$V = 0$$
$$\gamma = 0$$

$$\begin{bmatrix} \dot{V} \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} X_V & -g & X_\alpha & X_q \\ -Z_V & 0 & -Z_\alpha & -Z_q \\ Z_V & 0 & Z_\alpha & Z_q + 1 \\ M_V & 0 & M_\alpha & M_q \end{bmatrix} \cdot \begin{bmatrix} V \\ \gamma \\ \alpha \\ q \end{bmatrix} + \begin{bmatrix} X_\eta & X_{\delta T} \\ -Z_\eta & -Z_{\delta T} \\ Z_\eta & Z_{\delta T} \\ M_\eta & 0 \end{bmatrix} \cdot \begin{bmatrix} \eta \\ \delta_T \end{bmatrix}$$

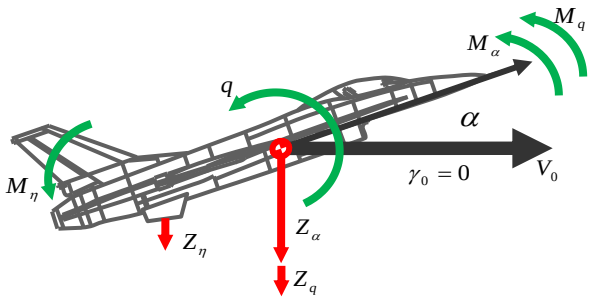
Primary control of α and q via η

Simplification of the short period

$$\begin{bmatrix} \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Z_\alpha & Z_q + 1 \\ M_\alpha & M_q \end{bmatrix} \cdot \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_\eta \\ M_\eta \end{bmatrix} \cdot [\eta]$$

Comparison Spring Mass Damper System

Short Period (Simplification):



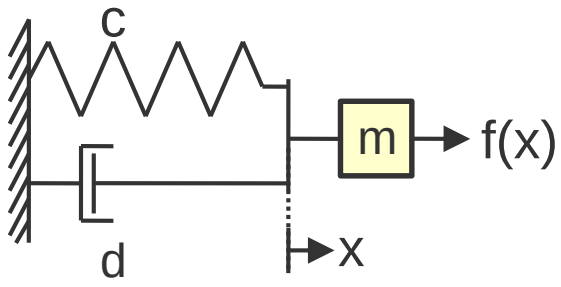
State space model:

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\eta$$

Characteristic equation:

$$\det(s\mathbf{I}-\mathbf{A}) = s^2 - (M_q + Z_\alpha)s + (Z_\alpha M_q - M_\alpha(Z_q+1))$$

Spring Mass Damper System:



State space model:

$$\begin{bmatrix}\dot{x}\\ \ddot{x}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{c}{m} & -\frac{d}{m}\end{bmatrix}\begin{bmatrix}\dot{x}\\ x\end{bmatrix}$$

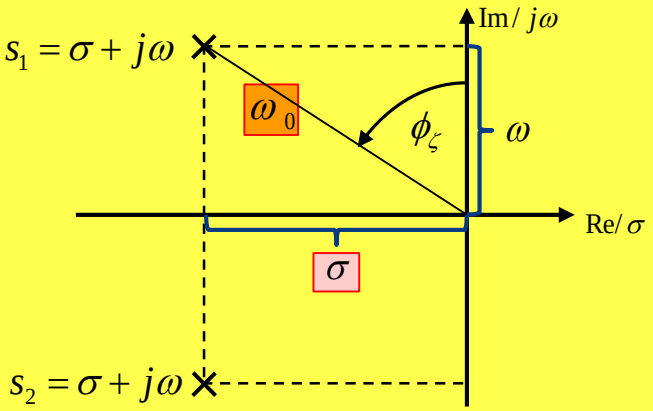
Characteristic equation:

$$\det(s\mathbf{I}-\mathbf{A}) = s^2 + \frac{d}{m}s + \frac{c}{m}$$

Characteristic equation:

(linear 2nd order system)

$$s^2 - 2\sigma\cdot s + \omega_0^2 = 0$$



Comparison Spring Mass Damper System (2)

Short Period (Simplification):

$$\det(s\mathbf{I}-\mathbf{A}) = s^2 - (M_q + Z_\alpha)s + (Z_\alpha M_q - M_\alpha(Z_q+1))$$

Spring Mass Damper System

$$\det(s\mathbf{I}-\mathbf{A}) = s^2 + \frac{d}{m}s + \frac{c}{m}$$

Characteristic Equation:

(linear 2nd order system)

$$s^2 - 2\sigma \cdot s + \omega_0^2 = 0$$

Comparison of coefficients:

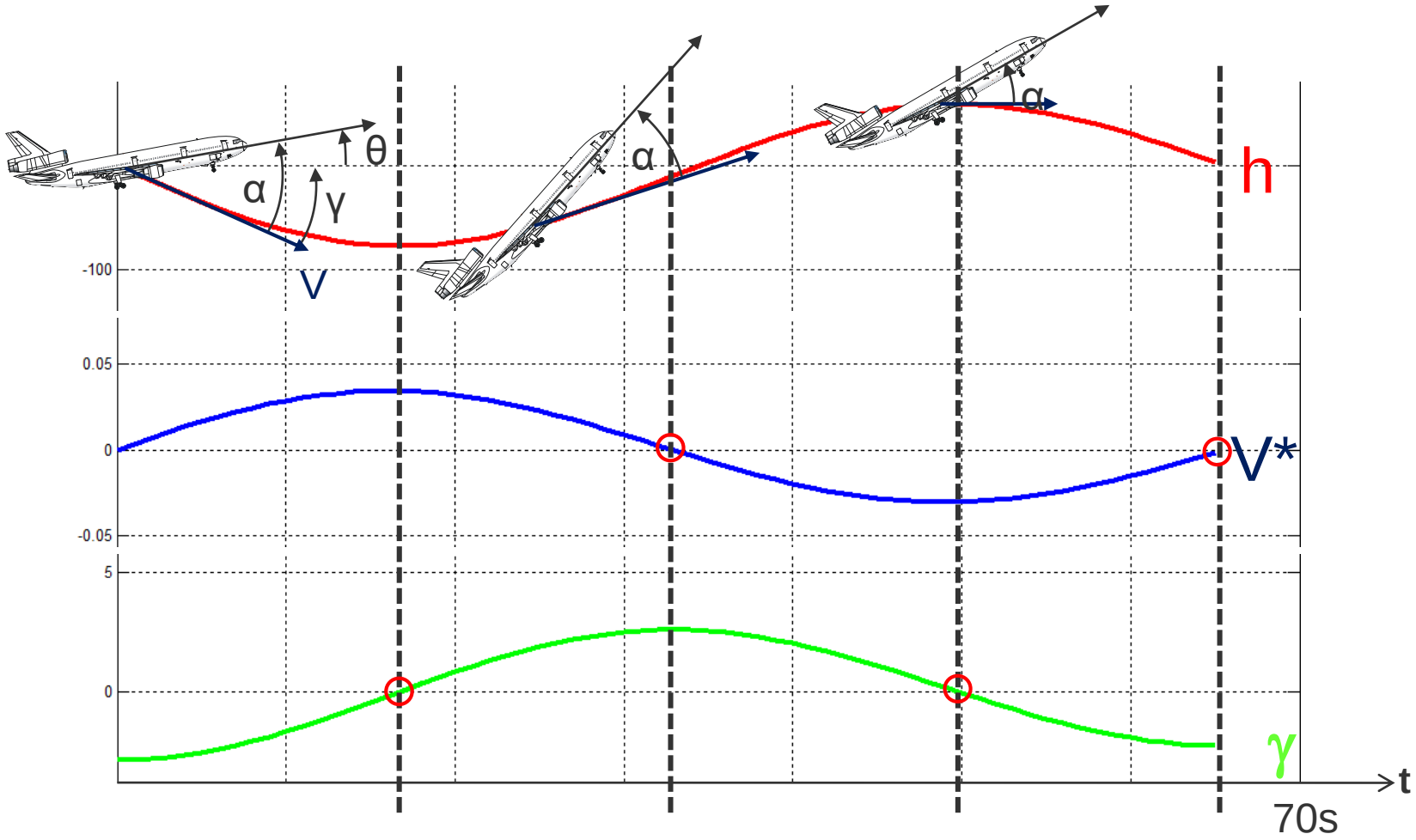
$\omega_{0,SP} = \sqrt{Z_\alpha M_q - M_\alpha(Z_q+1)} \approx \sqrt{-M_\alpha}$	$\Leftrightarrow$	$\omega_0 = \sqrt{\frac{c}{m}}$
$\sigma_{SP} = \frac{1}{2}(M_q + Z_\alpha) \approx \frac{1}{2}M_q$	$\Leftrightarrow$	$\sigma = -\frac{1}{2}\frac{d}{m}$
$\zeta_{SP} = -\frac{\sigma_{SP}}{\omega_{0,SP}} = -\frac{\frac{1}{2}(M_q + Z_\alpha)}{\sqrt{Z_\alpha M_q - M_\alpha(Z_q+1)}}$	$\Leftrightarrow$	$\zeta = -\frac{\sigma}{\omega_0} = \frac{d}{2\cdot\sqrt{m\cdot c}}$

Main Influences

$$Z_\alpha M_q - M_\alpha(Z_q+1) \triangleq \frac{c}{m}$$ Spring constant

$$(M_q + Z_\alpha) \triangleq -\frac{d}{m}$$ Damping coefficient

Process of the Phugoid



Energy Conservation

Approximately the Phugoid is an eigenmotion with constant (specific) energy:

$$\frac{V^2}{2} + gh = const.$$

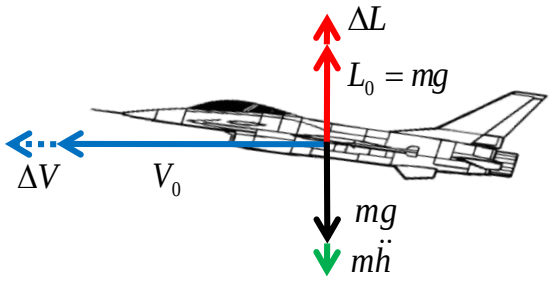
This leads to the following relation to the reference state 0:

$$V^2 - V_0^2 = -2g(h - h_0)$$

Equations of motion in vertical direction for constant angle of attack

$$m\ddot{h} = \Delta L = L - L_0 \approx \underbrace{C_{L0} \cdot \tfrac{1}{2}\rho_0 S}_{\frac{mg}{V_0^2}} \cdot \underbrace{(V^2 - V_0^2)}_{-2g(h-h_0)}$$

Assumption:
AoA does not contribute to the Phugoid
-> lift coefficient held constant at C_{L0}



$$\rightarrow \quad \ddot{h} + 0 \cdot \dot{h} + 2\left(\frac{g}{V_0}\right)^2 (h - h_0) = 0$$

This corresponds to an undamped oscillation with the solution

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = 0$$

$$(h - h_0) = |h - h_0|_{max} \sin(\omega_{0,PH} t)$$

and the natural frequency

$$\omega_{0,PH} = \frac{g}{V_0}\sqrt{2}$$

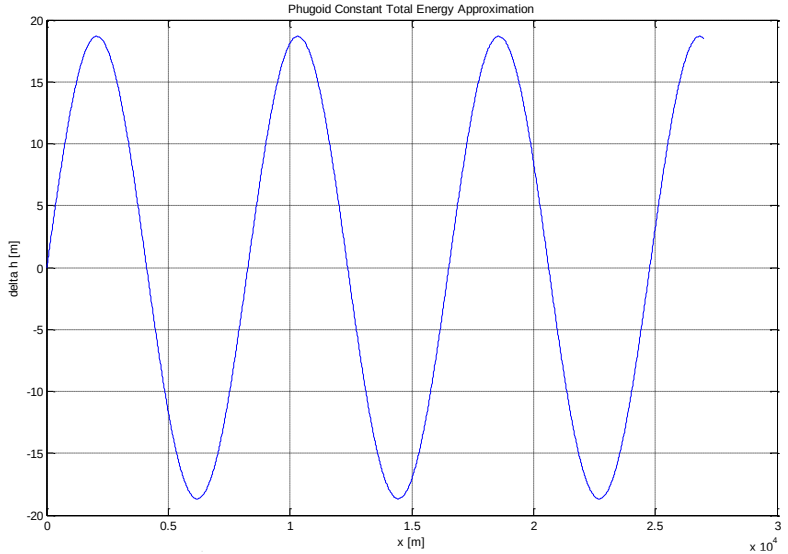
The **approximated natural frequency of the Phugoid** is:
- inversely proportional to the velocity
- independent of the altitude
- independent of the aircraft configuration

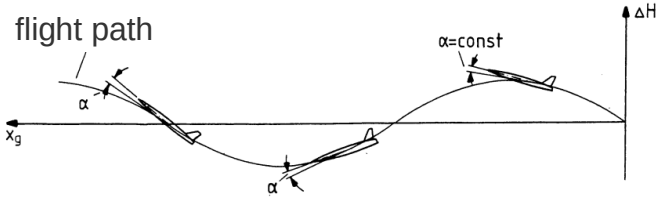
Energy Conservation (2)

Path of the center of gravity during the Phugoid

Relative motion of the COG for an observer on the ground

$$V_{observer} = 0$$

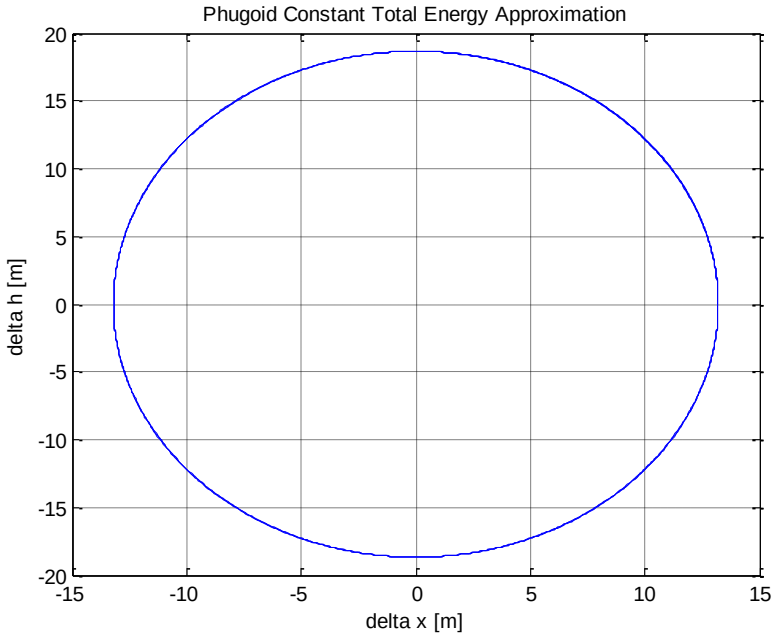




$$x_g = V_0 \cdot t + \Delta x_g = V_0 \cdot t + \frac{1}{\sqrt{2}} \frac{V_0}{g} \Delta V_{\max} \cdot \cos(\omega_{PH} \cdot t)$$

Relative motion of the aircraft for a moving observer

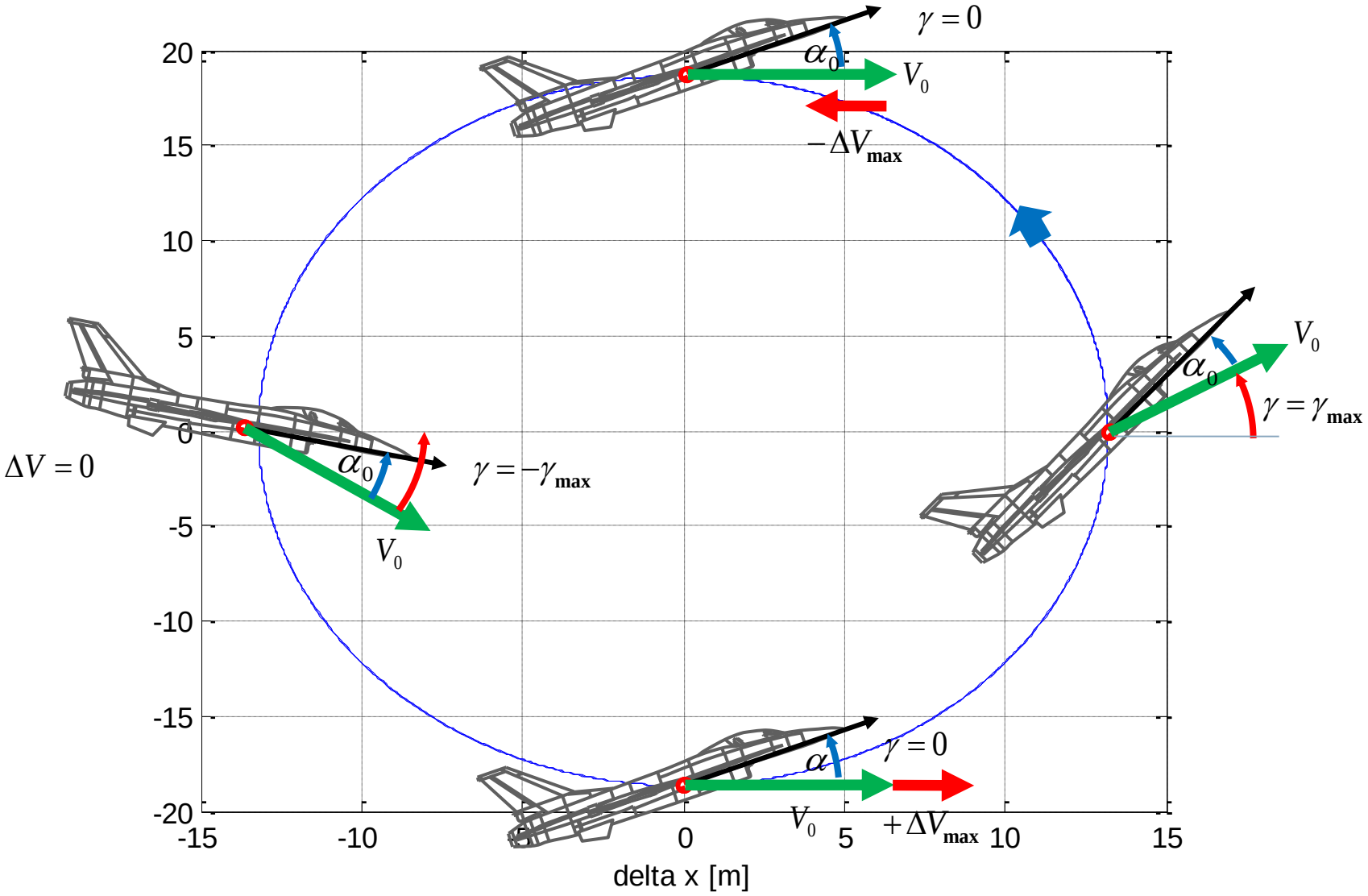
$$V_{observer} = V_0$$



Ellipse with an axis ratio:

$$\frac{\Delta x_{\max}}{\Delta h_{\max}} = \frac{1}{\sqrt{2}} = \frac{13{,}2032}{18{,}6709}$$

Energy Conservation (3)



Sample Exercise E.3

Two **identical** aircraft configurations A and B are considered. Aircraft A flies 4 times as fast as aircraft B.
How do the Phugoid frequencies of aircraft A and B compare?

$$\omega_{0,PH} = \frac{g}{V_0}\sqrt{2}$$

Two **different** aircraft configurations A and B are considered. Aircraft A flies 4 times as fast as aircraft B.
Can you still compare their Phugoid frequencies with the information given?

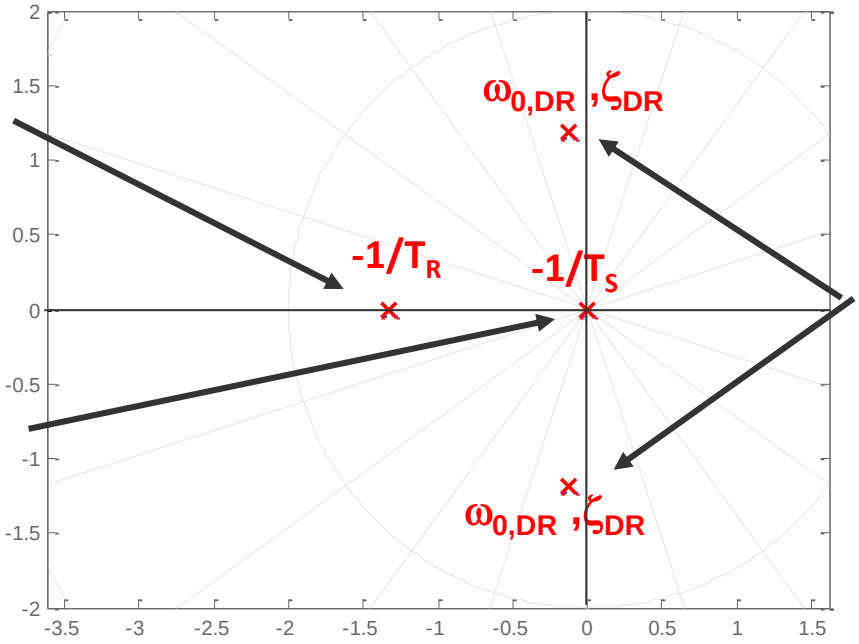
Hint: you can assume that the Phugoid is an eigenmotion of constant total energy.

Typical Eigenvalues and Eigenvectors(2)

Typical position of the

real eigenvalue $-1/T_R$: stable, aperiodic eigenmode

real eigenvalue $-1/T_S$: stable or instable, aperiodic eigenmode



complex conjugate eigenvalue pair $\omega_{0,DR}$,ζ_{DR}: stable, periodic eigenmode

	Re(λ) [-]	Im(λ) [-]	ζ [-]	ω_0 [rad/s]	
1	-1.3301	0	1	1.3301	fast, aperiodic — Roll Mode
2	-0.12742	-1.1906	0.10641	1.1974	fast, periodic — Dutch-Roll
	-0.12742	1.1906	0.10641	1.1974	
3	-0.0065518	0	1	0.0065518	slow, aperiodic — Spiral Mode

Roll Control (1)

State space model of the approximation of the Roll Mode

$$\begin{bmatrix}\dot{p}\\ \dot{\Phi}\end{bmatrix} = \begin{bmatrix}L_p & 0\\ 1 & 0\end{bmatrix}\cdot\begin{bmatrix}p\\ \Phi\end{bmatrix} + \begin{bmatrix}L_\xi\\ 0\end{bmatrix}\cdot\xi$$

$$\frac{p(s)}{\xi(s)} = \frac{L_\xi}{s - L_p} = -\frac{L_\xi}{L_p}\cdot\frac{1}{T_R\cdot s + 1}$$

Roll moment equation

$$\dot{p} = L_P\cdot p + L_\xi\cdot\xi$$

$$\xrightarrow[\text{Laplace Transformation}]{\text{L}}$$

$$s\cdot p(s) = L_p\cdot p(s) + L_\xi\cdot\xi(s)$$
$$p(s)\cdot(s - L_p) = L_\xi\cdot\xi(s)$$

Transfer function from aileron ξ to the roll rate p

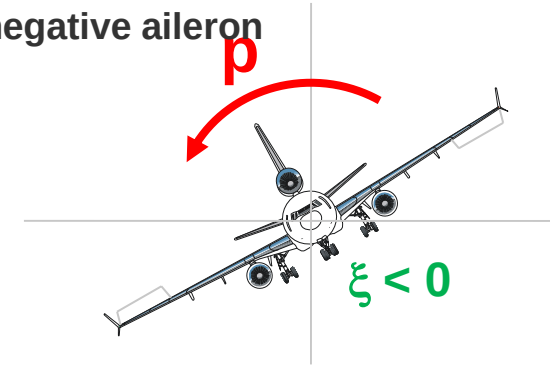
$$\frac{p(s)}{\xi(s)} = \frac{L_\xi}{s - L_p} = -\frac{L_\xi}{L_p}\cdot\frac{1}{T_R s + 1}$$

$$T_R = -\frac{1}{L_p}$$

PT_1 – behavior

Delay of first order as approximation of Roll Mode

Aircraft reaction to a negative aileron deflection (ξ<0):

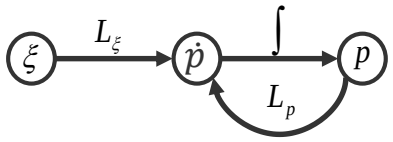


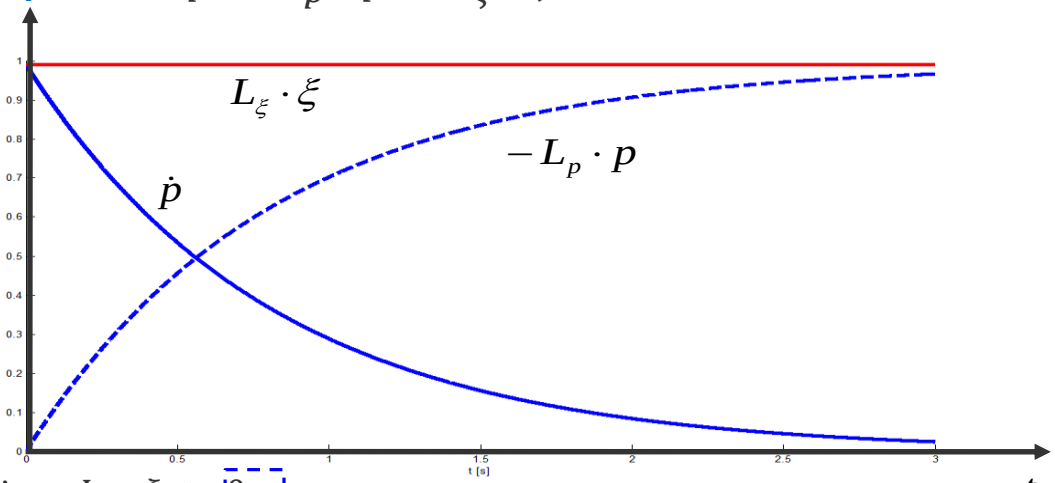
A negative aileron deflection (ξ<0) leads to a positive roll rate p

Negative aileron deflection (ξ<0) means:
- negative surface deflection on the right side
- positive surface deflection on the left side

Roll Control (2)

Explanation of the roll moment equation: $\dot{p} = L_p \cdot p + L_\xi \cdot \xi$





Negative aileron deflection $\xi_0 < 0$

< 0

➔ Positive roll moment: $L_\xi \cdot \xi_0 > 0$

➔ Positive initial roll acceleration: $\dot{p}_0 = L_\xi \cdot \xi_o > 0$

< 0

➔ Build up of a positive roll rate p: $p = \int_0^t \dot{p} dt > 0$

➔ Positive roll rate leads to a negative roll damping: $L_p \cdot p < 0$

➔ Increasing roll rate p leads to a constant increase of roll damping moment until the moment induced by roll damping equals the moment produced by the aileron deflection leading to a zero roll acceleration.

$L_p \cdot p = -L_\xi \cdot \xi$
$\Rightarrow \dot{p}_\infty = 0$

➔ **Stationary roll rate p_∞**

Step response of the roll rate

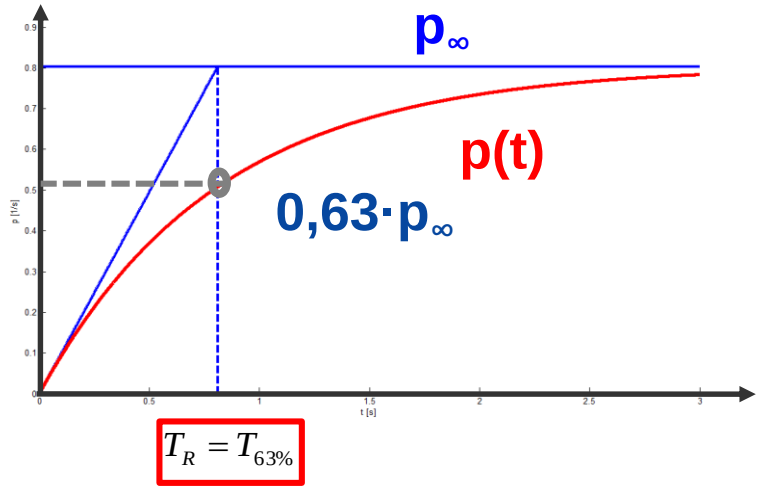
Aileron effectiveness

$$p(t) = -\frac{L_\xi}{L_p} \cdot \xi_0 \cdot \left(1 - e^{-\frac{t}{T_R}}\right)$$

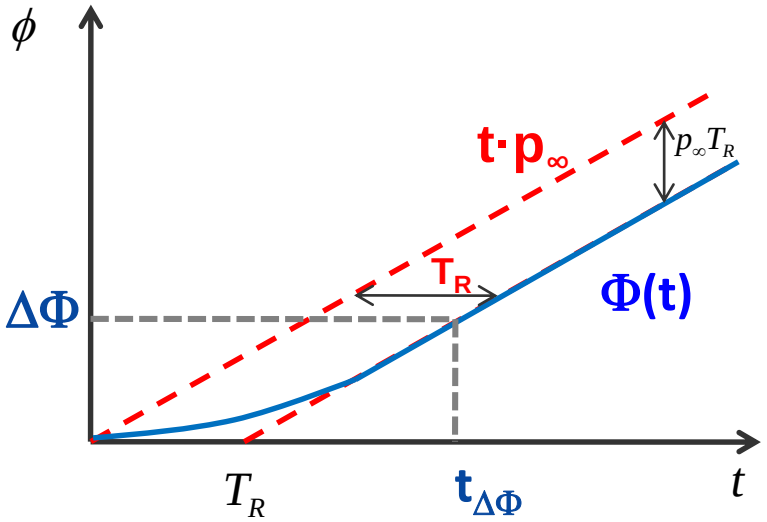
Roll damping

$$T_R = -\frac{1}{L_p}$$

Roll time constant from simplification of Roll Mode



- The roll time constant is the point in time, at which the roll rate equals 63% of the steady state roll rate (due to a step input of the aileron)
- The higher the roll time constant T_R, the longer the aircraft takes to achieve a rollrate after an aileron command
- The bank angle follows from integrating the roll rate p



The roll time constant T_R of an aircraft is 1 second. An initial aileron deflection ξ'_o of 2 degree caused p_∞ a steady state roll rate of 50 deg/s. What is the **initial roll acceleration out of level flight for an aileron deflection ξ_o of 10 degrees**?

$$\dot{p} = L_p \cdot p + L_\xi \cdot \xi \qquad \xi_o = \underbrace{10^\circ}$$

Initial roll acceleration (from level-flight): $\dot{p}_0 = 0 + L_\xi \cdot \xi_o$

Problem: L_ξ unknown

Steady-state roll rate for $\xi'_o = 2^\circ$: $p_\infty = 50°/s$

$$\dot{p}_\infty \overset{0}{=} L_p \cdot p_\infty + L_\xi \cdot \xi'_o \quad \rightarrow \quad L_\xi = -\frac{L_p p_\infty}{\xi'_o} = -\frac{-\frac{1}{T_R} p_\infty}{\xi'_o} = -\frac{-\frac{1}{1s} 50 \frac{^\circ}{s}}{2^\circ} = 25 \frac{1}{s^2}$$

$$\rightarrow \quad \dot{p}_0 = 0 + 25 \frac{1}{s^2} \cdot 10^\circ = 250^\circ \frac{1}{s^2}$$

Simplification as Pure Yaw Oscillation (1)

State Space Model Lateral Motion

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} N_r & N_\beta & N_p & 0 \\ -1 & Y_\beta & 0 & \frac{g}{V_0} \\ L_r & L_\beta & L_p & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} N_\xi & N_\zeta \\ Y_\xi & Y_\zeta \\ L_\xi & L_\zeta \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

Influence small

- The influence of N_p in the yaw moment equation and g/V_0 in the side force equation is comparably small
  - ➔ The Dutch Roll is mainly a yaw oscillation with a coupled and phase-shifted roll oscillation
  - ➔ Approximation of the Dutch Roll as pure yaw oscillation possible
- For aircrafts without dihydral wings and wing sweep the Dutch roll is almost a pure yaw oscillation



source: analitica.com

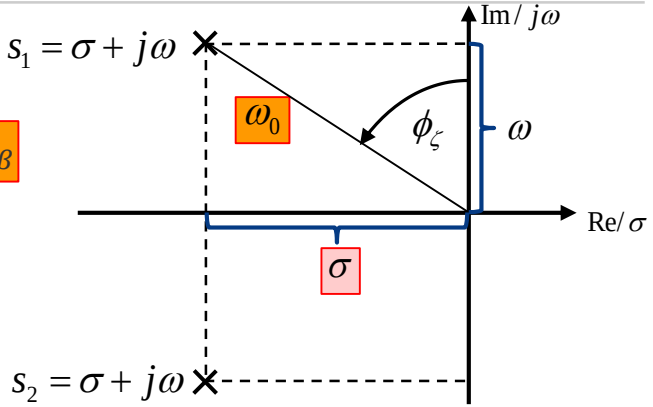
Simplification of the Dutch Roll as pure yaw oscillation

(⇒ Decoupling of roll and yaw states)

$$\begin{bmatrix}\dot{r}\\ \dot{\beta}\end{bmatrix} = \begin{bmatrix}N_r & N_\beta\\ -1 & Y_\beta\end{bmatrix} \cdot \begin{bmatrix}r\\ \beta\end{bmatrix} + \begin{bmatrix}N_\zeta\\ Y_\zeta\end{bmatrix} \cdot \zeta$$

Characteristic Equation

$$\begin{aligned}\det(sI - A) &= \det\begin{bmatrix}s - N_r & -N_\beta\\ 1 & s - Y_\beta\end{bmatrix} = s^2 - (N_r + Y_\beta)\cdot s + N_r Y_\beta + N_\beta\\ &= s^2 - 2\cdot\sigma_{DR}\cdot s + \omega_{0,DR}^2\\ &= s^2 + 2\cdot\zeta_{DR}\cdot\omega_{0,DR}\cdot s + \omega_{0,DR}^2\end{aligned}$$



Comparison of coefficients:

➔ **Natural Frequency** $\omega_{0,DR} = \sqrt{N_R Y_\beta + N_\beta}$

➔ **Absolute Damping** $\sigma_{DR} = 1/2 \cdot (N_R + Y_\beta)$

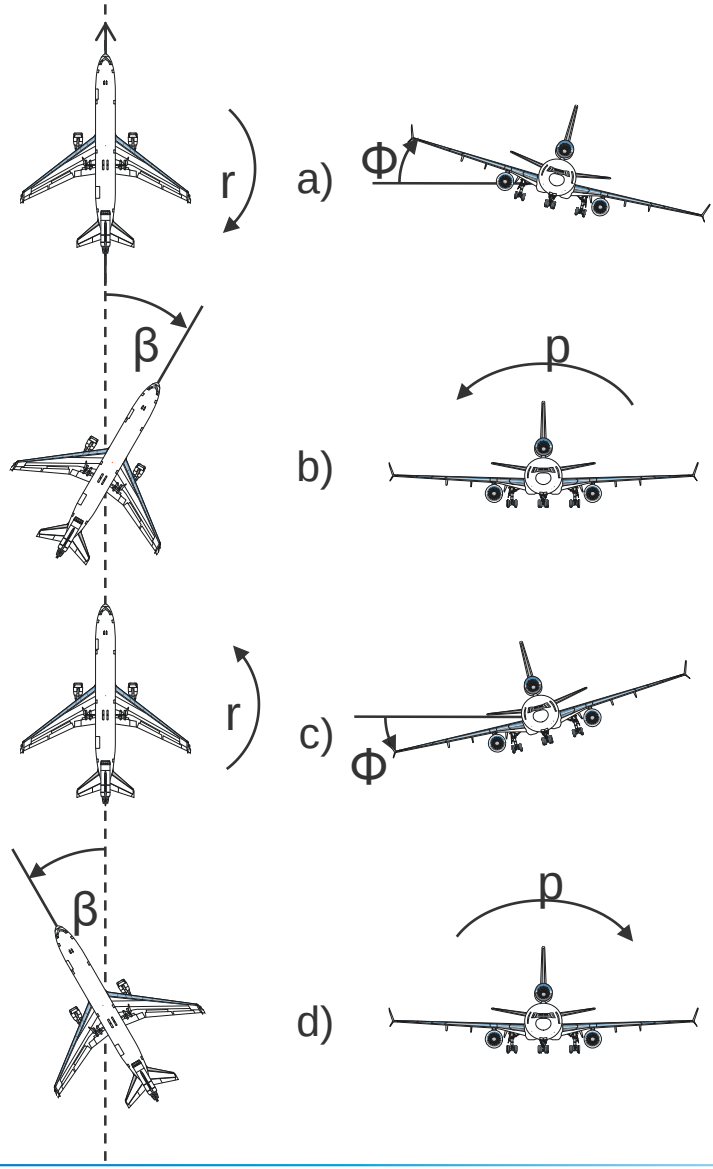
Main influences

➔ **Relative Damping** $\zeta_{DR} = \frac{-\sigma_{DR}}{\omega_{0,DR}} = \frac{-(N_r + Y_\beta)}{2\cdot\sqrt{N_r Y_\beta + N_\beta}}$

Discussion (1)

- Characteristic Dutch Roll motion through coupling of yaw and roll oscillation
- Process of the Dutch Roll
  a) Aircraft has a positive yaw rate r, the right wing points upward (negative bank angle Φ)
  b) Aircraft reaches minimum sideslip angle β and has a positive roll rate p at the horizontal position $\Phi = 0$
  c) Aircraft has a negative yaw rate r while being side slip free $\beta = 0$ and reaches maximum bank angle Φ
  d) Aircraft reaches maximum sideslip angle β and rotates with a negative roll rate p through the horizontal position $\Phi = 0$

- Oscillation is damped and decays over time



Stability Analysis (1)

The approximation for the spiral mode yields following eigenvalues for Roll and Spiral Mode:

$$\lambda_R = -\frac{1}{T_R} = L_p \cdot \left(1 + \frac{N_p - g/V_0}{L_p} \cdot \frac{L_\beta}{N_\beta}\right)$$

$$\lambda_S = -\frac{1}{T_S} = -T_R \cdot \frac{g}{V_0} \cdot \frac{1}{N_\beta} \cdot \left(N_r L_\beta - N_\beta L_r\right)$$

The Spiral Mode is stable if $\lambda_S < 0$:

$$T_R \cdot \frac{g}{V_0} \cdot \frac{1}{N_\beta} \cdot \left(N_r L_\beta - N_\beta L_r\right) > 0$$

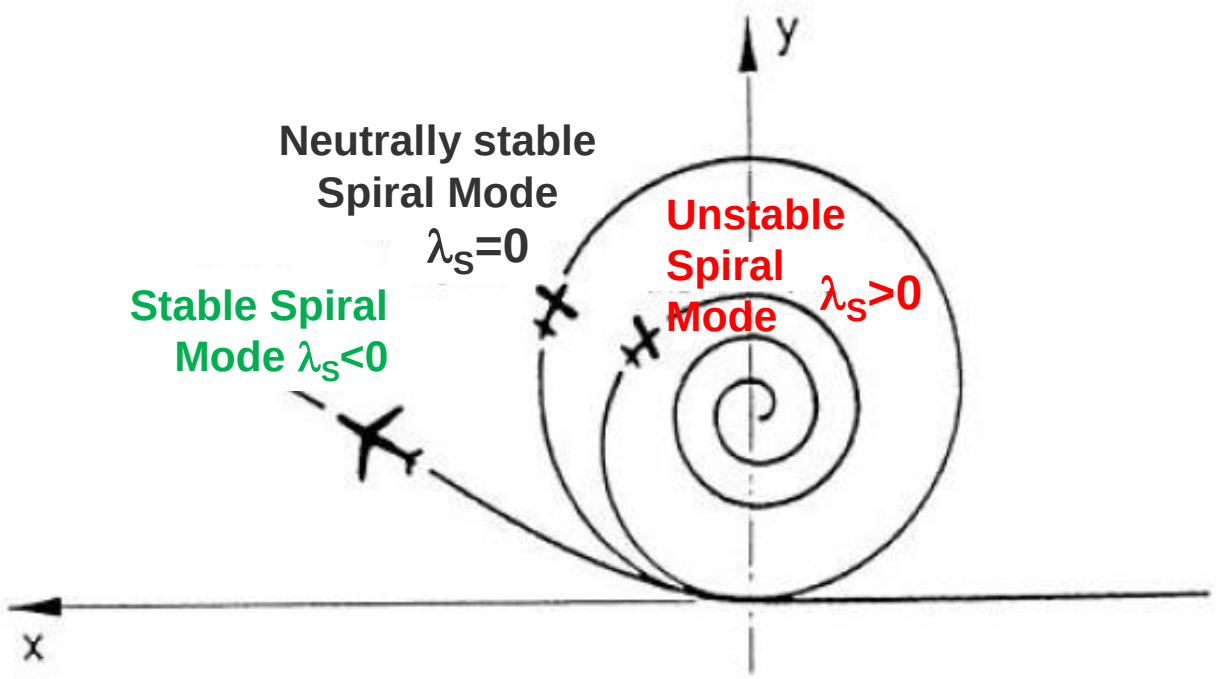
T_R, g, V_0 are greater than 0, and the yaw moment due to a sideslip N_β usually is greater than 0. This leads to the following criteria for stability of the Spiral Mode

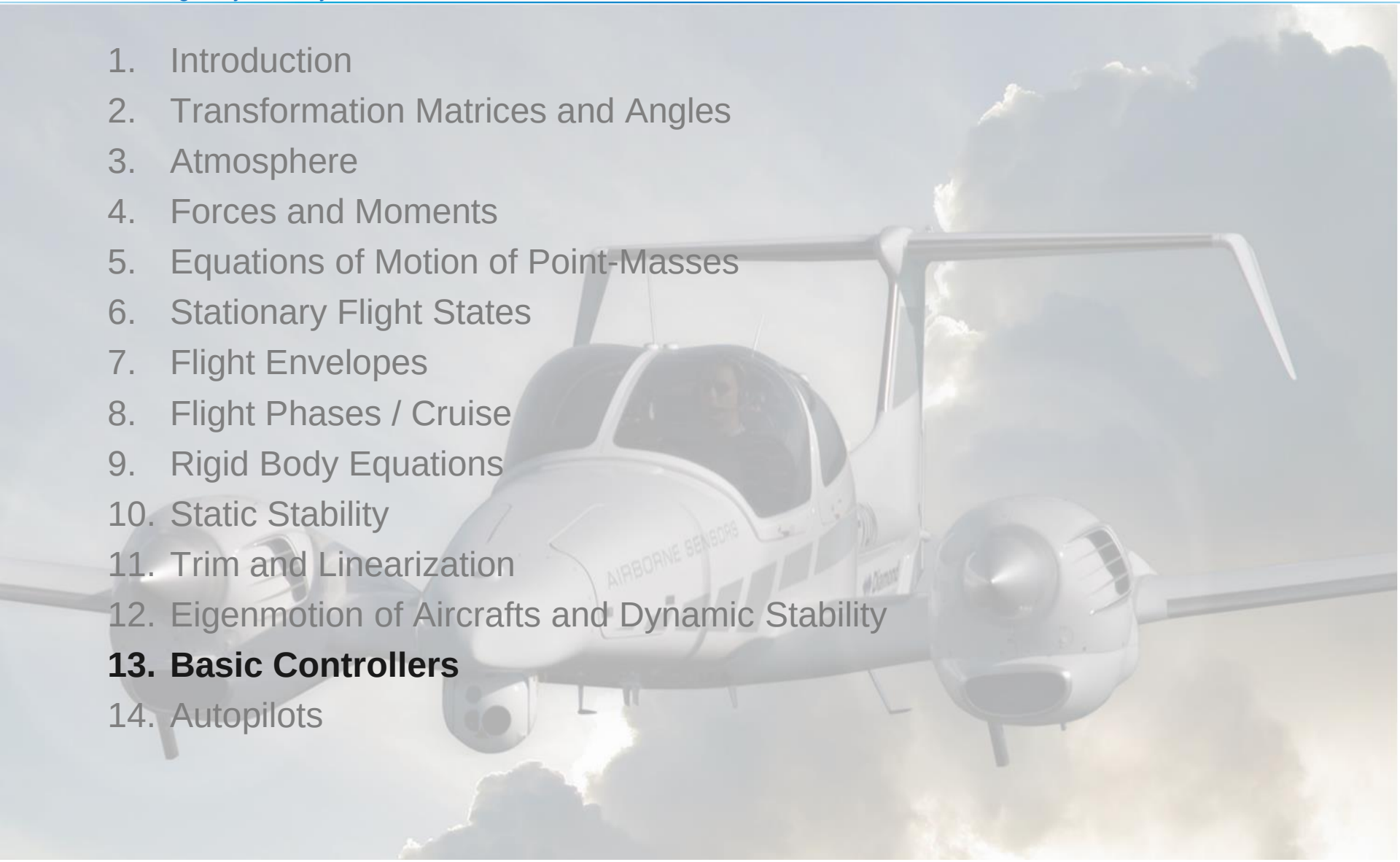
$$\underbrace{\underset{<0}{N_r} \cdot \underset{<0}{L_\beta}}_{>0\ \textbf{stabilizing}} - \underbrace{\underset{>0}{N_\beta} \cdot \underset{>0}{L_r}}_{>0\ \textbf{destabilizing}} > 0$$

The **Yaw Damping** N_r and the **Sideslip-Roll-Moment** L_β counteract turn and the Spiral Mode $\Rightarrow$ **stabilizing**

The **Sideslip-Yaw-Moment** N_β and the **Yaw-Roll-Moment** L_r support the turn and the Spiral Mode $\Rightarrow$ **destabilizing**

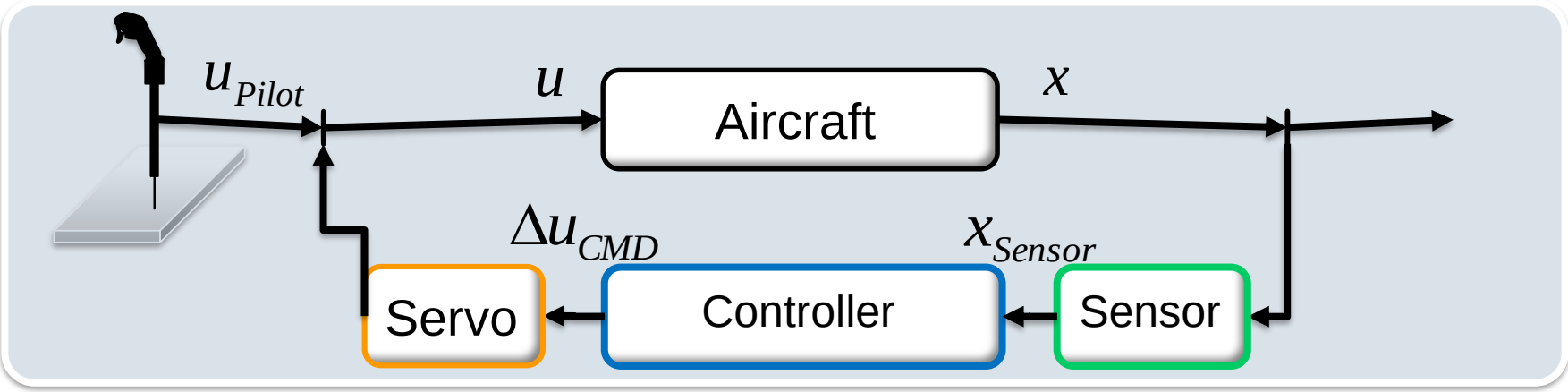
Trajectory of the Spiral Mode





1. Introduction
2. Transformation Matrices and Angles
3. Atmosphere
4. Forces and Moments
5. Equations of Motion of Point-Masses
6. Stationary Flight States
7. Flight Envelopes
8. Flight Phases / Cruise
9. Rigid Body Equations
10. Static Stability
11. Trim and Linearization
12. Eigenmotion of Aircrafts and Dynamic Stability
13. **Basic Controllers**
14. Autopilots

- **SAS Stability Augmentation System**
  - **General**: Changing eigencvectors and eigenvalues
    - Changing damping, natural frequency and time constants
    - Feedback of one ore more measured quantities to one ore more control inputs
    - Desired goals given by requirements for flight attributes
  - **Dampers**:
    - Lacking natural damping of eigenmode
    - Artificial increase via feedback of rates to the control input
      - Pitch Damper: Feedback pitch rate to elevator
      - Yaw Damper: Feedback yaw rate to rudder
      - Roll Damper: Feedback roll rate to aileron

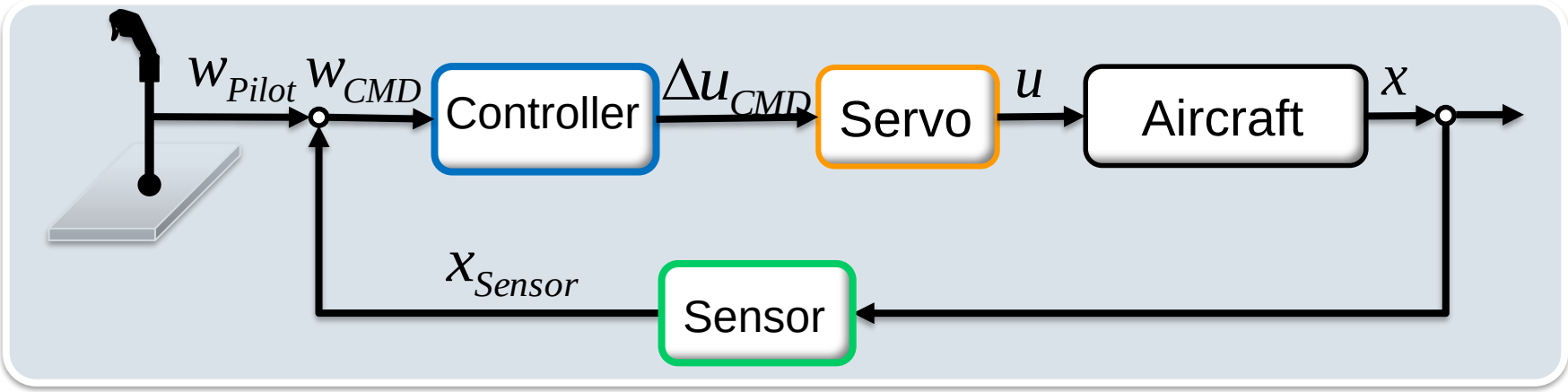


- **CAS Control Augmentation System**

  Pilot does not command surface deflection but desired values for defined quantities

Examples for CAS desired quantities	
Angle of attack	Pitch rate
Load factor	Yaw rate
Roll rate	Bank angle

Pilot inputs can be filtered in order to improve the response behavior for the pilot.



Aircraft Classes

The MIL-F-8785C defines four aircraft types:

Class I

Small, light airplanes

- Light utility
- Primary trainer
- Light observation

Class II

Medium-weight, low-to-medium maneuverability airplanes

- Heavy utility/search and rescue
- Medium Transporter
- Early warning/ electronic counter-measures/airborne command, control, or communications relay
- Antisubmarine
- Assault Transport
- Reconnaissance
- Tactical bomber
- Heavy Attack
- Trainer for Class II

Class III

Large, heavy, low-to-medium maneuverability airplanes

- Heavy transporter/ cargo/tanker
- Heavy bomber
- Trainer for Class III

Large passenger jets

Class IV

High-maneuverability airplanes

- Fighter/Interceptor
- Attack
- Tactical reconnaissance
- Observation
- Trainer for Class IV

Nonterminal Flight Phases

Category A

Require rapid maneuvering, precision tracking, or precise flight path control

- Air-to-air combat (CO)
- Ground attack (GA)
- Weapon delivery/launch
- Aerial recovery (AR)
- Reconnaissance (RC)
- In-flight refueling (receiver) (RR)
- Terrain following (TF)
- Antisubmarine search (AS)
- Close formation flying (FF)

Category B

Normally accomplished using gradual maneuvers and without precision tracking, although accurate flight-path control may be required

- Climb (CL)
- Cruise (CR)
- Loiter (LO)
- In-flight refuelling (tanker) (RT)
- Descent (D)
- Emergency descent (ED)
- Emergency deceleration (DE)
- Aerial delivery (AD)

Terminal Flight Phases

Category C

Normally accomplished using gradual maneuvers and usually require accurate flight-path control.

- Takeoff (TO)
- Catapult takeoff (CT)
- Approach (PA)
- Wave-off/ go-around (WD)
- Landing (L)

Motivation: Requirements for the Short Period

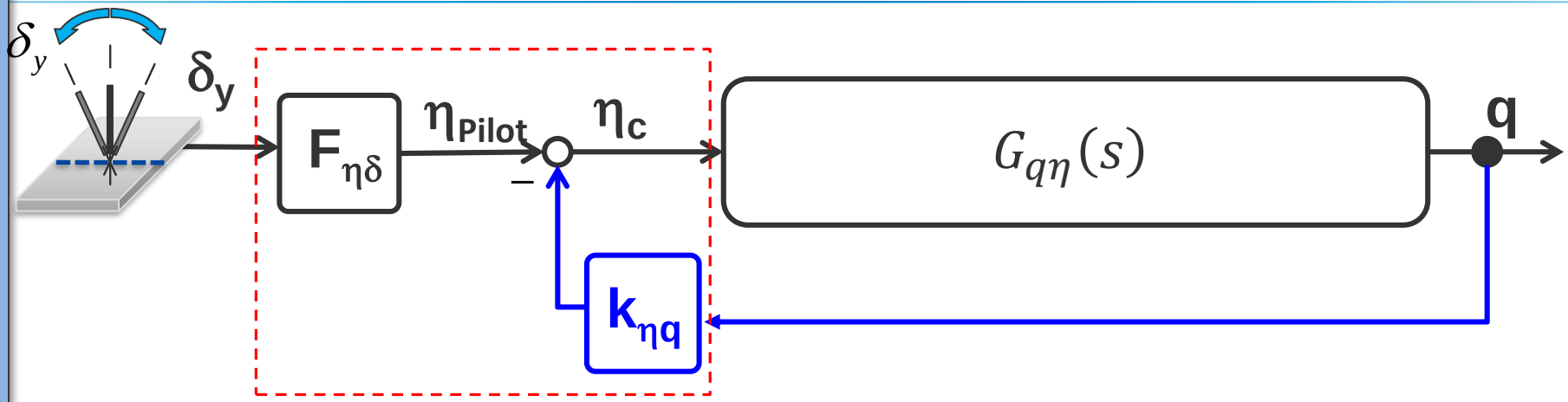
- Requirement for relative damping ζ_{SP} of the short period
  (MIL-F-8785C : Military Specification – Flying Qualities of Piloted Airplanes)

Level	Flight Phases A und C		Flight Phase B	
	Minimum	Maximum	Minimum	Maximum
1	0,35	1,30	0,30	2,00
2	0,25	2,00	0,20	2,00
3	0,15*	-	0,15*	-

Design goal optimal damping:

$$\zeta_{\mathrm{SP}} \approx \zeta_{\mathrm{opt}} = \frac{\sqrt{2}}{2}$$

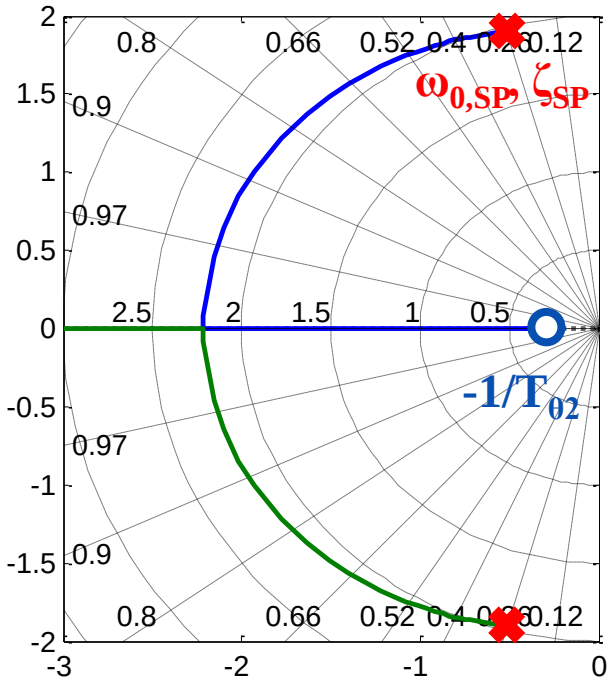
Controller Structure and Root Locus of a Pitch Damper



A feedback of the pitch rate to the elevator leads to the following root locus:

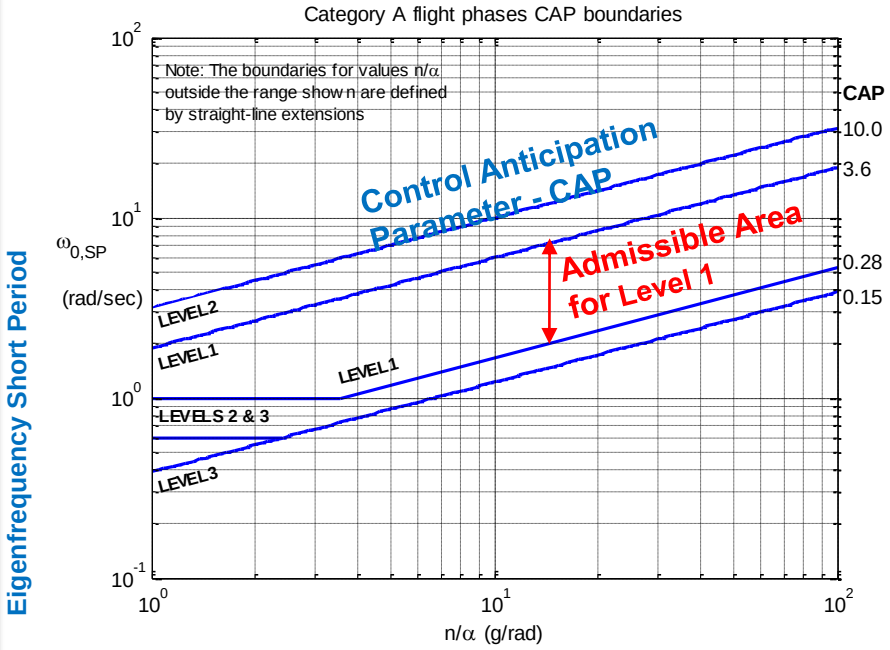
- The natural frequency of the short period remains almost constant.
- Increasing the feedback gain **increases the relative damping of the short period**

Reason: Production of an additional pitch moment proportional to the pitch rate → see next slide



Motivation: Requirements for the Short Period

There are no direct requirements for the eigenfrequency of the short period in the MIL-F-8785C. However, there is a requirement about the closely related CONTROL ANTICIPATION PARAMETER (CAP).



3.2.2 Longitudinal maneuvering characteristics

3.2.2.1 Short-period response. The short-period response of angle of attack which occurs at approximately constant speed, and which may be produced by abrupt pitch control inputs, shall meet the requirements of 3.2.2.1.1 and 3.2.2.1.2. These requirements apply, with the cockpit control free and with it fixed, for responses of any magnitude that might be experienced in sevice use. If oscillations are nonlinear with amplitude, the requirements shall apply to each cycle of the oscillation. In addition to meeting the numerical requirements of 3.2.2.1.1 and 3.2.2.1.2, the contractor shall show that the airplane has suitable response characteristics in atmospheric disturbances (3.7 and 3.8).

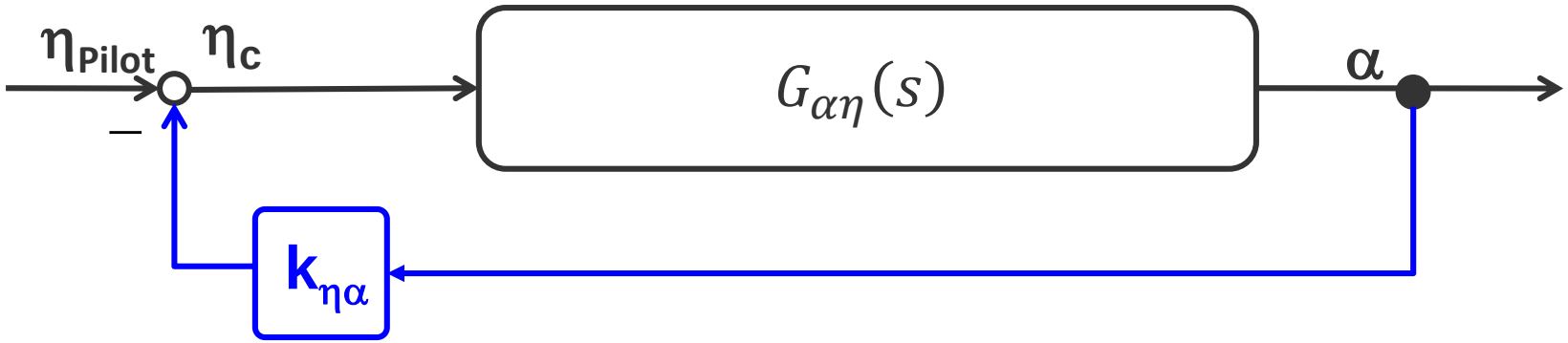
3.2.2.1.1 Short-period frequency and acceleration sensitivity. The equivalent short-period undamped natural frequency, ω_{nSP}, shall be within the limits shown on figures 1, 2, and 3. If suitable means of directly controlling normal force are provided, the lower bounds on ω_{nSP} and n/α of figure 3 may be relaxed if approved by the procuring activity.

(from MIL-F-8785C:
Military Specification – Flying Qualities of Piloted Airplanes)

$$\omega_{0,SP}^2 = -CAP \cdot \frac{V_0}{g} \cdot Z_\alpha$$

A more detailed discussion about the CAP can be found in the lecture “Fundamentals of Flight Control”. Here we just focus on changing the eigenfrequency of the short period.

Controller Structure/Root locus α-feedback

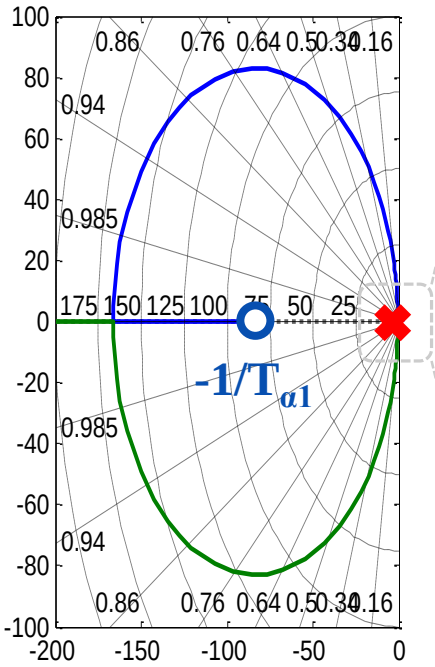


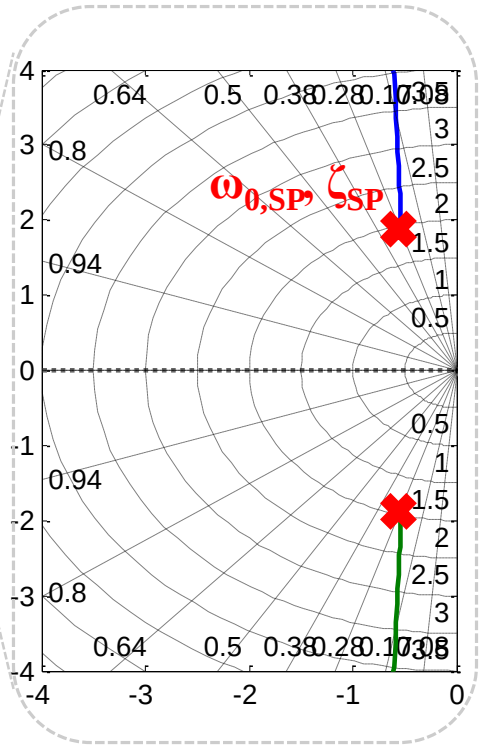
A feedback of the angle of attack to the elevator yields the following root locus:

- For increasing the feedback gain **the eigenfrequency of the short period increases.**
- For reasonable feedback gains, the relative damping remains almost constant.

Reason: Production of an additional moment proportional to the angle of attack

→ See next slide



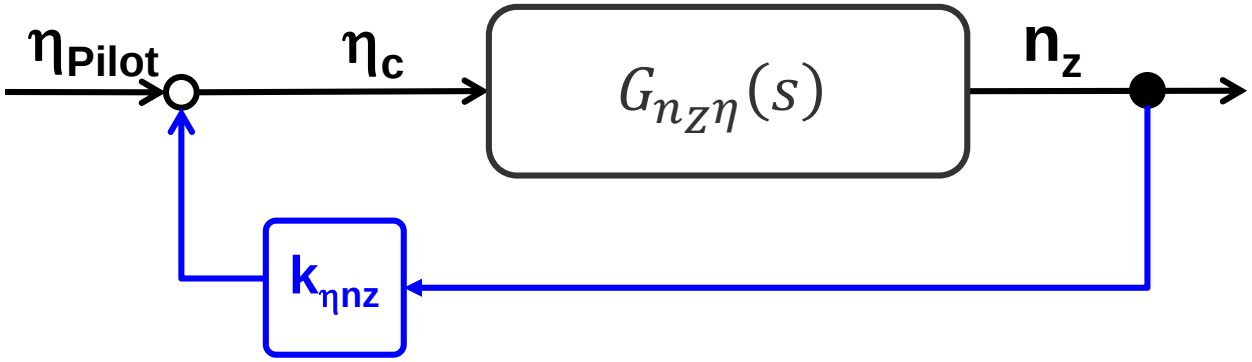


Discussion α-Feedback

- With a pitch stabilizing and pitch damping feedback (pitch controller) natural frequency and relative damping of the short period can be set
- **In the shape of this lecture, the controller design was demonstrated for the simplified 2x2-model**. For **practical application** the **whole state space model (including actuators) has to be analyzed and taken into account**
- Because of the working principle of the sensors, the angle of attack α cannot be measured with a high bandwidth or the measurement is not available at all. As measuring the angle of attack is important for setting the frequency of the short period, we introduce alternative feedback quantities for changing the natural frequency of the short period:

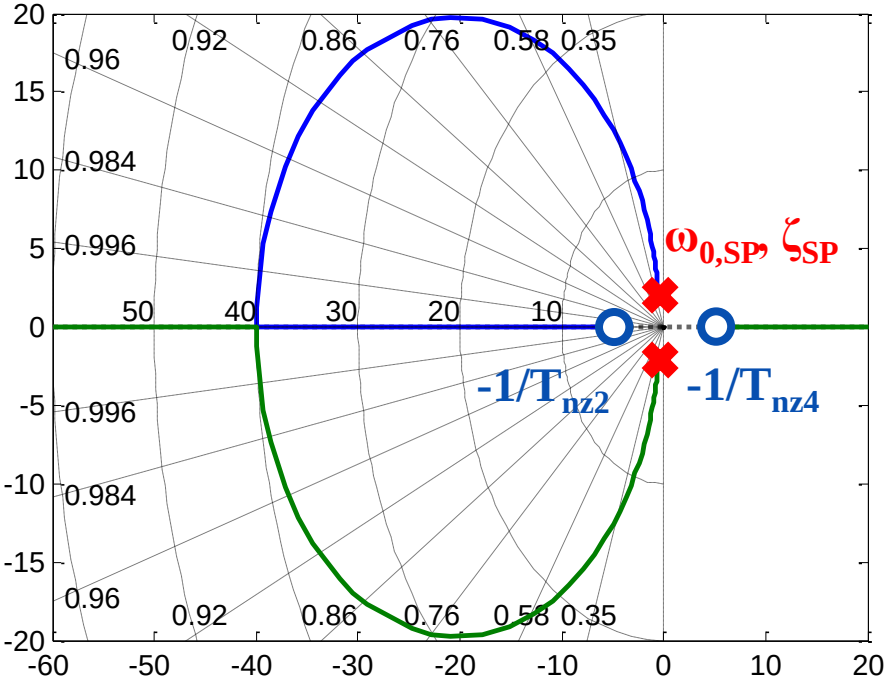
1. **Feedback of the load factor to the elevator**
2. **Pitch controller with a lead lag filter**

Controller Structure/Locus n_z-Feedback

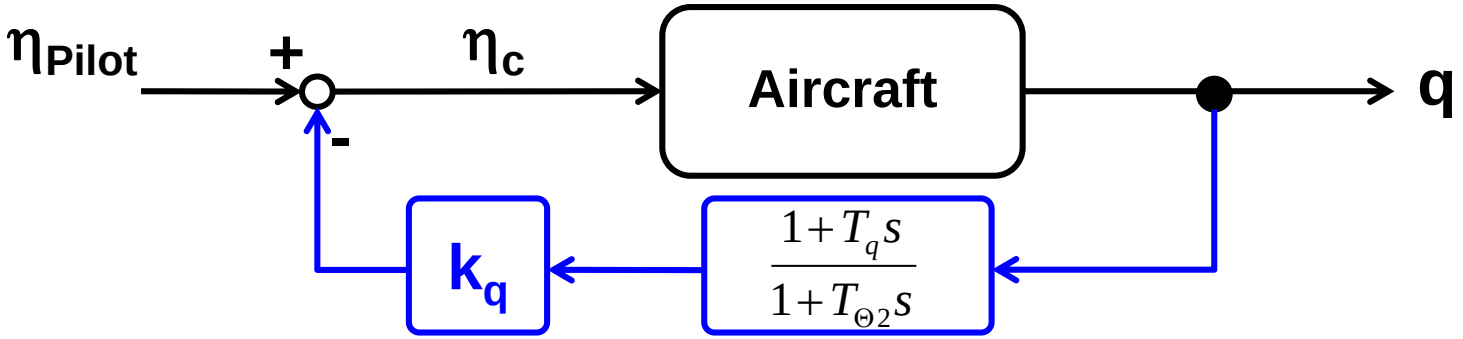


A feedback of the loadfactor leads to the following root locus:

- Increasing the feedback gain of the load factor **increases the natural frequency of the short period.**
- The relative damping remains constant for reasonable feedback gains.



Pitch controller with lead lag filter in the feedback



Lead lag filter to estimate α from q:

(see recap on next slide)

$$\delta\alpha = \frac{T_{\Theta 2}}{(T_{\Theta 2}s + 1)} \cdot \delta q$$

$$T_{\Theta 2} \approx -\frac{1}{Z_\alpha}$$

Derivation Pitch Controller

Another feedback possibility can be obtained by looking at the relation between angle of attack and pitch rate from the 2x2 approximation:

$$\dot{\alpha} = Z_\alpha \alpha + q \qquad \text{with} \qquad Z_q = Z_\eta = \dot{\gamma}_0 = \gamma_0 = 0$$

Laplace Transformation:

$$s\alpha = Z_\alpha \alpha + q \longrightarrow (s - Z_\alpha)\alpha = q$$

Transfer function $G_{\alpha q}(s)$:

$$G_{\alpha q}(s) = \frac{\alpha(s)}{q(s)} = \frac{1}{s - Z_\alpha} = \frac{-1/Z_\alpha}{-1/Z_\alpha \cdot s + 1} = \frac{T_{\Theta_2}}{T_{\Theta_2} s + 1}$$

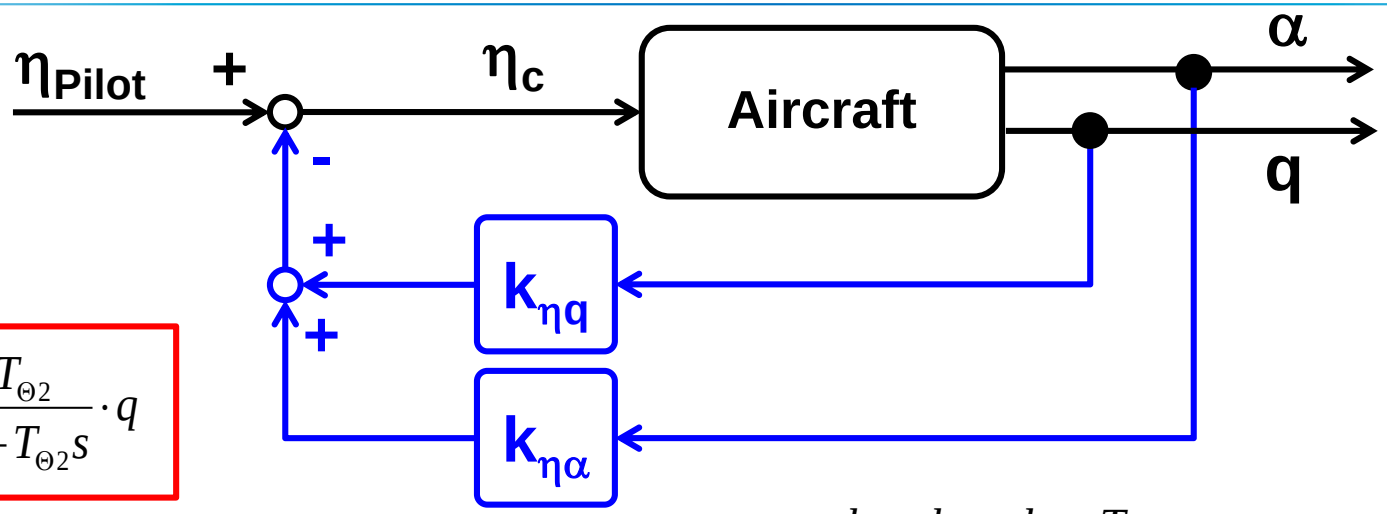
$$\alpha(s) = \frac{T_{\Theta_2}}{T_{\Theta_2} s + 1} q(s)$$

$$T_{\Theta_2} = -1/Z_\alpha$$

Reminder: The assumptions for these equations are:

1. Steady state cruise: $\dot{\gamma}_0 = 0, \gamma_0 = 0$
2. $Z_\eta = 0$
3. $Z_q + 1 = 1$

Controller Structure Pitch Controller



$$\alpha = \frac{T_{\Theta 2}}{1 + T_{\Theta 2} s} \cdot q$$

$$\Delta\eta = -\left(k_{\eta q} q + k_{\eta\alpha} \Delta\alpha\right) = -\left(k_{\eta q} + k_{\eta\alpha} \frac{T_{\Theta 2}}{1 + T_{\Theta 2} s}\right) q = -k_q \cdot \frac{1 + T_q s}{1 + T_{\Theta 2} s} \cdot q$$

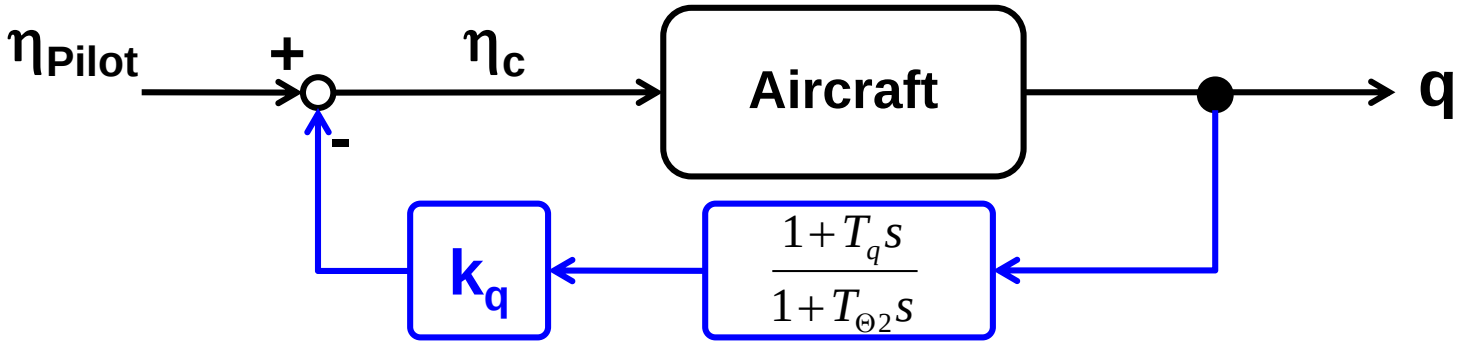
$$k_q = k_{\eta q} + k_{\eta\alpha} \cdot T_{\Theta 2}$$

$$T_q = \frac{k_{\eta q} \cdot T_{\Theta 2}}{k_{\eta q} + k_{\eta\alpha} \cdot T_{\Theta 2}}$$

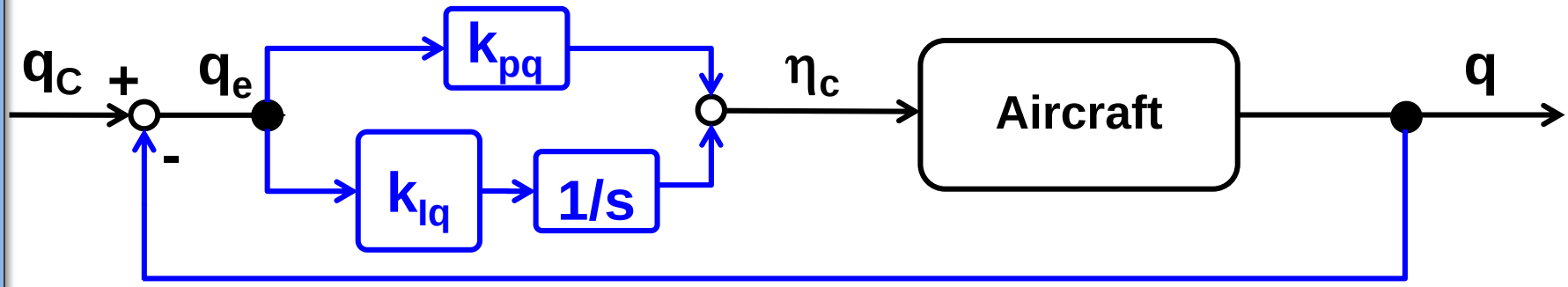
k_q and T_q can be chosen to set the damping and the frequency of the short period

Lead lag filter as a replacement for angle of attack feedback

Pitch controller with lead lag fitler in the feedback

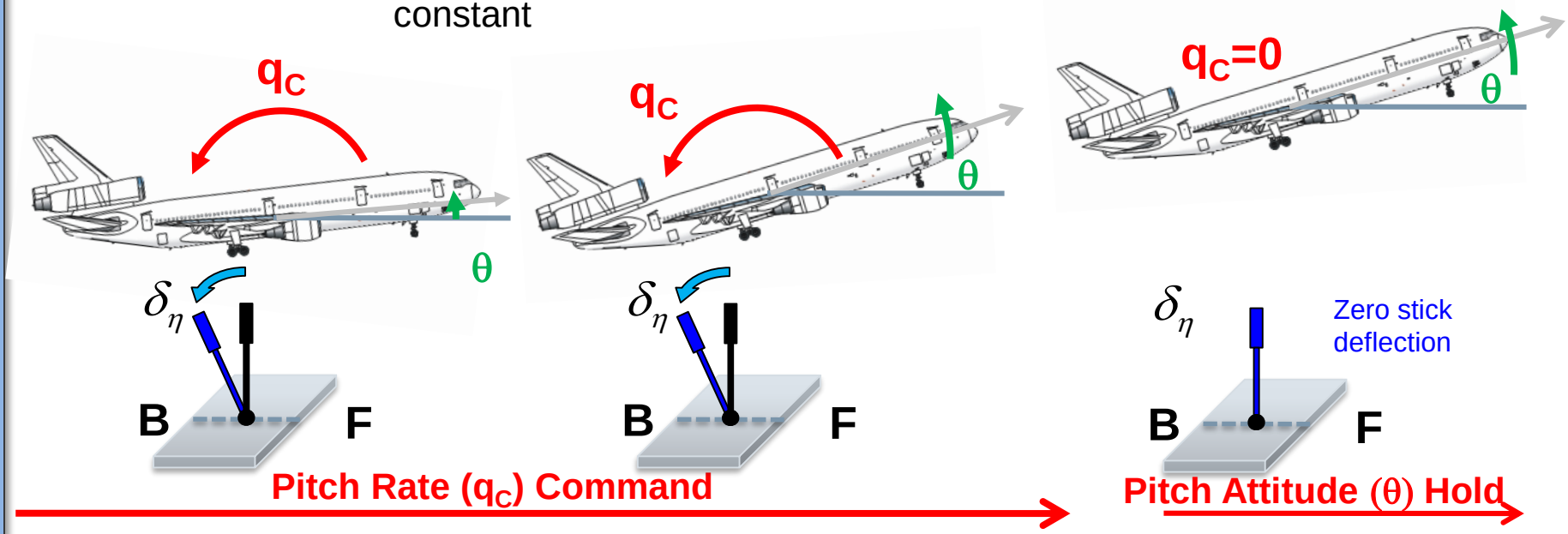


Pitch Rate Command and Attitude Hold System



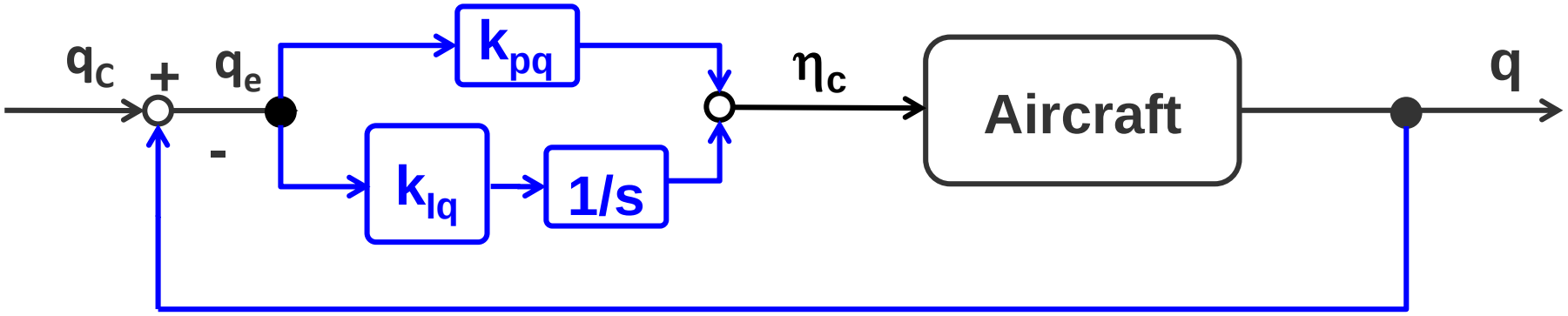
RATE COMMAND The pilot stick commands a pitch rate q_C

ATTITUDE HOLD Zero stick position leads to a pitch rate q=0 and keeps the pitch angle θ constant



Pitch RCAH System (2)

Pitch rate command and attitude hold system



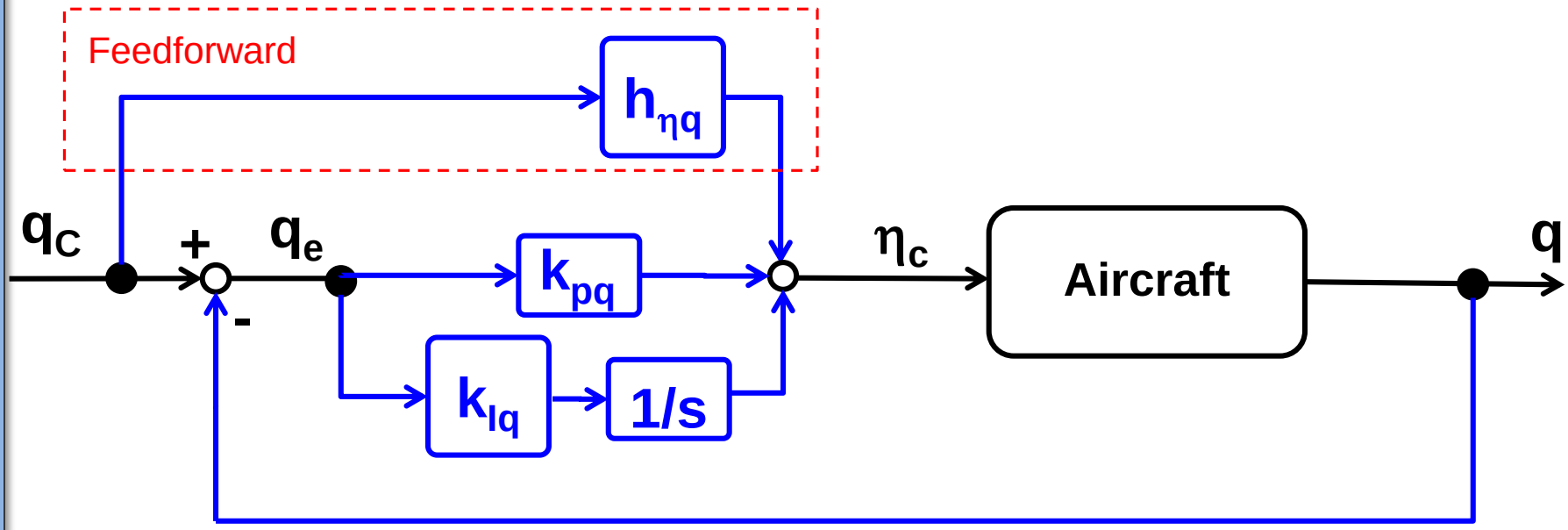
Proportional Part:

- Improves short period damping

Integrator Part:

- Via error integration, this part ensures that the stationary commanded value is reached (commanded pitch rate)
- Compensates disturbances
- Stabilizes the attitude
- Worsens the dynamic stability

Pitch Rate Command and Attitude Hold System with Feedforward



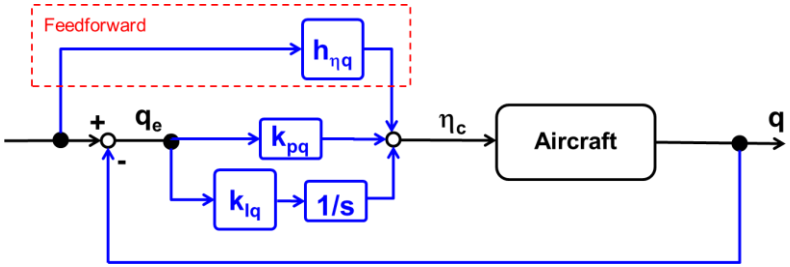
Meaning of the feedforward:

- Direct feedthrough of the command signal q_c to the elevator deflection
- In a stationary state the pitch rate q equals the commanded pitch rate q_c (even without integrator) → see next slide
- The integrator now only handles disturbances

Pitch RCAH System with Feedforward(2)

With the help of transfer functions, we would like to understand which effect is cause by using an RCAH structure with a feedforward. Strarting from the function $(q_c - q) \rightarrow \eta_c$:

$$\eta_c = (q_c - q) \cdot \left(k_{pq} + \frac{1}{s} \cdot k_{Iq}\right) + \mathrm{h}_{\eta q} \cdot q_c$$



Dividing into q_c and q

$$\eta_c = q_c \cdot \left(k_{pq} + \frac{1}{s} \cdot k_{Iq} + h_{\eta q}\right) - q \cdot \left(k_{pq} + \frac{1}{s} \cdot k_{Iq}\right)$$

Place the zero in the location of the CL integrator pole:
-> from the pilot‘s perspective, the integrator is not „visible“

$$\eta_c = q_c \cdot \left(\frac{k_{Iq} + s \cdot (k_{pq} + h_{\eta q})}{s}\right) - q \cdot \left(\frac{k_{Iq} + k_{pq} \cdot s}{s}\right)$$

The feedforward produces a second zero, which can be placed by the additional degree of freedom $h_{\eta q}$. This is chosen so that the stationary output of the system equals the commanded value. The integrator part is then only required for disturbances.

One zero is introduced by the RCAH system

Motivation: Model Reduction

The same reasoning for the simplification of the longitudinal motion holds for the lateral motion:

Main Contributors Dutch Roll

Main Contributors Roll Motion

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} N_r & N_\beta & N_p & 0 \\ -1 & Y_\beta & 0 & \frac{g}{V_0} \\ L_r & L_\beta & L_p & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} N_\xi & N_\zeta \\ Y_\xi & Y_\zeta \\ L_\xi & L_\zeta \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

Reduced model for the Roll Motion (The spiral pole is approximated with an integrator):

$$\begin{bmatrix} \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} L_p & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} p \\ \Phi \end{bmatrix} + \begin{bmatrix} L_\xi \\ 0 \end{bmatrix} \cdot \xi$$

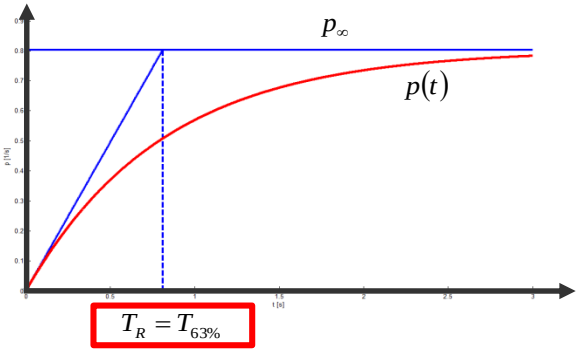
Reduced model for the Dutch Roll:

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \end{bmatrix} = \begin{bmatrix} N_r & N_\beta \\ -1 & Y_\beta \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \end{bmatrix} + \begin{bmatrix} N_\zeta \\ Y_\zeta \end{bmatrix} \cdot \zeta$$

IMPORTANT: For reducing complexity we only focus on the poles which are changed by the corresponding feedback in the first place.

Motivation: Requirements Roll Motion

The **MIL-F-8785C** gives **maximum roll time constants** depending on the category of flight phase, aircraft class and the level of handling quality



MIL-F-8785C

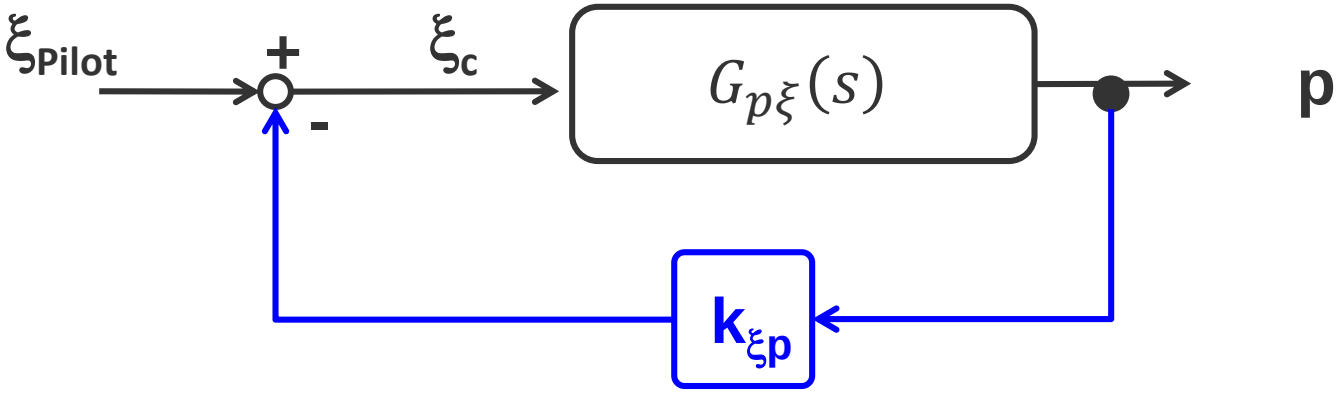
3.3.1.2 Roll mode. The roll-mode time constant, τ_R, shall be no greater than the appropriate value in table VII.

Maximum Roll Time Constants T_R [s] (MIL-F-8785C)

Flight Phase Category	Class	Level 1	Level 2	Level 3
A	I, IV	1.0	1.4	10
	II, III	1.4	3.0	10
B	AII	1.4	3.0	10
C	I,II-C,IV	1.0	1.4	10
	II-L, III	1.4	3.0	10

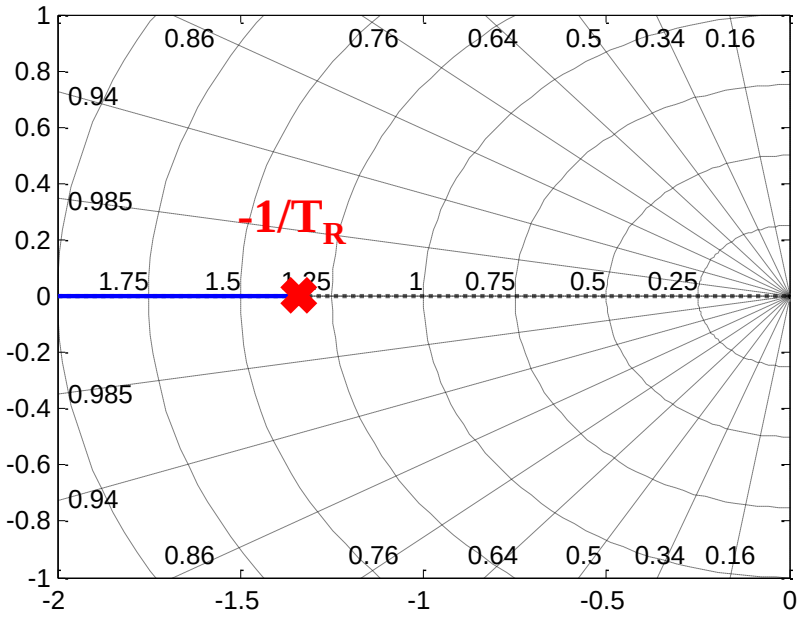
Maximum roll time constant $T_R = -\frac{1}{L_p}$ ⇨ **Minimum roll damping** L_p

Controller Structure/Root Locus Roll Damper



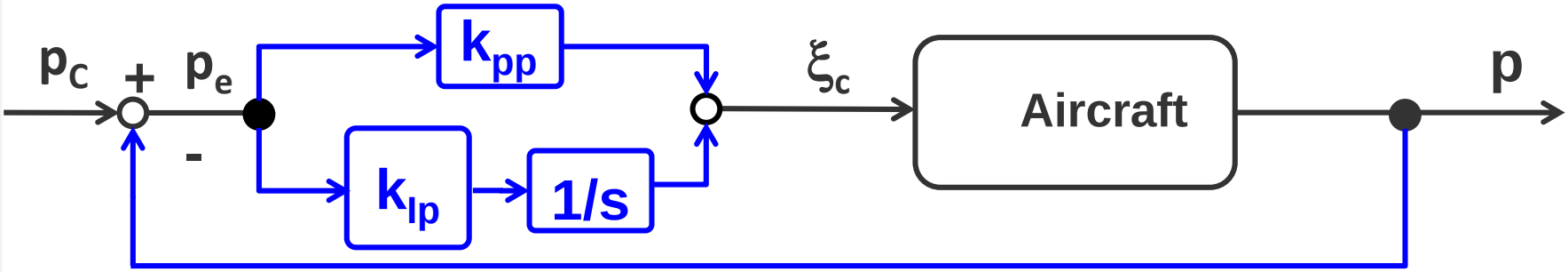
A feedback of the roll rate to the aileron leads to the following root locus:

- Increasing the feedback gain moves the roll pole to the left. The roll time constant T_R becomes smaller and the damping higher



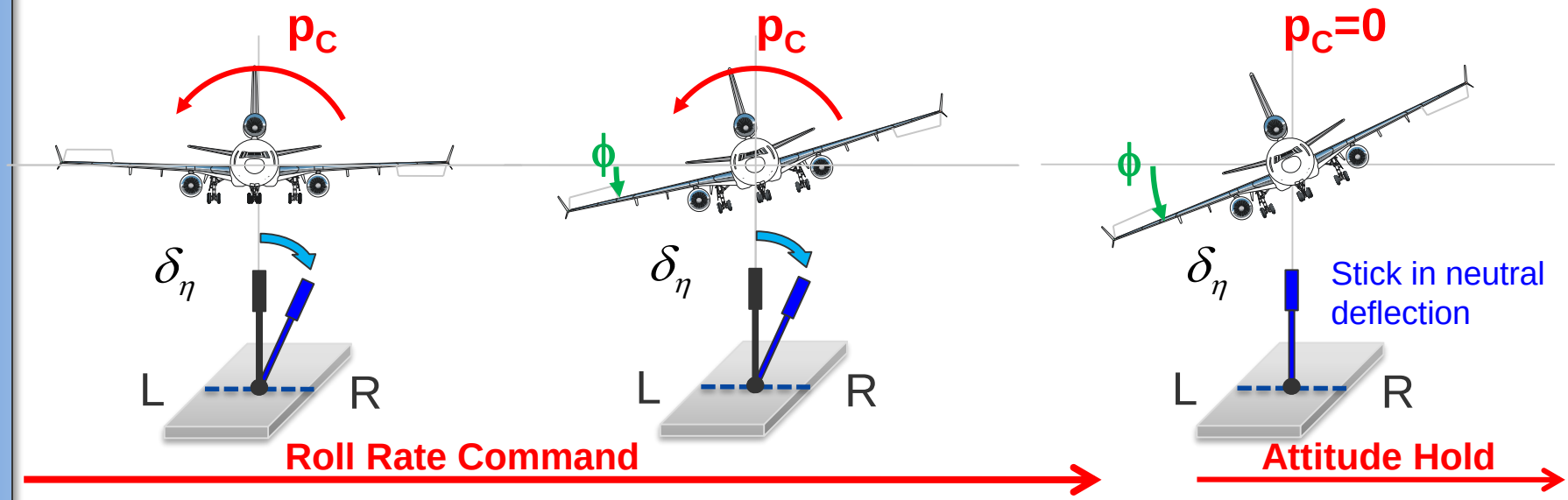
Controller Structure RCAH Roll Motion

Roll Rate Command and Attitude Hold System

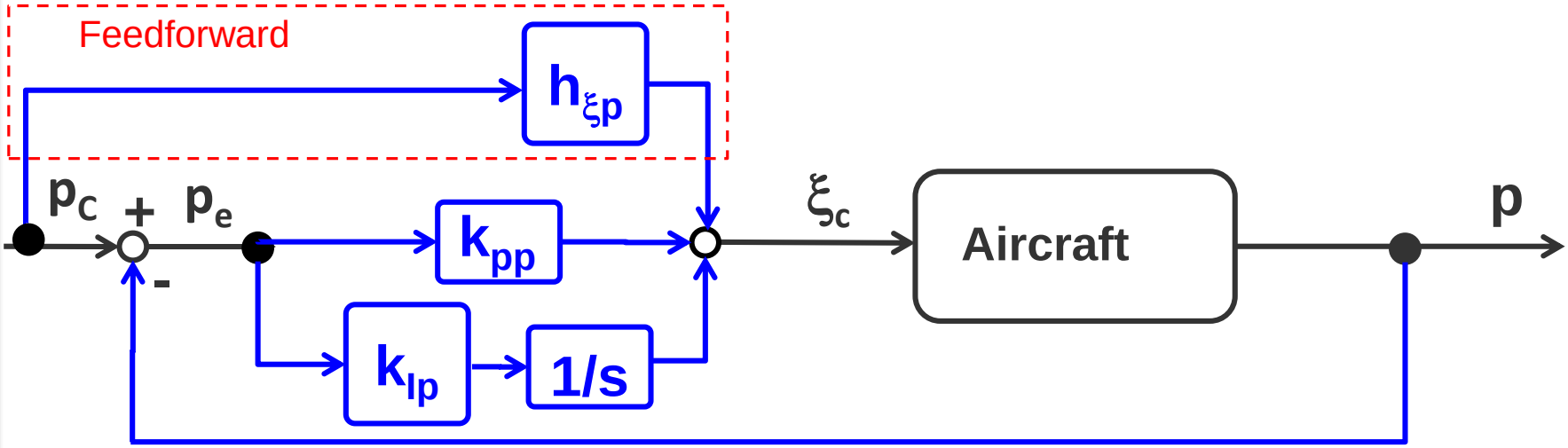


RATE COMMAND The pilot stick commands a roll rate p_C

ATTITUDE HOLD Leaving the stick at neutral position leads to a pitch rate $p=0$ and a hold of the attitude Φ (bank angle)

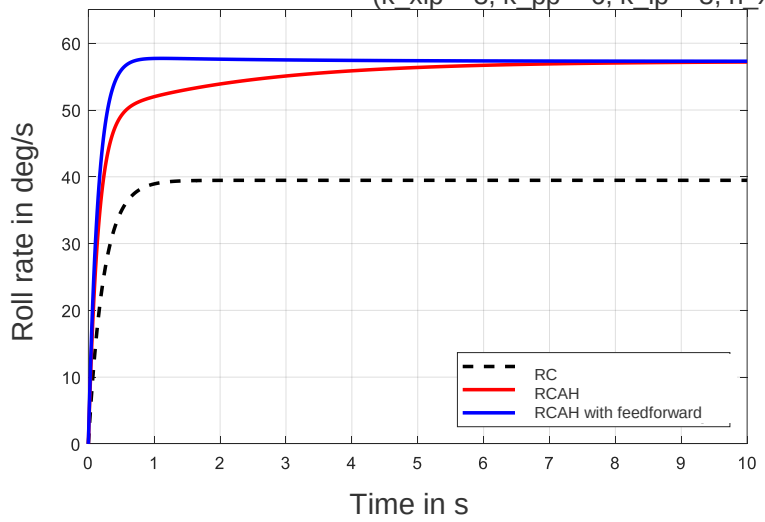


Roll Rate Command and Attitude Hold System with Feedforward



Comparison of Step Responses

Step response of the roll rate to a roll rate command for an RCAH system
(k_xip = 3, k_pp = 6, k_Ip = 3, h_xip = 1)



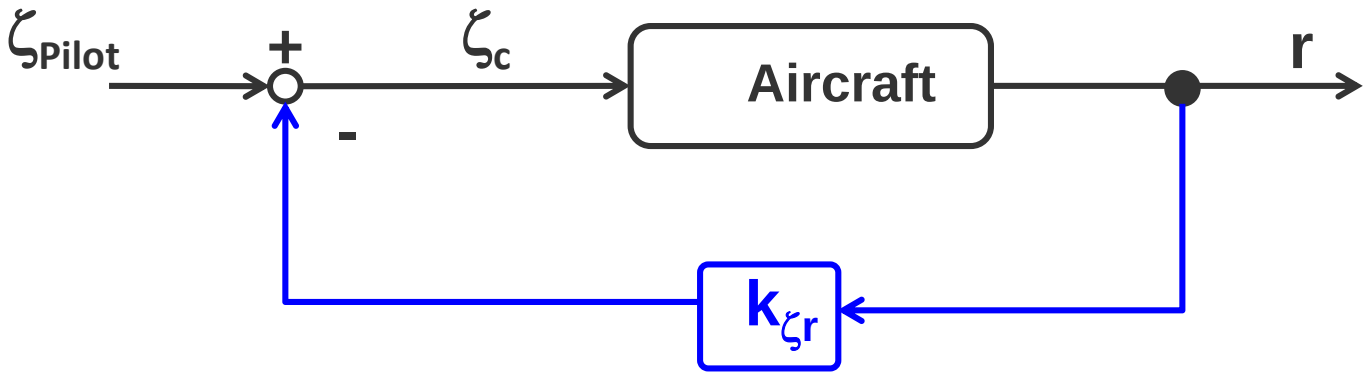
Motivation: Requirements Dutch Roll

- Requirements for the **relative damping ζ_{DR}**, the absolute damping σ_{DR} and the natural frequency $\omega_{0,DR}$ of the Dutch Roll
  (MIL-F-8785C : Military Specification – Flying Qualities of Piloted Airplanes)

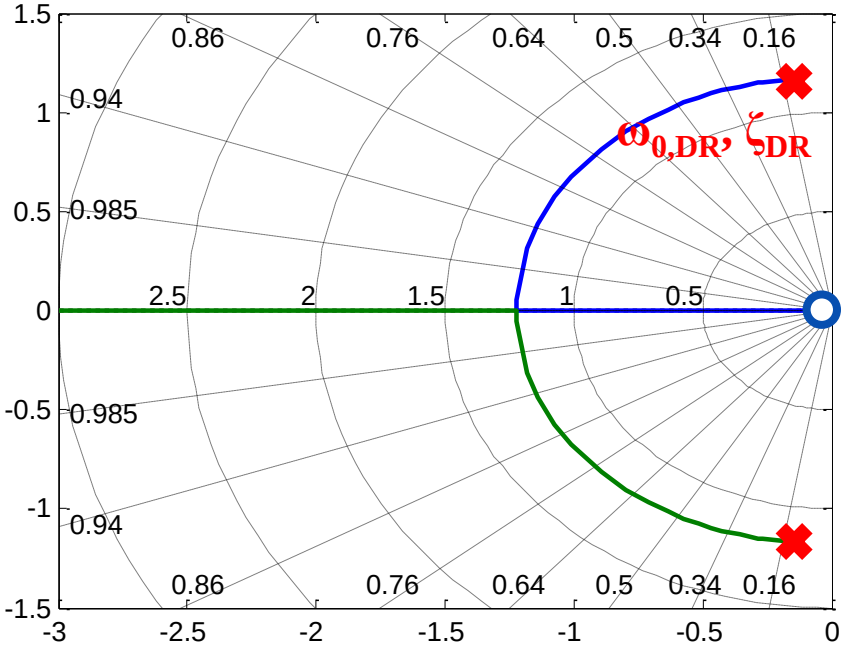
Level	Flight Phase	Class	Min ζ_d*	Min $\zeta_d\omega_{0d}$ * rad/sec	Min ω_{0d} rad/sec
1	A	II/III	0,19	0,35	0,4
	B	All	0,08	0,15	0,4
	C	II-L/III	0,08	0,10	0,4
2	All	All	0,02	0,05	0,4
3	All	All	0	-	0,4

* The main criteria leads to a higher value for ζ_d. The optimal ζ_d is 0.7 for class III.

Controller Structure/Root Locus Yaw Damper



- For modern aircraft the damping of the Dutch roll is quite small
- The damping can be increased by a feedback of yaw rate r to a rudder deflection ζ (see root locus)
- The frequency of the Dutch roll almost remains constant
- This structure is also referred to as yaw damper

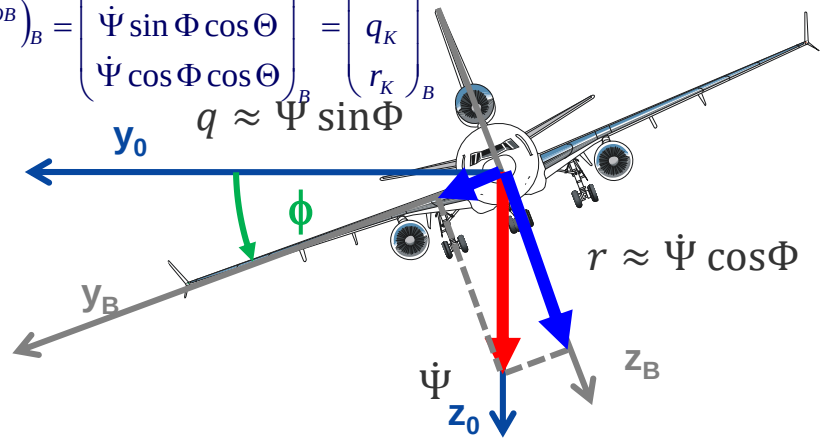


Discussion Yaw Damper

Body fixed angular rates, horizontal turn

$$\Rightarrow \left(\vec{\omega}^{OB}\right)_B = \begin{pmatrix} -\dot{\Psi}\sin\Theta \\ \dot{\Psi}\sin\Phi\cos\Theta \\ \dot{\Psi}\cos\Phi\cos\Theta \end{pmatrix}_B = \begin{pmatrix} p_K \\ q_K \\ r_K \end{pmatrix}_B$$

$$q \approx \dot{\Psi}\sin\Phi$$



$$r \approx \dot{\Psi}\cos\Phi$$

Attitude Propagation

$$\left(\omega_K^{OB}\right)_B = \begin{bmatrix} \dot{\Phi} - \dot{\Psi}\sin\Theta \\ \dot{\Theta}\cos\Phi + \dot{\Psi}\sin\Phi\cos\Theta \\ \dot{\Theta}\sin\Phi + \dot{\Psi}\cos\Phi\cos\Theta \end{bmatrix}$$

Stationary Turn: $\dot{\Phi} = 0, \dot{\Theta} = 0$

$$\left(\omega_K^{OB}\right)_B = \begin{bmatrix} \dot{\Psi}\sin\Theta \\ \dot{\Psi}\sin\Phi\cos\Theta \\ \dot{\Psi}\cos\Phi\cos\Theta \end{bmatrix}$$

Small Pitch angle: $\Theta = const.$

$$p_K \approx 0 \qquad q_K \approx \dot{\Psi}\sin\Phi$$

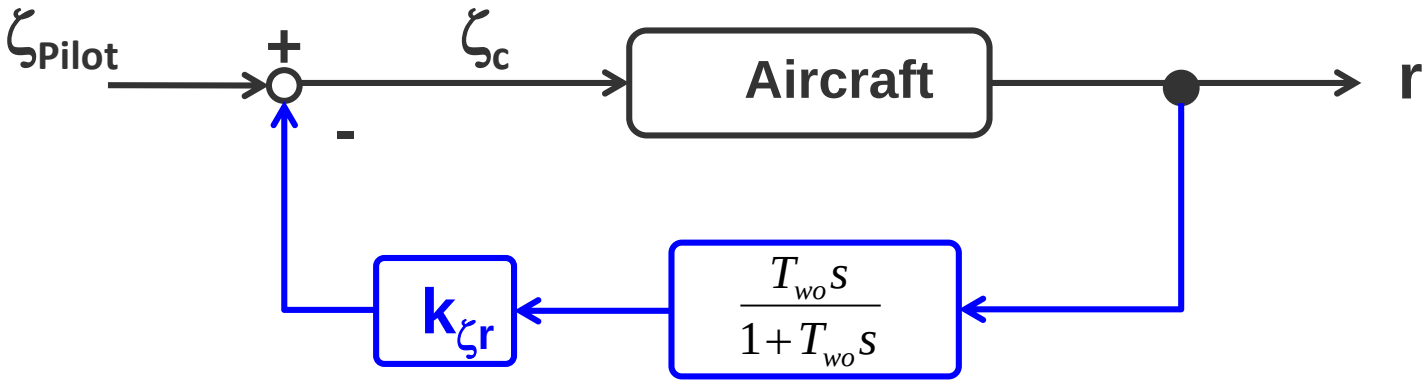
$$r_K \approx \dot{\Psi}\cos\Phi$$

Problem of a yaw rate feedback

- For horizontal turn there is a yaw rate *r* and a pitch rate *q* in the body fixed system.
- A feedback of the yaw rate *r* to the rudder z does not only suppress the Dutch roll, but also low frequent yaw motion (and rotation around the geodetic z-axis), so that stationary turn is suppressed too

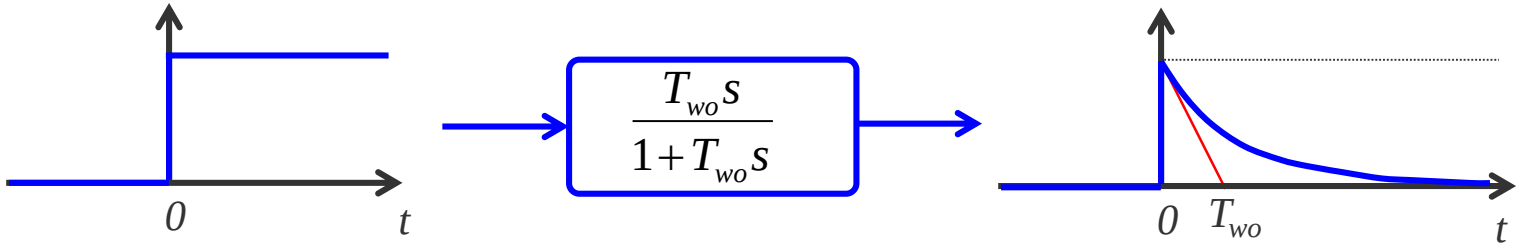
Solution: High pass filter in the feedback of the yaw rate, so that the yaw damper becomes ineffective for low frequencies.

Yaw Damper with High Pass



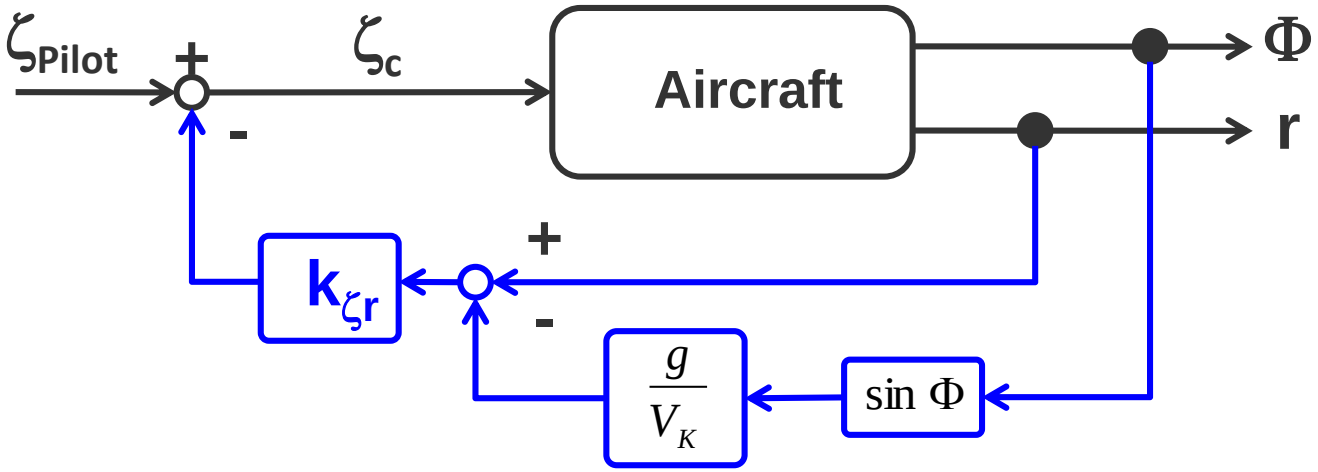
- **Feedback of the yaw rate over a high pass filter to the rudder, so the the yaw damper becomes inactive for stationary signals (small frequency). This avoids the suppression of stationary turn and maneuver flight via the feedback of the yaw rate r.**

High Pass Filter DT_1 (Wash-Out-Filter)

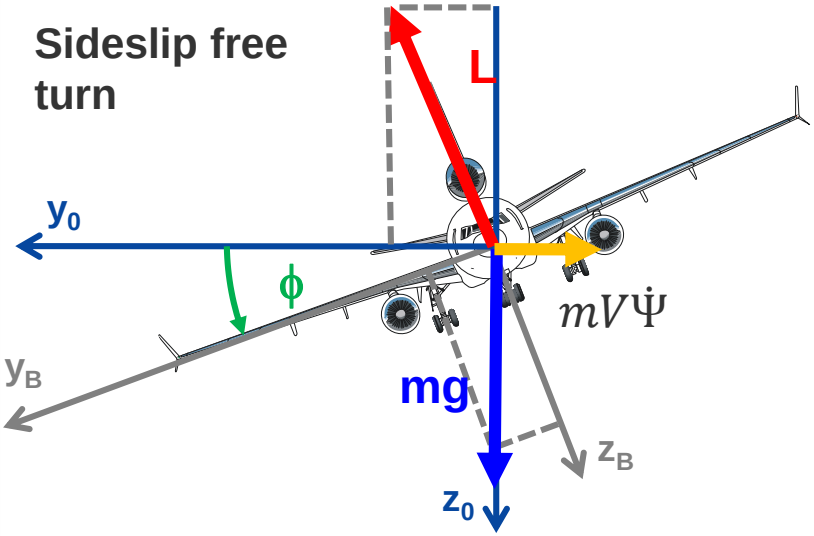


Alternative for the High Pass Filter

Instead of using a highpass filter, the yaw rate due to turn g/V·sinϕ can be subtracted from the measured yaw rate.



Sideslip free turn



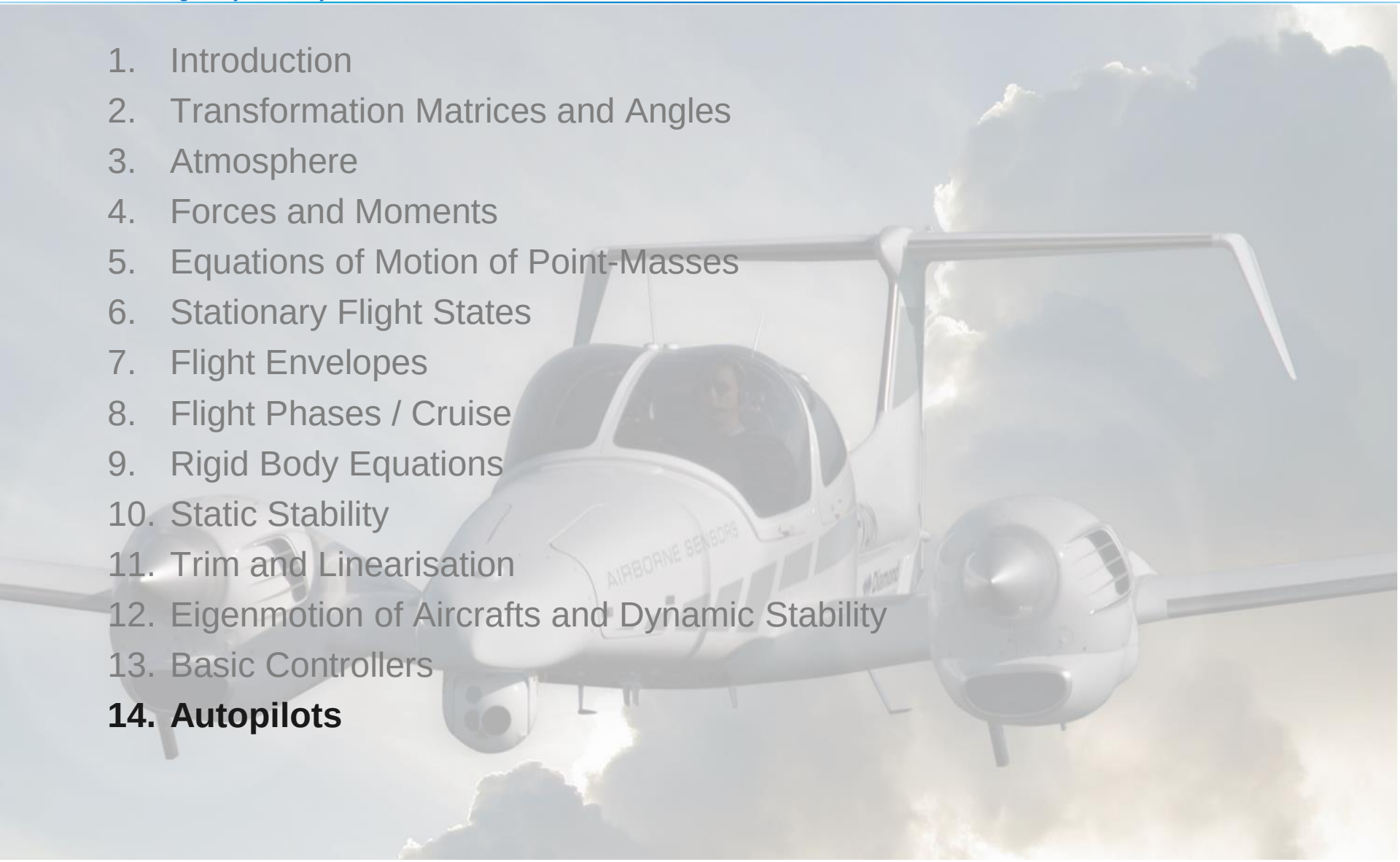
Force balance

z_0-direction: $L \cdot \cos\Phi = mg$ $\Rightarrow L = \frac{mg}{\cos\Phi}$

y_0-direction: $mV\dot{\Psi} = L\sin\Phi$

$$\dot{\Psi} = \frac{g}{V\cos\Phi}\sin\Phi$$

$$r = \dot{\Psi}\cos\Phi\cos\Theta \approx \dot{\Psi}\cos\Phi \Rightarrow \boxed{r = \frac{g}{V}\cdot\sin\Phi}$$

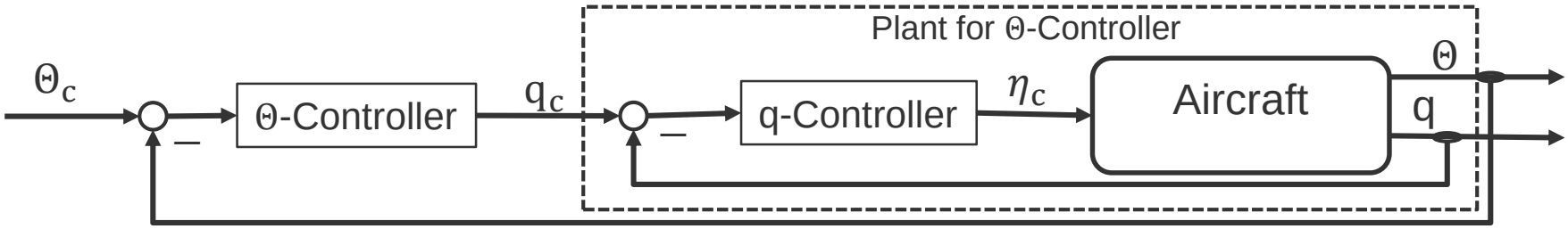


1. Introduction
2. Transformation Matrices and Angles
3. Atmosphere
4. Forces and Moments
5. Equations of Motion of Point-Masses
6. Stationary Flight States
7. Flight Envelopes
8. Flight Phases / Cruise
9. Rigid Body Equations
10. Static Stability
11. Trim and Linearisation
12. Eigenmotion of Aircrafts and Dynamic Stability
13. Basic Controllers
14. **Autopilots**

Cascade Structure

Besides requirements for the stability of all control loops, the MIL-DTL-9490-E poses other requirements on performance and accuracy.
The stability requirements can be achieved by adapting the position of poles in the complex plane.

We will see, that for some types of operation, a simple feedback of the controlled variable, is not enough to reach a stable as well as accurate response in the desired time. We need to feedback multiple signals. For this we use the principle of **"Cascade Control":**



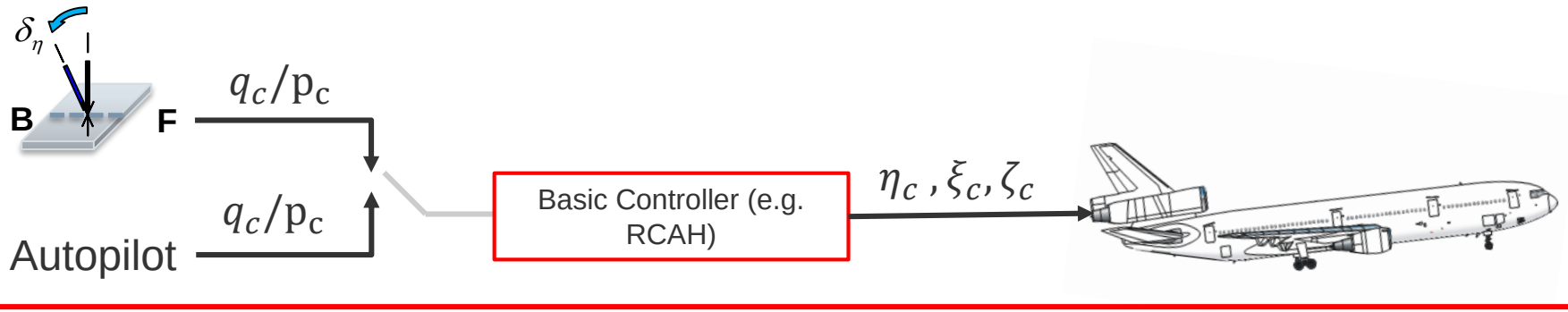
Cascade structure for a pitch angle controller

The cascade divides the controller according to the individual feedbacks, the individual feedback loops built up on each other.

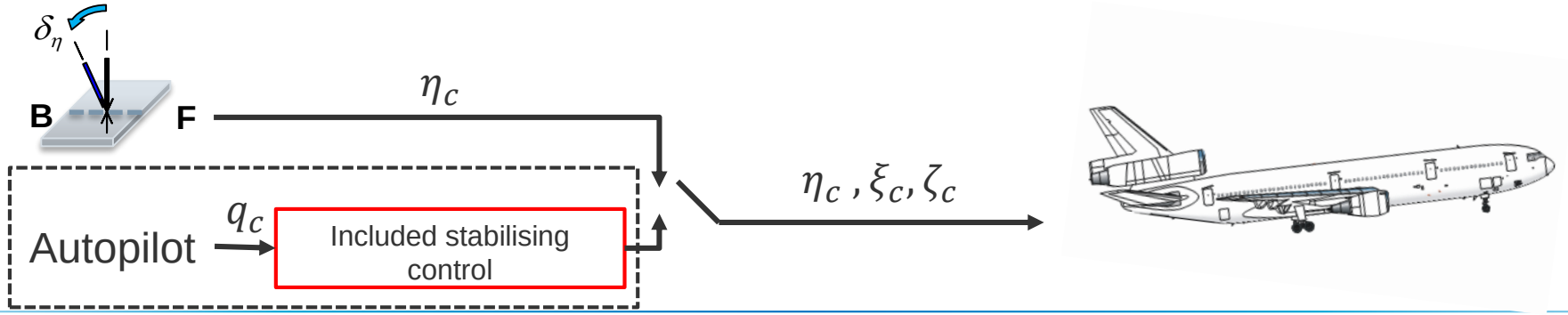
Advantage: We can analyze the individual loops of the cascade individually. For the outer loops, more inner loops are seen as part of the plant.

The structure of an autopilot depends on whether and what type of basic control system is available.

Possibility A: The basic control system is a SAS or a CAS. The autopilot then uses these systems.

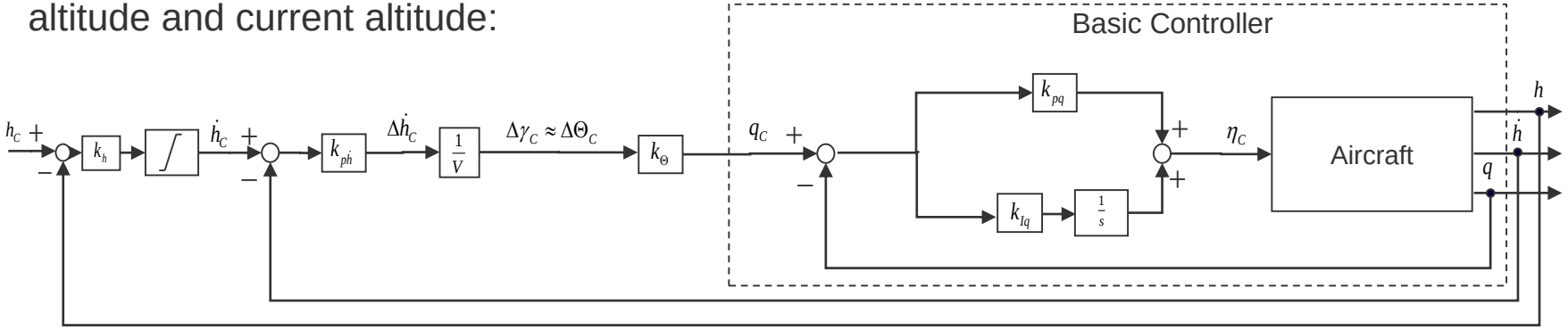


Possibility B: The autopilot directly controls surface deflections. For this case the autopilot requires its own basic controller.



Controller Structure

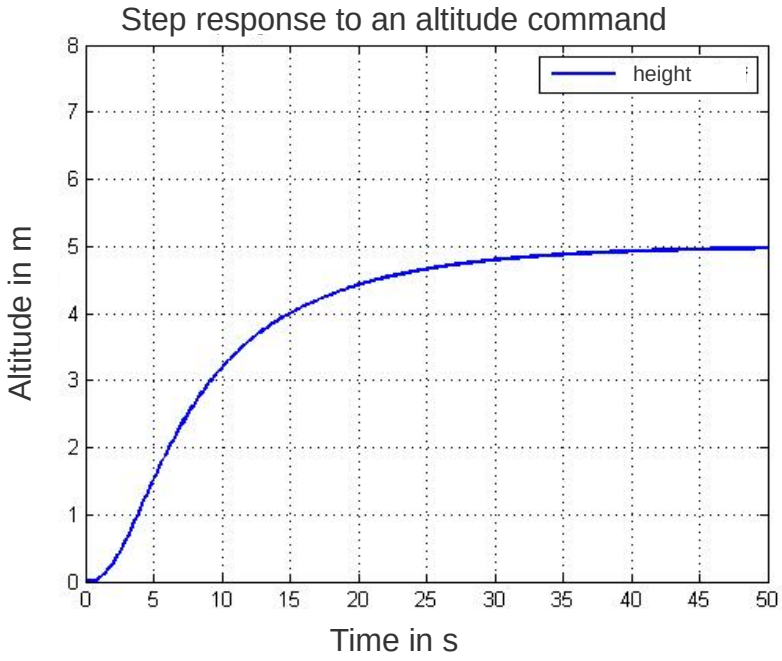
The controller structure follows the attribute of the system, that h is a pure integration of $\dot{h}$. We can generate a command for the $\dot{h}$-controller via a comparison of commanded altitude and current altitude:



The gain is chosen so that an aperiodic response is reached:

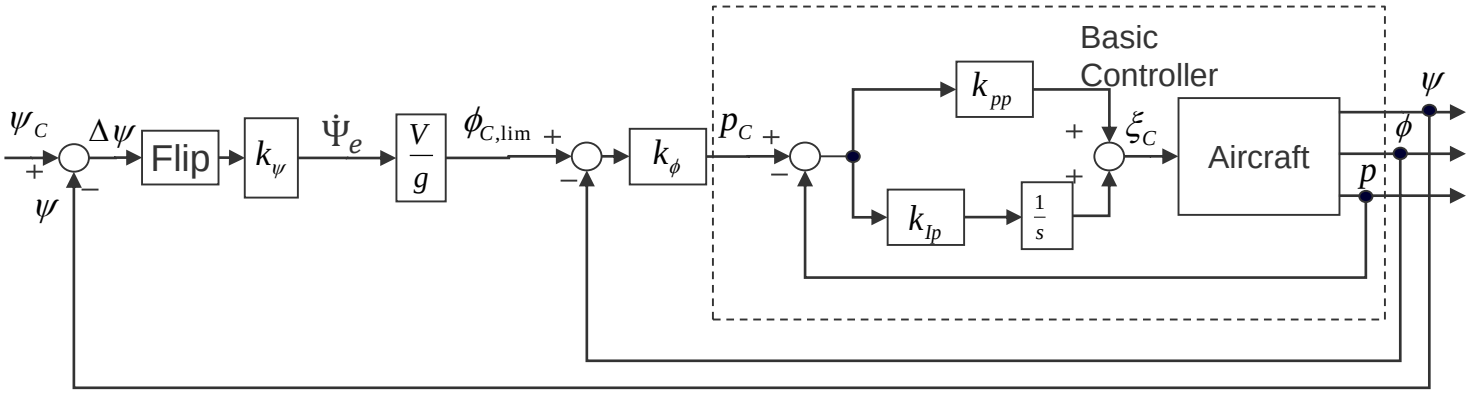
- An oscillation (even for small amplitudes) would be uncomfortable for the passengers.
- Slower response is accepted due to this.

Following figure shows the step response of our B777 example system:



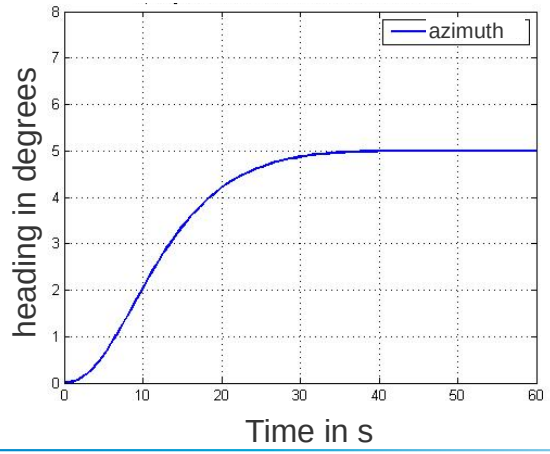
Controller Structure

We now have a relation between $\dot{\Psi}$ and Φ. This leads to the following structure:



From the azimuth control error $\Delta\Psi$ we generate a $\dot{\Psi}$ command, which can be transformed in a command for the bank angle.
For doing the minimum difference for the whole heading range 360° to 0° we require a “Flip“-block,

Response of the heading angle to a heading angle step command



The figure shows the step response of the example B777 system:

IMPORTANT: Unlike for the longitudinal motion, we do not command increments here, but absolute values Φ_c.

We can derive turn limitations for the derived controller structure:

1. We would like to fly a turn with maximum 30° bank angle.
2. A standard turn has $\Delta\dot{\chi} \approx \Delta\dot{\Psi} = 3\,\frac{\mathrm{deg}}{\mathrm{s}}$.

Both quantities are given as command quantities in the controller structure and therefore can be limited:

